Class 6

Mathematics

Solutions

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1. Number System-1

Estimating and Comparing Numbers

LEVEL – I

1. Comparison: 102394 ___ 99887

102394 is a 6-digit number
99887 is a 5-digit number
Any 6-digit number is always greater than any 5-digit number.
So,
102394 > 99887
Correct Answer: A) >

2. Comparison: 2507324 ___ 2517324

Compare digit by digit:
Both have 7 digits
Compare from left:
2 = 2
5 = 5
0 < 1
So,2507324 < 2517324
Correct Answer: B) <

3. Numbers:63547201, 102345694, 6539542, 8354208, 12345678

Step-by-step:
6539542 (7 digits) → smallest)
8354208
12345678
63547201
102345694 (largest)
Ascending order:6539542, 8354208, 12345678, 63547201, 102345694
Correct Answer: B)

4. Numbers:056944000, 56943201, 5695440, 56943300, 5694437

(Remove leading zero: 056944000 = 56,944,000)
Compare:
56,944,000 (largest)
56,943,300
56,943,201
5,695,440
5,694,437
Descending order:056944000, 56943300, 56943201, 5695440, 5694437
Correct Answer: C)

5. Greatest and smallest 5-digit numbers

Digits: 5, 0, 2, 8, 1
Greatest: arrange in descending (no leading zero)
→ 8 5 2 1 0 = 85,210
Smallest: smallest non-zero first, then ascending
→ 1 0 2 5 8 = 10,258
Correct Answer: C) 85,210; 10,258

6. Greater number and difference

71,23,400 = 7,123,400
7,12,340 = 712,340
Greater number: 71,23,400
Difference:
7,123,400 − 712,340 = 6,411,060 = 64,11,060
Correct Answer: D) 71,23,400; 64,11,060

7. Number of 5-digit natural numbers

Smallest 5-digit number = 10000
Largest = 99999
Total = 99999 − 10000 + 1 = 90000
Correct Answer: B) 90000

8. Smallest 4-digit palindrome

A number that remains same when reversed = palindrome
Check options:
1001
1111
1010 (becomes 0101)
1011
Smallest valid = 1001
Correct Answer: D) 1001

9. Estimate sum (nearest tens)

Round each:
21396 → 21400
27808 → 27810
42504 → 42500
Sum = 21400 + 27810 + 42500 = 91710
Correct Answer: A) 91710

10. Sum by estimation (greatest place)

Round each number to its greatest place:

i) 1335 + 8657
1335 → 1000
8657 → 9000
Sum = 10000

ii) 2675 + 9087
2675 → 3000
9087 → 9000
Sum = 12000

Correct Answer: B) 10000; 12000

11. Difference by estimation (greatest place)

i) 275 − 167
275 → 300
167 → 200
Difference = 100

ii) 9312 − 7603
9312 → 9000

7603 → 8000
Difference = 1000
Correct Answer: D) 100; 1000

12. Place value of '0' in 120945349

Number: 120945349
The digit 0 is in the ten lakhs place (1,000,000 place)
But place value = digit × place = 0 × 10,00,000 = 0
Correct Answer: D) 0

13. Sum of place values of 9 in 54,936,919

Digits 9 occur at:
Hundred thousands place → 900000
Hundreds place → 900
Ones place → 9
Sum = 900000 + 900 + 9 = 900909
Correct Answer: A) 900909

14. Expanded form of 7,303,206

Break into place values:
7,303,206 =
7×1000000 + 3×100000 + 0×10000 + 3×1000 + 2×100 + 0×10 + 6×1
Correct Answer: A)

15. Greatest mass when rounded to nearest 1000

If rounded to 17000, actual values range:
16500 to 17499
Greatest possible value = 17499 g
Correct Answer: C) 17499 g

16. Total distance in a week

Distance one way = 1 km 875 m = 1.875 km
Daily (both ways):
= 1.875 × 2 = 3.75 km
For 7 days:
= 3.75 × 7 = 26.25 km
Correct Answer: A) 26.25 km

17. Descending order

Descending means largest to smallest
Check options:
A) 345 < 456 < 567 (ascending)
B) 567 > 456 > 345
C) 345 > 567 > 456
D) 456 < 567 < 345
Correct Answer: B) 567 > 456 > 345

18. Round off to nearest thousand: 48,672

Check hundreds digit:
48,672 → hundreds part is 672 (≥ 500, so round up)
So,
48,672 ≈ 49,000
Correct Answer: B) 49,000

19. Estimate the difference (nearest thousand)

Round each number to nearest thousand:
85,764 → 86,000
27,489 → 27,000
Now subtract:
86,000 − 27,000 = 59,000
Correct Answer: B) 59,000

20. Expanded form of 3,705,090

Break into place values:
3,705,090 =
3,000,000
700,000
5,000
90
So expanded form:
3,000,000 + 700,000 + 5,000 + 90
Correct Answer: A)

LEVEL – II

Single Answer Questions

1. Smallest and largest 8-digit numbers

Digits in 20089405: 2, 0, 0, 8, 9, 4, 0, 5
Smallest number: smallest non-zero first, then ascending
→ 2 0 0 0 4 5 8 9 = 20004589
Largest number: descending order
→ 9 8 5 4 2 0 0 0 = 98542000
Correct Answer: C) 20004589; 98542000

2. 3-digit numbers using 0, 2, 9 (no repetition)

Total permutations = 3! = 6
But numbers starting with 0 are not allowed.
Valid numbers:
209, 290, 902, 920 → 4 numbers
Correct Answer: A) 4

3. Interchanging digits

Rule:
Tens ↔ Hundreds
Ones ↔ Thousands
A) 7865 transforms to 5687
B) 8865 transforms to 5688
C) 8765 transforms to 5678
D) 7865 transforms to 5687
Comparing the resulting new numbers, 5688 is the largest.
Correct Answer: B) 8865

4. Place value comparison

In 239766, digit 3 is in ten-thousands place → 30,000
In 45655, digit 6 is in hundreds place → 600

Ratio:
30000 ÷ 600 = 50
Correct Answer: B) 50

5. Total toy soldiers
Boxes = 70
Each box → 45 packages
Each package → 25 soldiers
Total = 70 × 45 × 25
First: 45 × 25 = 1125
Then: 1125 × 70 = 78750
Correct Answer: A) 78750

6. Estimated quotient (nearest hundreds)
2838 → 2800
125 → 100
2800 ÷ 100 = 28
Correct Answer: C) 28

7. Product by estimation (greatest place)
i) 102 × 25
102 → 100
25 → 30
Product = 100 × 30 = 3000
ii) 23502 × 23
23502 → 20000 (ten-thousands place)
23 → 20 (tens place)
Product = 20000 × 20 = 400000
Correct Answer: A) 3000; 400000

8. Total pages printed
Copies = 10,420
Pages = 24
10420 × 24 = 250080
Correct Answer: C) 250080

9. Total mass
15 kg 250 g = 15.25 kg
15.25 × 28 = 427 kg
Correct Answer: B) 427 Kg

10. Rounded to nearest tens
839 → 840
Correct Answer: C) 839

One or More Than One Correct Answer Questions

11. Which of the following are correct?
A) 1173 > 73 (true)
B) 8110 < 9000 (true)
C) 117 > 977 (false)
D) 234 > 237 (false)
Correct Answers: A, B

12. Correct place values of digit 8

A) In 8,23,000 → 8 lakh (correct)
B) In 58,000 → 8000 (8 is in thousands place)
C) In 804 → 800 (8 is in hundreds place)
D) In 4,80,000 → 80000 (8 is in ten-thousands place)
Correct Answers: A, B, C, D

13. Numbers greater than 7,50,000

A) 7,49,999 (less)
B) 7,50,001 (greater)
C) 7,60,000 (greater)
D) 7,05,000 (less)
Correct Answers: B, C

14. Which numbers round up to nearest hundred?

Rule:
A number rounds up to the next hundred if its last two digits are 50 or more.
Check each:
A) 1450 → last two digits = 50 → rounds up to 1500
B) 1499 → last two digits = 99 → rounds up to 1500
C) 1449 → last two digits = 49 → rounds down to 1400
D) 1501 → last two digits = 01 → rounds down to 1500
Correct Answers: A, B

Comprehension Type Questions

Passage – I:

Given numbers: 8973, 8192, 8937, 8312, 8729
Step 1: Arrange in ascending order (smallest → biggest)
8192, 8312, 8729, 8937, 8973

15. 2nd number in ascending order

Ascending list:
8192, 8312, 8729, 8937, 8973
Correct Answer: B) 8312

16. Middle number in descending order

Descending order (biggest → smallest):
8973, 8937, 8729, 8312, 8192
Middle (3rd number) = 8729
Correct Answer: A) 8729

17. Difference between 4th and 2nd numbers (ascending order)

Ascending order:
8192, 8312, 8729, 8937, 8973
4th number = 8937
2nd number = 8312
Difference = 8937 − 8312 = 625
Correct Answer: C) 625

Matching Type Questions

18. Matching Type Questions

a) 10 millimetres → Q) 1 centimetre
b) 1 metre → R) 100 centimetres
c) 1 kilometre → P) 1000 metres
d) 1 kilogram → S) 1000 grams
Final Answer:a–Q, b–R, c–P, d–S

Integer Type Questions

19. Thousands place in 2354125

Number: 2,354,125
Thousands digit = 4
Answer: 4

20. Solution: We need to form a 3-digit number using the digits 0, 5, and 7 without repetition.

- Hundreds place: A 3-digit number cannot start with 0. So, the hundreds place can only be filled by either 5 or 7. (2 options)
- Tens place: Out of the three given digits, one is used in the hundreds place. So, there are 2 digits remaining for the tens place (including 0). (2 options)
- Ones place: After filling the hundreds and tens places, only 1 digit remains for the ones place. (1 option)
- Total number of ways $=2 \times 2 \times 1 = 4$

The possible numbers are: 507, 570, 705, 750.
Answer: 4

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Comparison: 3572014 ___ 1025310

3572014 is greater than 1025310
Correct Answer: A) >

2. Ascending order

Numbers:
1808088, 181888, 190909, 1808090, 16060666
Arrange from smallest to largest:
181888 < 190909 < 1808088 < 1808090 < 16060666
So:
181888, 190909, 1808088, 1808090, 16060666
Correct Answer: D)

3. Descending order

Numbers:
1020216, 1020308, 1021430, 893425, 893245
Arrange largest to smallest:
1021430 > 1020308 > 1020216 > 893425 > 893245
Correct Answer: B)

4. Smallest 3-digit number with unique digits

Check options:
101 → digits repeat
012 → not a 3-digit number

102 → valid and smallest
010 → not valid
Correct Answer: C) 102

5. Difference

Greatest 7-digit number = 9,999,999
Smallest 5-digit number = 10,000
Difference = 9,999,999 − 10,000 = 9,989,999
Correct Answer: D) 99,89,999

6. Difference of place value and face value of 7 in 73810924

Number: 73,810,924
Digit 7 is in crores place → place value = 7,00,00,000
Face value of 7 = 7
Difference = 7,00,00,000 − 7 = 6,99,99,993
Correct Answer: D) 6,99,99,993

7. Digit with place value of thousand in 5,68,731

Number: 5,68,731
Places:
5 (lakhs)
6 (ten-thousands)
8 (thousands)
7 (hundreds)
3 (tens)
1 (ones)
Digit in thousand place = 8
Correct Answer: A) 8

8. Thousand place in 14,512

Number: 14,512
Places:
1 (ten-thousands)
4 (thousands)
5 (hundreds)
1 (tens)
2 (ones)
Digit in thousand place = 4
Correct Answer: D) 4

One or More Than One Correct Answer Questions

9. Place value in 7,45,32,109

Break the number:
7 crore 45 lakh 32 thousand 109
A) 7 → 7,00,00,000
B) 4 → 40,00,000, not 4,00,000
C) 5 → 5,00,000
D) 2 → 2,000
Correct Answers: A, C, D

10. Ascending order (smallest → largest)

A) 2,03,456 < 2,30,456 < 2,34,056

B) 1,98,765 > 1,89,765

C) 4,56,789 < 4,65,789 < 4,75,689

D) 3,21,456 > 3,12,456

Correct Answers: A, C

11. Descending order (largest → smallest)

A) 5,43,210 > 5,34,201 > 5,30,412

B) 6,12,345 < 6,21,345

C) 7,89,654 > 7,88,999 > 7,80,000

D) 8,45,210 < 8,54,120

Correct Answers: A, C

Comprehension Type Questions

Passage

12. Round 468 to the nearest tens

Check the ones digit (8 ≥ 5, so round up):

468 → 470

Correct Answer: B) 470

13. Round 7,234 to the nearest hundreds

Check the tens digit (3 < 5, so round down):

7,234 → 7,200

Correct Answer: A) 7,200

14. Round 12,567 to the nearest thousands

Check the hundreds digit (5 ≥ 5, so round up):

12,567 → 13,000

Correct Answer: B) 13,000

Matching Type Questions

a) Largest 4-digit number → T (9999)

b) Largest 5-digit number → R (99999)

c) Largest 6-digit number → S (999999)

d) Largest 3-digit number → P (999)

Final Answer: a–T, b–R, c–S, d–P

Integer Type Questions

16. Thousands place in 138463

Number: 1,38,463

Thousands place digit = 8

Answer: 8

17. Greatest number rounding to 4800 (nearest hundred)

Numbers that round to 4800 lie between:

4750 to 4849

Greatest number = 4849

Answer: 4849

Indian and International System of Numeration

LEVEL – 1

1. Write in digits

i) Four million five hundred twenty thousand six hundred seventy-eight
= 4,520,678

ii) Eight million nine thousand eight
= 8,009,008

Correct Answer: C) 4,520,678; 8,009,008

2. International system commas

Number: 4032897

International system → groups of 3 digits:

= 4,032,897

Correct Answer: B) 4,032,897

3. How many lakhs make ten million?

1 lakh = 100,000

10 million = 10,000,000

$$\frac{10,000,000}{100,000} = 100$$

Correct Answer: C) 100

4. Equivalent of 10 crores

1 crore = 10 million

So, 10 crores = 100 million

Correct Answer: B) 100 million

5. How many lakhs make a billion?

1 billion = 1,000,000,000

1 lakh = 100,000

$$\frac{1,000,000,000}{100,000} = 10,000$$

Correct Answer: A) 10000

6. Convert to International system

7,54,32,109 → 75,432,109

Correct Answer: A) 75,432,109

7. Correct Indian system writing

Rule: first comma after 3 digits, then every 2 digits

A) 123,45,678

B) 12,345,678 (International system)

C) 1,23,45,678

D) 1234,56,78

Correct Answer: C) 1,23,45,678

8. Write in Indian system

Seven crore five lakh four thousand sixty

= 7,00,00,000 + 5,00,000 + 4,000 + 60

= 7,05,04,060

Correct Answer: B) 7,05,04,060

9. International system commas for 8945673

International system groups digits in threes from right:
8945673 → 8,945,673
Correct Answer: A) 8,945,673

10. Convert to International system

Thirty-five lakh sixty-two thousand nineteen
= 35,00,000 + 62,000 + 19
= 35,62,019 (Indian)
= 3,562,019 (International)
Correct Answer: A) 3,562,019

11. Roman numeral for 44

44 = 40 + 4
= XL + IV
= XLIV
Correct Answer: A) XLIV

12. XCIX represents

XC = 90
IX = 9
Total = 99
Correct Answer: B) 99

13. Roman numeral for 275

275 = 200 + 70 + 5
= CC + LXX + V
= CCLXXV
Correct Answer: A) CCLXXV

14. Value of CDXLII

Break it:
CD = 400
XL = 40
II = 2
Total = 400 + 40 + 2 = 442
Correct Answer: A) 442

15. Value of MCMXCIX

Break it:
M = 1000
CM = 900
XC = 90
IX = 9
Total = 1000 + 900 + 90 + 9 = 1999
Correct Answer: B) 1999

16. Value of XL

XL = 50 − 10 = 40
Correct Answer: A) 40

17. Value of CM

CM = 1000 − 100 = 900
Correct Answer: B) 900

18. Value of DCCXLII

Break it:
D = 500
CC = 200
XL = 40
II = 2
Total = 500 + 200 + 40 + 2 = 742
Correct Answer: A) 742

19. Value of CDXL

Break it:
CD = 400
XL = 40
Total = 440
Correct Answer: A) 440

20. Value of DCCC

D = 500
CCC = 300
Total = 800
Correct Answer: B) 800

Level – 2

1. Convert to International system

Four crore twelve lakh nine thousand seventy
= 4,00,00,000
12,00,000
9,000
70
= 4,12,09,070 (Indian)
= 41,209,070 (International)
Correct Answer: A) 41,209,070

2. Write in digits

Three million two hundred five thousand four hundred six
= 3,000,000
205,000
406
= 3,205,406
Correct Answer: B) 3,205,406

3. Correct Indian comma placement

Number: 87595762
Indian system → 8,75,95,762
Correct Answer: B) 8,75,95,762

4. International comma placement

Number: 78125643
International system → 78,125,643
Correct Answer: A) 78,125,643

5. Correct comma placement (Indian system)

Rule: first comma after 3 digits, then every 2 digits

A) 1,234,5678

B) 12,34,56,78

C) 1,23,45,678

D) 123,45,678

Correct Answer: C) 1,23,45,678

6. Value of CMXL + XL

CMXL = 900 + 40 = 940

XL = 40

Total = 940 + 40 = 980

Correct Answer: A) 980

7. Largest Roman numeral

A) MMCD = 2400

B) MCMXC = 1990

C) MMXC = 2090

D) MMD = 2500

Largest = 2500

Correct Answer: D) MMD

8. Roman numeral for 944

944 = 900 + 40 + 4

= CM + XL + IV

= CMXLIV

Correct Answer: A) CMXLIV

One or more than one correct answer type

9. Indian system statements

A) First comma after 3 digits

B) Next commas after every 2 digits

C) Uses millions and billions

D) Uses lakhs and crores

Correct Answers: A, B, D

10. Number: 8,00,25,048

A) Indian system

B) Contains crore period (8 crore)

C) International system

D) Has 3 commas

Correct Answers: A, B, D

11. Which of the following are correct representations?

(A) 90 = XC: Correct. (100 - 10 = 90).

(B) 400 = CD: Correct. (500 - 100 = 400).

(C) 900 = DM: Incorrect. (900 is represented as CM, i.e., 1000 - 100).

(D) 40 = XL: Correct. (50 - 10 = 40).

Correct choices: (A), (B), and (D).

12. Which statements are correct?

(A) V can be repeated: Incorrect. (The symbols V, L, and D are never repeated).

(B) I can be subtracted from V and X: Correct. (IV = 4, IX = 9).

(C) X can be subtracted from L and C: Correct. (XL = 40, XC = 90).
(D) D can be repeated: Incorrect. (The symbols V, L, and D are never repeated).
Correct choices: (B) and (C).

13. Let's check each Roman numeral using the rules of subtraction:
A) IL → Invalid (I cannot be subtracted from L; correct for 49 is XLIX)
B) XC → 100 − 10 = 90 (valid)
C) IC → Invalid (I cannot be subtracted from C; correct for 99 is XCIX)
D) CD → 500 − 100 = 400 (valid)
Correct Answers: B, D

Comprehension Type Questions

Passage

14. International system form of 7,50,000
7,50,000 = 750,000
In words (International system):
= Seven hundred fifty thousand
Correct Answer: B) Seven hundred fifty thousand

15. Indian system form of 3,200,000
3,200,000 = 32,00,000
In words (Indian system):
= Thirty-two lakh
Correct Answer: A) Thirty-two lakh

16. Correct comma placement (Indian system)
Rule: first comma after 3 digits, then every 2 digits
A) 45,00,000
B) 4,500,000 (International system)
C) 450,0000
D) 4,50,00,0
Correct Answer: A) 45,00,000

Matrix matching type

17. Match Column A with Column B
Convert Roman numerals:
CXL = 100 + 40 = 140 → B
CCCXL = 300 + 40 = 340 → A
DCCXC = 500 + 200 + 90 = 790 → C
CMX = 900 + 10 = 910 → D
Final Matching: 1–B, 2–A, 3–C, 4–D

Integer Type Questions

18. Value of LXXXVII
L = 50
XXX = 30
VII = 7
Total = 50 + 30 + 7 = 87
Answer: 87

19. Roman numeral for 68

68 = 50 + 10 + 8
= L + X + VIII
= LXVIII
Answer: LXVIII

20. Repetition rule

A symbol can be repeated maximum 3 times consecutively
Answer: 3

STUDENT PRACTICE QUESTIONS

1. International comma placement

Number: 87594652
International system → groups of 3 digits:
87,594,652
Correct Answer: B) 87,594,652

2. Convert to International system

Ninety-one lakh twenty-three thousand eight hundred four
= 91,23,804 (Indian)
= 9,123,804 (International)
Correct Answer: A) 9,123,804

3. International form of 7,25,43,000

7,25,43,000 = 72,543,000
In words (International):
= Seventy-two million five hundred forty-three thousand
Correct Answer: B)

4. Indian form of 23,456,789

International → Indian grouping:
23,456,789 → 2,34,56,789
Correct Answer: A) 2,34,56,789

5. Meaningless Roman numeral

Rules: A symbol cannot be repeated more than 3 times consecutively
LXVI
XXX (valid, 30)
XLVII
XXXX (invalid repetition)
Correct Answer: d) XXXX

6. Sum of XXVI and XXXV

XXVI = 26
XXXV = 35
Sum = 26 + 35 = 61
Roman numeral = LXI
Correct Answer: A) LXI

One or more than one correct answer type

7. The number is: 9,87,65,432

A) It follows Indian system: Correct. The Indian numbering system uses commas to group digits as 3, 2, 2, 2, etc., starting from the right (e.g., 9,87,65,432).

B) It represents a number less than 1 crore: Incorrect. The number is 9 crore 87 lakh 65 thousand 432, which is much greater than 1 crore.
C) It has lakh and thousand periods: Correct. In the Indian system, the periods are: Crores (9), Lakhs (87), Thousands (65), and Ones (432).
D) It has 3 commas: Correct. As written (9,87,65,432), there are exactly three commas.
Correct Statements: A, C, and D.

8.

XLIX = 40 (XL) + 9 (IX) = 49
IL (invalid Roman numeral form)
XXXXIX (not standard; repetition rule violated)
XLVIIII (not standard; should use IX instead of VIIII)
Correct Answer: A

9. Correct subtraction rule in Roman numerals:

A) IV = 4 (correct subtraction)
B) VX (invalid)
C) XC = 90 (correct subtraction)
D) LC (invalid)
Correct options: A, C

Comprehension Type

Passage

10. Which symbol represents the greatest value?

X = 10, L = 50, V = 5, C = 100 (greatest)
Answer: B) C

11. Which numeral follows the subtraction rule?

IX = 10 − 1 = 9 (subtraction rule)
Others are additive forms
Answer: C) IX

12. Which of the following is correctly written?

XIV = 10 + (5 − 1) = 14
IIV (invalid repetition)
VX (invalid order)
IC (incorrect subtraction form)
Answer: C) XIV

Matrix matching type

P) 1,00,000 → B) One Hundred Thousand
Q) 10,00,000 → A) One Million
R) 1,00,00,000 → C) Ten Million
S) 10,00,00,000 → D) One Hundred Million

Integer type

14. Find the value of (XLVIII + LXII)

XLVIII = 48
LXII = 62
48 + 62 = 110
Answer: 110 (Roman: CX)

15. Convert into Roman numeral: 3888

3000 = MMM
800 = DCCC
80 = LXXX
8 = VIII
Answer: MMMDCCCLXXXVIII

Whole Numbers

LEVEL – 1

1. Which of the following will not represent zero?

$0 \times 5 = 0$, $0/11 = 0$, $(15 - 15) \div 2 = 0$, but $11 + 0 = 11 \neq 0$
Correct Answer: C

2. How many whole numbers are there between 397 and 5321?

Numbers between = $5321 - 397 - 1 = 4923$
Correct Answer: B

3. Difference between successor of 3678 and predecessor of 2897

Successor of 3678 = 3679
Predecessor of 2897 = 2896
Difference = $3679 - 2896 = 783$
Correct Answer: C

4. Product of predecessor of 369 and successor of 24

Predecessor of 369 = 368
Successor of 24 = 25
Product = $368 \times 25 = 9200$
Correct Answer: B

5. 198 ÷ 0 =

Division by 0 is not defined
Correct Answer: D

6. Three consecutive even numbers preceding 809

Even numbers before 809: 808, 806, 804
Correct Answer: D

7. Consecutive prime numbers → only pair is 2 and 3

Answer: D) 2, 3

8. Check valid operation in whole numbers:

A: False
B: False
C: $0 + 8 = 8$ (True)
D: False (since $3 \times 1 = 1 \times 3$)
Answer: C) $0 + 8 = 8$

9. Prime numbers between 50 and 100:

53, 59, 61, 67, 71, 73, 79, 83, 89, 97 → 10 primes
Answer: A) 10

10. Composite numbers strictly between 1 and 20:

4, 6, 8, 9, 10, 12, 14, 15, 16, 18 → 10 numbers
Answer: D) 10

11. a + b = c (result is also a whole number) → Closure property

Answer: A) Closure property over addition

12. $a + b = b + c$**(rearranging terms) → Commutative property of addition**
Answer: B

13. **Commutative property of multiplication:** $a \times b = b \times a$
Answer: D

14. $793 \times _ = 19 \times 793$
Using commutative property → blank = 19
Answer: A

15. $5 \times 87 \times 20 = (5 \times 20) \times 87 = 100 \times 87$
Answer: A

16. Commutative property: $a + b = b + a$
Only option matching: $3 + 7 = 7 + 3 = 10$
Answer: C

17. $16 \times 9 + 16 \times 6 = 16(9 + 6)$
Blank = **9**
Answer: B

18. Ali: $16 \times 20 = 320$km
Sam: $20 \times 16 = 320$km
Both same distance
Answer: D

19. **16 dots can form rectangle (4×4), square, or line, but not a triangle**
Answer: C

20. **Check pairs:**
3 → Triangle
5 → Line
7 → Rectangle (7 cannot form a rectangle arrangement)
2 → Line
Incorrect match: 7 – Rectangle
Answer: C

LEVEL – 2

SINGLE OPTION CORRECT TYPE

1. **Top line: jumps from 8 to 6 → (8 − 6)**
Bottom line: jumps from **1 to 7 → (1 + 6)**
Answer: C

2. **Number line shows jumps of 2 units each, total 6 jumps**
So, $2 \times 6 = 12$
Answer: B

3. **6 papers =** $\mathbf{6 \times 2 = 12}$**pages (no numbering)**
Numbered pages = $532 - 12 = 520$
Last page number = **520**
Answer: C

4. **Difference =** $\mathbf{8350 \times (76 - 67) = 8350 \times 9 = 75,150}$
Answer: B

5. **Meera:** $\mathbf{12 \times 8 = 96}$**km**
Sunita: $8 \times 12 = 96$km
Both same
Answer: C

6. **Total cost = $5(15 + 5 + 4) = 5 \times 24 = 120$**
Answer: B

7. **Total litres = $61 + 34 + 25 = 120$**
Cost = $120 \times 45 = 5400$
Answer: C

8. **Rupa: $7 \times 4 = 28$**
Doli: $5 \times 4 = 20$
Difference = $28 - 20 = 8$
Answer: B

9. **Aarthi: $5 \times 10 = 50$tickets**
Hari: $10 \times 5 = 50$tickets
Both same
Answer: C

10. **A number that is both a square and rectangle → perfect square**
Among options: $36 = 6 \times 6$
Answer: D

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. $34 - (16 - 3 \times 4) = 34 - (16 - 12) = 34 - 4 = 30$
Correct options: C, D

12. **Triangular numbers: $1, 3, 6, 10, 15, \ldots$**
From options → **1, 6, 15**
Correct options: **A, C, D**

13. (A) Smallest whole number is 0
(B) No largest whole number
(C) Every whole number has a successor
(D) Every whole number has a predecessor (0 has no predecessor)
Correct options: **A, B, C**

14. **Identity elements:**
Addition → 0
Multiplication → 1
Correct options: **A, B**

COMPREHENSION TYPE

Passage I:

15. $2018 + _ = 2018$→ additive identity is **0**
Answer: A

16. $2020 \times _ = 2020$→ multiplicative identity is **1**
Answer: B

17. $2 \times (3 + 4) = 2 \times 3 + 2 \times 4$(distributive property)
Answer: C

MATRIX MATCHING TYPE

18. **Match the following:**
a) Commutative property → $a + b = b + a$→ **Q**
b) Multiplicative identity property → $a \times 1 = a$→ **S**
c) $0 \div 7 = 0$→ **P**
d) $18 \times 1 = 18$→ **R**
Answer: a–Q, b–S, c–P, d–R

INTEGER TYPE

19. $\frac{7-7}{2} = \frac{0}{2} = 0$
Answer: 0

20. Greatest two-digit number = 99
Smallest two-digit number = 10
Difference = 99 − 10 = 89
Answer: 89

STUDENT PRACTICE CORNER

1. Between 11 and 24 → 12 to 23
Count = **12**
Answer: A

2. **The prime numbers up to 50 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, and 47.**
Prime numbers from 1 to 50 = **15**
Answer: C

3. Total numbers = 90 - 50 - 1 = 39 numbers.
Next, find how many of these are prime.
The prime numbers between 50 and 90 are: 53, 59, 61, 67, 71, 73, 79, 83, and 89. That is 9 prime numbers.
Composite numbers = Total numbers - Prime numbers
39 - 9 = 30
Correct Option: A) 30

4. Commutative property: order changes $(2 \times 3 \times 4) = (4 \times 3 \times 2)$
Answer: C

5. Number line in steps of 5 → 0, 5, 10, 15, 20, ...
P is at 55
Answer: D

6. Cost per sandwich = 225 ÷ 15 = 15
For 21 sandwiches = 21 × 15 = 315
Answer: D

7. Triangular numbers: 1,3,6,10,15, ...
From options → 10
Answer: C

8. 189 × _ = 189→ multiplicative identity = **1**
Answer: B

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

9. Successor of 678 = 678 + 1 = 679
Correct options: B, C

10. Whole numbers: 0,1,2,3, ...
(A) 0 is a whole number.
(B) -1 is an integer, but not a whole number.
(C) 5 is a whole number.
(D) 3/2 is a fraction (rational number).
Valid → **0, 5**
Correct options: A, C

11. Closed operations on whole numbers:
Addition, Multiplication
(Subtraction & Division are not always whole numbers)
Correct options: A, C

COMPREHENSION TYPE

PASSAGE-1

12. Square numbers: 4,9,16→ 12 is not a square
Answer: C

13. 6 can form:
Triangle (1+2+3 = 6)
Rectangle (2×3 = 6)
Answer: D

14. Increasing rows pattern → triangular numbers
Answer: C

MATRIX MATCHING TYPE

15.

a) $7 \times 1 = 7$→ Multiplicative identity → **S**
b) $0 \div 19 = 0$→ **P**
c) $3 + (7 + 8) = (3 + 7) + 8$→ Associative property → **T**
d) $8 + 13 = 13 + 8$→ Commutative property → **R**

INTEGER TYPE

16. Predecessor of 10,000 = $10{,}000 - 1 = 9{,}999$
Answer: 9999

17. $x + 15 = 15 + 7$
By commutative property: $15 + x = 15 + 7 \Rightarrow x = 7$
Answer: 7

2. NUMBER SYSTEM – 2

Integers

LEVEL-I

1. Correct Answer: B) −41
Explanation: Successor means +1 → −42 + 1 = −41.

2. Correct Answer: C) −3
Explanation: (−10) − (−7) = −10 + 7 = −3.

3. Correct Answer: B) −28
Explanation: Additive inverse of 28 is −28.

4. Correct Answer: B) −1
Explanation: Predecessor means −1 → 0 − 1 = −1.

5. Correct Answer: A) −490
Explanation: Let other number = x
x + 135 = −355 → x = −355 − 135 = −490.

6. Correct Answer: C)- 9
Explanation: Pairing:
(1−2)+(3−4)+(5−6)+…+(17−18)
= (−1)×9 = −9 → but check carefully:
Actually sequence gives **9 terms of −1 → −9**

7. Correct Answer: B) >
Explanation: −14 is greater than −24 → so −14 > −24 → but blank is written as −14 ___ −24 → correct is **>**

8. Correct Answer: B) 17
Explanation: |5| + |−12| = 5 + 12 = 17

9. Correct Answer: B) Kashmir
Explanation: −12°C is colder than −8°C.

10. Correct Answer: B) Positive
Explanation: Product of two negative numbers is always positive.

11. Correct Answer: C) 24
Explanation:
Geetha's score = $45 \times 5 - 5 \times 3 = 225 - 15 = 210$
Sita's score = $42 \times 5 - 8 \times 3 = 210 - 24 = 186$
Difference = $210 - 186 = 24$

12. Correct Answer: B) −60
Explanation:
Other integer = $-1800 \div 30 = -60$

13. Correct Answer: A) 75 m
Explanation:
Height from seabed = 50 + 25 = 75m

14. Correct Answer: B) −250
Explanation:
Score = $10 \times 100 - 25 \times 50 = 1000 - 1250 = -250$

15. Correct Answer: C) 108

Explanation:

Highest one-digit positive = 9

Highest two-digit number = 99 → its negative = −99

$9 - (-99) = 9 + 99 = 108$

16. Correct Answer: C) -64

Explanation:

Expression: $56 - \{52 \times 2 - (13 - 6 \div 2) \times (4 - 6)\} + 4$

$6 \div 2 = 3$, so $13 - 3 = 10$

$4 - 6 = -2$

$10 \times (-2) = -20$

$52 \times 2 = 104$

Bracket: $104 - (-20) = 124$

Final: $56 - 124 + 4 = -64$

17. Correct Answer: A) a − 3

Explanation:

If $a > 3$, then $a - 3$is positive

So $| a - 3 | = a - 3$

18. Correct Answer: D) −15419

Explanation:

$$-87362 + 71943 = -(87362 - 71943) = -15419$$

19. Correct Answer: B) 20 ÷ 4

Explanation:

A: $20 \div (-4) = -5$

B: $20 \div 4 = 5$

C: $(-20) \div (-5) = 4$

D: $20 \div (-5) = -4$

Greatest = 5 → actually **B**

Comparing the quotients we found:

-5, 5, 4, 4

The greatest value among these is **5**.

20. Correct Answer: B) Even

Explanation:

Product of three integers is odd ⇒ all are odd

Odd + Odd + Odd = Odd

But sum of 3 odd numbers = Odd (not even)

Correct answer should be A) Odd

LEVEL – 1

Single Answer Questions

1. Correct Answer: D) −150

Explanation:

Input = −2500

Add 5000 → $-2500 + 5000 = 2500$

Subtract 3000 → $2500 - 3000 = -500$

Multiply by 6 → $-500 \times 6 = -3000$

Divide by 20 → $-3000 \div 20 = -150$

2. Correct Answer: B) - 50

Explanation:

Series: $1 - 2 + 3 - 4 + \cdots + 99 - 100$

Pairing: $(1 - 2) + (3 - 4) + \cdots + (99 - 100)$

Each pair = -1, total 50 pairs → sum = $50 \times (-1) = -50$

3. Correct Answer: C) −52

Explanation:

Let other number = x

$$x + 18 = -34$$
$$x = -34 - 18 = -52$$

4. Correct Answer: D) 62

Explanation:

$$(-3) \times 9 + 4 \times (-5) - 15 = -27 - 20 - 15 = -62$$

5. Correct Answer: B) x − |x| = 0

Explanation:

If $x > 0$, then $| x | = x$

So $x - x = 0$

6. Correct Answer: B) x − 2

Explanation:

If $x > 2$, then $| 2 - x | = x - 2$

7. Correct Answer: C) 9

Explanation:

Given $a\Delta b = a - b - 2$

$7\Delta(-4) = 7 - (-4) - 2 = 7 + 4 - 2 = 9$

One or More Than One Correct Answer Questions

8. Correct Answer: A, B

Explanation:

Integers are whole numbers (positive, negative, zero) without fractions.

5 and −2 are integers, but $\frac{3}{2}, \frac{1}{2}$ are not.

9. Correct Answer: B, C

Explanation:

B) 98 < 100

C) −2 > −200 (−2 is greater)

10. Correct Answer: B) 49, 69

Explanation:

Natural numbers between −20 and 50 → from 1 to 49 → 49 numbers

Integers between −20 and 50 → from −19 to 49 → total 49 + 20 = 69

Comprehension Type Questions

11. Correct Answer: B) 140

Explanation:

Given: $a\square b = -a + b - 5$

$$105\square 250 = -105 + 250 - 5 = 140$$

12. Correct Answer: D) 690

Explanation:

Given: $a \triangle b = a - b - 15$

$$1230 \triangle 525 = 1230 - 525 - 15 = 690$$

13. Correct Answer: D) −735

Explanation:

$1510 \square 780 = -1510 + 780 - 5 = -735$

Matching Type Questions

14. Matching Answer:

a) $|-7| \rightarrow$ R) 7

b) $p \times 1 = 1 \times p = p \rightarrow$ S) 1 is multiplicative identity

c) $x \times (y + z) = (x \times y) + (x \times z) \rightarrow$ P) Distributive

d) $x \div 0 \rightarrow$ T) Not defined

Integer Type Questions

15. Correct Answer: 9

Explanation:

Pair terms: $(-1 + 2) + (-3 + 4) + (-5 + 6) + \cdots + (-17 + 18)$

Each pair = 1, total 9 pairs → sum = $9 \times 1 = 9$

16. Correct Answer: 7

Explanation:

Additive inverse of −7 is +7 (number which gives 0 when added)

17. Correct Answer: 8

Explanation:

Difference = $27 - 19 = 8$

18. Correct Answer: 0

Explanation:

$$3 + (-3) + (-3) + 3 + (-3) + 3$$

Group: $(3 - 3) + (-3 + 3) + (-3 + 3) = 0 + 0 + 0 = 0$

19. Correct Answer: 3

Explanation:

$$(78 - 77) + (18 - 17) + (28 - 27) = 1 + 1 + 1 = 3$$

20. Correct Answer: 1

Explanation:

$81 - 91 - 101 + 100 + 5 + 7$

$= -10 - 101 + 100 + 5 + 7$

$= -111 + 100 + 5 + 7 = -11 + 5 + 7 = 1$

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Correct Answer: A) −48

Explanation:

Sum = $-23 + 33 = 10$

Subtract from $-38 \rightarrow -38 - 10 = -48$

2. Correct Answer: B) 15

Explanation:

$$0 - (-15) = 0 + 15 = 1$$

3. **Correct Answer: A) −4**
Explanation:
8 more than −12 → $-12 + 8 = -4$

4. **Correct Answer: C) 723**
Explanation:
$450 - (-273) = 450 + 273 = 723$

5. **Correct Answer: A) −731**
Explanation:
$(-43) \times 17 = -731$

6. **Correct Answer: C) 45360 m**
Explanation:
Peak = +6300m, Mine = −39060m
Distance = $6300 - (-39060) = 6300 + 39060 = 45360$m

7. **Correct Answer:** D) $0 + 5 + -3 + -2 + 4$
Explanation:
So the correct expression is 0+ 5 + (−3) + (-2) + 4

8. **Correct Answer: C) −1301**
Explanation:
Other integer = $15612 \div (-12) = -1301$

One or More Than One Correct Answer Questions

9. Correct Answer: B, C
Explanation:
$15 - (-5) - 2 = 15 + 5 - 2 = 18$
A) 18
B) $15 - (-3) = 18$
Correct options are A, B

10. Correct Answer: B) −63
Explanation:
Let other number = x
$x + 40 = -23$
$x = -23 - 40 = -63$

11. **Correct Answer: A, B**
Explanation:
B) $17 + (-2) = 15 < 17$
C) $19 < 20 - (-2) = 22$
D) $15 + 5 = 20 < 13 - (-10) = 23$
Correct options are B, C, D

Comprehension Type Questions

Passage – I:

12. **Correct Answer: C) -117**
Explanation:
$(-2)^3 = -8$, $(-4)^2 = 16$, $(-5)^3 = -125$
Sum = $-8 + 16 - 125 = -117$

13. Correct Answer: S) −166

Explanation:

$(-3)^4 = 81$, $(-2)^5 = -32$, $(-6)^3 = -216$, $(-1)^8 = 1$

Sum = $81 - 32 - 216 + 1 = -166$

14. Matching Answer:

a) $(-a) \times b = b \times (-a)$→ Q) Commutative property

b) $(9 + 5) + (-2) = 9 + [5 + (-2)]$→ T) Associative property

c) $0 + 5 = 5 + 0 = 5$→ R) 0 is additive identity

d) $(-709) \div (+709)$→ P) −1

Integer Type Questions

15. Correct Answer: 0

Explanation:

Given $a\Delta b = -a + b - 2$

$1\Delta 3 = -1 + 3 - 2 = 0$

16. Correct Answer: 5

Explanation:

$1 - 10 + 9 - 7 + 6 + 2$

$= -9 + 9 - 7 + 6 + 2$

$= 0 - 7 + 6 + 2 = -7 + 6 + 2 = 1$

Divisibility rules:

Level – 1

1. Number of factors of 36

Prime factorization: $36 = 2^2 \times 3^2$

Number of factors = $(2 + 1)(2 + 1) = 9$

Correct Answer: B) 9

2. Number of factors of 54, 62

$54 = 2 \times 3^3 \Rightarrow (1 + 1)(3 + 1) = 8$

$62 = 2 \times 31 \Rightarrow (1 + 1)(1 + 1) = 4$

Correct Answer: C) 8; 4

3. Difference between number of factors of 105, 63

$105 = 3 \times 5 \times 7 \Rightarrow 2 \times 2 \times 2 = 8$

$63 = 3^2 \times 7 \Rightarrow 3 \times 2 = 6$

Difference = $8 - 6 = 2$

Correct Answer: A) 2

4. Sum of factors of 49

Factors: 1, 7, 49 → Sum = 57

Correct Answer: B) 57

5. Identify perfect number

Proper divisors: 1, 2, 4, 7, 14

Sum = 1 + 2 + 4 + 7 + 14 = 28

Correct Answer: C) 28

6. Add ___ to sum of factors of 97 to make a perfect number

97 is prime → factors: 1, 97 → sum = 98

98 + 96 = 194 (not perfect), none works

Correct Answer: D) None

7. Number of multiples of 23
Multiples continue infinitely
Correct Answer: D) Infinite

8. Prime number among the following
157 = prime
Correct Answer: D) 157

9. Which is NOT a prime number?
103 = prime
239 = prime
337 = prime
963 = divisible by 3 → not prime
Correct Answer: D) 963

10. Composite number among the following
1129 = prime
1453 = prime
1944 = composite
1787 = prime
Correct Answer: C) 1944

11: The pair of prime and composite numbers
Check each option:
A) 34; 193
34 = 2 × 17 → composite
193 → not divisible by 2, 3, 5, 7, 11, 13 → prime

12. Count primes between 50 and 100 → 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 (10 primes)
Correct Answer: A) 10

13. 39 is divisible by 13
Correct Answer: A) 13

14. Answer: B) 31
Explanation:
21 = 3 × 7 (not prime)
31 has only two factors (1 and 31) → prime
39 = 3 × 13 (not prime)
51 = 3 × 17 (not prime)

15. Answer: A) 4, 9
Explanation:
4 = 2 × 2 (composite), 9 = 3 × 3 (composite)
5, 7, 11, 13, 3, 17 are all prime numbers

16. $315 = 3 \times 3 \times 5 \times 7 = 3^2 \times 5 \times 7$
Correct Answer: D) $3^2 \times 5 \times 7$

17. $2^3 \times 5^2 \times 7 = 8 \times 25 \times 7 = 1400$
Correct Answer: B) 1400

18. Divisible by 2 → even number → 492 ends with 2
Correct Answer: C) 492

19. 5286 divisible by 6 since it is divisible by 2 (even) and 3 (sum of digits = 21)
Correct Answer: B) 6

20. **Divisible by 4 → last two digits divisible by 4 → 68 ÷ 4 = 17**
Correct Answer: D) 14468

LEVEL – 2

1. Prime factorization of 2145:
$2145 = 3 \times 5 \times 11 \times 13$→ contains twin primes (3,5) = P and (11,13) = R
Correct Answer: D) P and R

2. Check which number is NOT a divisor of any given numbers (41160, 5040, 10080, 2016, 9240):
27 (3^3) does NOT divide all
Correct Answer: C) 27

3. A number divisible by 24 must be divisible by 3 and 8.
301632 → sum of digits = 15 (divisible by 3) and last three digits 632 divisible by 8
Correct Answer: C) 301632

4. Dividend = Divisor × Quotient = $19 \times 50 = 950$
Correct Answer: B) 950

5. Correct relation: Dividend = Divisor × Quotient + Remainder
Correct Answer: B) Quotient × Divisor + Remainder = Dividen

6. $\frac{\text{Dividend} - \text{Remainder}}{\text{Divisor}} = \text{Quotient}$
Correct Answer: C) Quotien

7. Number between 50 and 75, multiple of 6 → 60, 66, 72
Check ÷9 remainder 3:
66 ÷ 9 → remainder 3
Correct Answer: D) 66

8. Divisible by 3 → sum of digits divisible by 3
486 → 18
Correct Answer: B) 486

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Factors of 30 → 1, 2, 3, 5, 6, 10, 15, 30
Given options → 3, 10, 15, 30
Correct Answer: A, B, C, D

10. **Correct Answer: A) 90 C) 1350 D) 9045**
Number must be divisible by both 9 and 5 → divisible by 45.
90 ÷ 45 = 2
C) 1 + 3 + 5 + 0 = 9
D) 9 + 0 + 4 + 5 = 18
Others are not divisible by 45.

11. Multiples of 14 → divisible by 2 and 7.
2002 ÷ 14 = 143
Others are not divisible by 14.
Correct Answer: D) 2002 A) 3640

12. Multiples of any natural number continue endlessly.
Correct Answer: D) Infinit

13. 9A82 divisible by 2 → last digit 2
For divisibility by 3 → sum = $9 + A + 8 + 2 = 19 + A$must be divisible by 3

$$19 + A \equiv 0(\text{mod}3) \rightarrow A \equiv 2(\text{mod}3)$$

Possible values from options → 2, 5, 8

Correct Answer: A, B, C

MATRIX MATCHING TYPE

14. Matrix Matching

a) 14157 → sum of digits = 18 → divisible by 3 and 9 → R) 3, 9, 11

b) Divisible by 49 → 49 = 7 × 7→ divisible by 7 → T) 7

c) Any even palindrome → always divisible by 11 → P) 11

d) Smallest 6-digit number divisible by 111 → 100011 → Q) 100011

Correct Matching: a–R, b–T, c–P, d–Q

COMPREHENSION TYPE

15. Even number → divisible by 2 → ends with 0,2,4,6,8

74

Correct Answer: A) 74

16. Divisible by 2 → even number → 1300

Correct Answer: C) 1300

INTEGER TYPE

17. Explanation:

For divisibility by 9, sum of digits must be a multiple of 9.

Sum = $4 + 5 + 6 + x + 8 = 23 + x$

Nearest multiple of 9 after 23 is 27 → $23 + x = 27 \Rightarrow x = 4$

Correct Answer: 4

18.

Divisible by 5 ⇒ last digit $y = 0$or 5

Divisible by 4 ⇒ last two digits $8y$divisible by 4 → only $y = 0$works

Divisible by 9 ⇒ sum $2 + 4 + x + 8 + 0 = 14 + x$divisible by 9 → $x = 4$

Sum $x + y = 4 + 0 = 4$

Correct Answer: 4

19.

Divisible by 4 ⇒ last two digits $3A$divisible by 4 → possible A = 2, 6

Divisible by 6 ⇒ divisible by 2 and 3

→ A must be even and sum divisible by 3

Check:

A=2 → sum = 2+5+9+3+2=21

Correct Answer: 2

20. Units digit of $2 \times 5 \times 11^2 \times 17$

Units: $2 \times 5 = 10 \rightarrow 0$, multiplied by anything remains 0

Correct Answer: 0

STUDENT PRACTICE QUESTIONS

1. Sum = 1229 + 937 + 1059 = 3225→ ends with 5 ⇒ divisible by 5

Correct Answer: D) 5

2. Divisible by 6 ⇒ divisible by 2 and 3

9456 is even and sum of digits = 24 (divisible by 3)

Correct Answer: C) 9456

3. 21492 → last three digits 492 divisible by 4 but check 8: 492 ÷ 8 not integer; sum = 18 ⇒ divisible by 9
Correct Answer: C) 9

4. Divisible by 10 ⇒ ends with 0 → 50520
Correct Answer: D) 50520

5. Divisible by 8 ⇒ last three digits divisible by 8
11848 → 848 ÷ 8 = 106
Correct Answer: B) 11848

6. Divisible by 11 ⇒ difference of alternate digit sums is multiple of 11
894036 → (8+4+3) − (9+0+6) = 15 − 15 = 0
Correct Answer: D) 894036

One or More Than One Correct Answer Questions

7. Divisible by 2 → even numbers (last digit 0,2,4,6,8)
2468, 980
Correct Answer: A, C

8. Divisible by both 2 and 3 ⇒ divisible by 6
36, 72
Correct Answer: A, C

9. Divisible by both 8 and 2 ⇒ must be divisible by 8
256, 144, 128
Correct Answer: A, B, C

COMPREHENSION TYPE

10. Divisible by both 5 and 10 ⇒ must end with 0
250
Correct Answer: A) 250

11. Divisible by 4 ⇒ last two digits divisible by 4
1236 → 36
Correct Answer: A,

Matching Type

12. a) $2,97,675 = 3^5 \times 5^2 \times 7^2$ → **S**
b) Common factors of 182, 266 → 1, 2, 7, 14 → **T**
c) Prime numbers between 1–100 → 25 → **P**
d) Sum of digits divisible by 9 → number divisible by 9 → **R**
e) Number of factors of any number → finite → **Q**
Correct Matching: a–S, b–T, c–P, d–R, e–Q

Integer Type Questions

13. To be divisible by 5 → last digit must be 0 or 5
4786 → add 4 → 4790
Correct Answer: 4

14. Divisible by 4 ⇒ last two digits divisible by 4
9632 → 32 is already divisible by 4 → subtract 0
Correct Answer: 0

HCF & LCM

LEVEL – 1

1. Common multiple → smallest common multiple (LCM) of 36 and 108 = 108
Correct Answer: B) 108

2. Common factor → highest common factor of 72 and 108 = 36
Correct Answer: A) 36

3. $63 = 3^2 \times 7$, $98 = 2 \times 7^2$→ common factor = 7
Number of factors of 7 = 2 (1, 7)
Correct Answer: D) 2

4. GCD of co-primes is always 1
Correct Answer: A) 1

5. To leave remainder 2 in each case, the number must divide:
$98 - 2 = 96$and $168 - 2 = 166$
Now find HCF of 96 and 166:
$96 = 2^5 \times 3$
$166 = 2 \times 83$
Common factor = 2
Correct Answer: 2

6. $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, $54 = 2 \times 3^3$
LCM = $2^3 \times 3^3 = 8 \times 27 = 216$
Correct Answer: A) 216

7. LCM × HCF = product ⇒ LCM = $180 \div 6 = 30$
Correct Answer: A) 30

8. $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$, $168 = 2^3 \times 3 \times 7$
HCF = $2^3 \times 3 = 24$
Correct Answer: B) 24

10. LCM of co-primes = product = $3 \times 5 \times 7 = 105$
Correct Answer: A) 105

11. LCM of 8, 12, 15 = $2^3 \times 3 \times 5 = 120$
Correct Answer: A) 120

12. $96 = 2^5 \times 3$, $144 = 2^4 \times 3^2$
HCF = $2^4 \times 3 = 48$
Correct Answer: B) 48

13. LCM of 6, 8, 9 = $2^3 \times 3^2 = 72$
Correct Answer: A) 72

14. Required number divides (306−9)=297 and (657−9)=648
HCF(297, 648) = 27
Correct Answer: A) 27

15. LCM of 14, 21, 28 = $2^2 \times 3 \times 7 = 84$
Correct Answer: A) 84

16. HCF of 45, 60, 75 = $3 \times 5 = 15$
Correct Answer: C) 15

17. LCM of 36 and 60 = 180
Correct Answer: A) 180

18. LCM of 12, 15, 20 = 60
Correct Answer: A) 60

19. Product = LCM × HCF = 180 × 6 = 1080
Other number = 1080 ÷ 30 = 36
Correct Answer: A) 36

20. $121 = 11^2$, $242 = 2 \times 11^2$→ common factor = $11^2 = 121$
Correct Answer: C) 121

LEVEL – 2

SINGLE OPTION CORRECT TYPE

1. Required number divides (138−3)=135, (183−3)=180, (243−3)=240
HCF(135, 180, 240) = 15
Correct Answer: A) 15

2. Greatest equal fruits per heap = HCF(153, 119) = 17
Number of heaps = 153/17 = 9and 119/17 = 7→ total heaps = 16
Correct Answer: D) 17, 16

3. Convert to cm: 5m60cm = 560 cm, 2m40cm = 240 cm
Largest tile = HCF(560, 240) = 80 cm
Correct Answer: B) 80 cm

4. Time when all ring together = LCM(6, 8, 12) = 24
Correct Answer: B) 24 min

5. Required number = HCF(187, 221, 323)
187 = 11 × 17, 221 = 13 × 17, 323 = 17 × 19→ common = 17
Correct Answer: A) 17

6. LCM(3, 4, 5) = 60
Correct Answer: A) 60

7. HCF(25, x) = 5 ⇒ x must be a multiple of 5 but not 25
10, 15, 20 (30 also works but single correct expected → smallest valid)
Correct Answer: A) 10

8. HCF(14, 21) = 7
Correct Answer: A) 7

9. $48 = 2^4 \times 3$, $72 = 2^3 \times 3^2$
HCF = $2^3 \times 3 = 24$
Correct Answer: B) 24

One or More Than One Correct Answer Questions

10. Factors of 60 → 1,2,3,4,5,6,10,12,15,20,30,60
From options → 4, 6, 15
Correct Answer: A, B, C

11. Multiples of both 6 and 8 ⇒ LCM (6, 8) = 24 → multiples: 24, 48, 72, 96
Correct Answer: A, B, C, D

12. HCF = 1
(14,25) → 1
(8,15) → 1
Correct Answer: A, C

13. A) True
C) True
D) True
Correct Answer: A, C, D

Passage – I

14. Product = HCF × LCM ⇒ 248 × 868 = HCF × 1736
Since 1736 = 2 × 868, HCF = 248 ÷ 2 = 124
Correct Answer: B) 124

15. LCM = Product ÷ HCF = 15870 ÷ 23 = 690
Correct Answer: D) 690

16. **Matching**
a) LCM(15, 20, 30) = 60 → **P**
b) HCF(84, 126) = 42 → **Q**
c) LCM(9, 12, 15) = 180 → **R**
d) HCF(48, 180) = 12 → **S**
Correct Matching: a–P, b–Q, c–R, d–S

17. LCM of 48, 72, 100
$48 = 2^4 \times 3$, $72 = 2^3 \times 3^2$, $100 = 2^2 \times 5^2$
LCM = $2^4 \times 3^2 \times 5^2 = 16 \times 9 \times 25 = 3600$seconds = 60 minutes
8:00 a.m. + 60 min = 9:00 a.m.
Correct Answer: 9:00 a.m.

18. Maximum capacity = HCF (117, 81)
$117 = 3^2 \times 13$, $81 = 3^4 \rightarrow$ HCF = $3^2 = 9$
Correct Answer: 9 litres

19. HCF(12, 16, 1, 28) = 1 (since 1 divides all numbers)
Correct Answer: 1

20. Product = LCM × HCF = 180 × 6 = 1080
Correct Answer: 1080

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. $36 = 2^2 \times 3^2$, $54 = 2 \times 3^3 \rightarrow$ HCF = $2 \times 3^2 = 18$
Correct Answer: C) 18

2. LCM(9, 6) = 18
Correct Answer: B) 18

3. LCM = Product ÷ HCF = 300 ÷ 5 = 60
Correct Answer: A) 60

4. HCF (24, 60) = 12
Correct Answer: B) 12

5. LCM (4, 5, 6) = $2^2 \times 3 \times 5 = 60$
Correct Answer: A) 60

6. HCF (45, 75) = 15
Correct Answer: C) 15

7. HCF (16, 20) = 4
Correct Answer: B) 4

One or More Than One Correct Answer Questions

8. Divisible by both 4 and 6 ⇒ divisible by LCM (4, 6) =12
24, 36, 48 (54 not divisible by 4)
Correct Answer: A, B, C

9. HCF must divide both 18 and 24
Common factors: 1, 2, 3, 6
From options → 2, 3, 6
Correct Answer: A, B,

10. A) LCM(4,5)=20
B) LCM(6,8)=24
C) LCM(9,12)=36
Correct Answer: A, B, C

Comprehension Type Questions

11. LCM(12,15,20) = 60 minutes
Correct Answer: B) 60

12. 8:00 AM + 60 minutes = 9:00 AM
Correct Answer: B) 9:00 AM

13. **Matching**
a) LCM = Product ÷ HCF = 300 ÷ 5 = 60→ **P**
b) Product = LCM × HCF = 180 × 6 = 1080→ **Q**
c) HCF(121, 242) = 121 → **R**
d) LCM(7, 11, 13) = 7 × 11 × 13 = 1001→ **S**
Correct Matching: a–P, b–Q, c–R, d–S

Integer Type Questions

14. LCM(14, 21) = 42
Correct Answer: 42

15. HCF(12, 18, 24) = 6
Correct Answer: 6

3.Rational Numbers, Fractions and Decimals

LEVEL – 1

1. $\frac{4}{-11}+\frac{7}{11}=\frac{-4+7}{11}=\frac{3}{11}$

Correct Answer: B) $\frac{3}{11}$

2. $-\frac{8}{3}-\frac{1}{4}-\frac{11}{6}+\frac{3}{8}-3$

LCM = 24

$-\frac{64}{24}-\frac{6}{24}-\frac{44}{24}+\frac{9}{24}-\frac{72}{24}=\frac{-177}{24}=-\frac{59}{8}$

Correct Answer: C) $-\frac{59}{8}$

3. Compare $-\frac{3}{4}$and $-\frac{5}{6}$

Step 1: Find a common denominator.

LCM of 4 and 6 = 12

Step 2: Convert both fractions:

$-\frac{3}{4}=-\frac{3\times3}{4\times3}=-\frac{9}{12}$

$-\frac{5}{6}=-\frac{5\times2}{6\times2}=-\frac{10}{12}$

Step 3: Compare numerators.

Since $-9>-10$(closer to zero),

$-\frac{9}{12}>-\frac{10}{12}$

$-\frac{3}{4}>-\frac{5}{6}$

Correct Option: A. $-\frac{3}{4}>-\frac{5}{6}$

4. $\frac{9}{10}-\left(-\frac{3}{5}\right)=\frac{9}{10}+\frac{3}{5}$

$=\frac{9}{10}+\frac{6}{10}=\frac{15}{10}=\frac{3}{2}$

Correct Answer: B) $\frac{3}{2}$

5. Subtract $-\frac{7}{13}$from $-\frac{6}{13}$

Step 1: Write the expression

$-\frac{6}{13}-\left(-\frac{7}{13}\right)$

Step 2: Change subtraction to addition

$-\frac{6}{13}+\frac{7}{13}=\frac{-6+7}{13}=\frac{1}{13}$

Correct Answer: D) $\frac{1}{13}$

6. The sum of two numbers is $-\frac{4}{3}$. One number is $-\frac{15}{7}$

Step 1: Let the other number be x

$x + \left(-\frac{15}{7} \right) = -\frac{4}{3}$

Step 2: Solve

$x = -\frac{4}{3} + \frac{15}{7}$

LCM = 21

$x = -\frac{28}{21} + \frac{45}{21} = \frac{17}{21}$

Correct Answer: A) $\frac{17}{21}$

7. The number should be subtracted from $\left(\frac{3}{4} - \frac{2}{3}\right)$to get $-\frac{1}{6}$

Step 1: Compute inside bracket

$\frac{3}{4} - \frac{2}{3} = \frac{9-8}{12} = \frac{1}{12}$

Step 2: Let number be x

$\frac{1}{12} - x = -\frac{1}{6}$

Step 3: Solve

$x = \frac{1}{12} + \frac{1}{6} = \frac{1}{12} + \frac{2}{12} = \frac{3}{12} = \frac{1}{4}$

Correct Answer: B) $\frac{1}{4}$

8. Which rational number lies between $\frac{1}{3}$and $\frac{1}{2}$?

$\frac{1}{3} \approx 0.33, \frac{1}{2} = 0.5$

$\frac{2}{5} = 0.4$ (lies between)

Correct Answer: A) $\frac{2}{5}$

9. The number should be subtracted from $\frac{3}{7}$to get $\frac{5}{4}$

Step 1: Let number be x

$\frac{3}{7} - x = \frac{5}{4}$

Step 2: Solve

$x = \frac{3}{7} - \frac{5}{4}$

LCM = 28

$x = \frac{12}{28} - \frac{35}{28} = -\frac{23}{28}$

Correct Answer: B) $-\frac{23}{28}$

10. Multiplicative inverse of $\left(-\frac{4}{3} \times \frac{5}{2}\right)$

Step 1: Multiply

$-\frac{4}{3} \times \frac{5}{2} = -\frac{20}{6} = -\frac{10}{3}$

Step 2: Inverse

$$-\frac{3}{10}$$

Correct Answer: C) $-\frac{3}{10}$

11. Simplest form of $\frac{70}{-84}$

Step 1: Simplify

$$\frac{70}{-84} = -\frac{70}{84}$$

Divide by 14:

$$-\frac{5}{6}$$

Correct Answer: B) $-\frac{5}{6}$

12. Property of rational numbers

Given:

$$\left(\frac{2}{3}+\frac{1}{2}\right)+\frac{1}{5}=\frac{2}{3}+\left(\frac{1}{2}+\frac{1}{5}\right)$$

Grouping of numbers changes, order remains same → Associative property

Correct Answer: C) Associative

13. Let's evaluate each option by finding a common denominator to compare the numerators:

A) $-\frac{7}{10} > \frac{8}{-15} \Rightarrow -\frac{21}{30} > -\frac{16}{30}$

(False, $-21 < -16$)

B) $-\frac{4}{9} > \frac{5}{-12} \Rightarrow -\frac{16}{36} > -\frac{15}{36}$

(False, $-16 < -15$)

C) $\frac{2}{-3} < -\frac{4}{9} \Rightarrow -\frac{6}{9} < -\frac{4}{9}$

(True, $-6 < -4$)

D) $\frac{8}{-15} > \frac{3}{5}$

A negative number cannot be greater than a positive number (False)

Correct Option: C) $\frac{2}{-3} < -\frac{4}{9}$

14. Multiply $\frac{-22}{9} \times \frac{51}{-88}$

Signs: negative × negative = positive

$$\frac{22}{9} \times \frac{51}{88}$$

Simplify:

$$\frac{22}{88} = \frac{1}{4}$$

$$= \frac{1 \times 51}{9 \times 4} = \frac{51}{36} = \frac{17}{12}$$

Correct Answer: C) $\frac{17}{12}$

15. Simplify $\left(-\frac{9}{4} \times \frac{5}{3}\right) + \left(\frac{13}{2} \times \frac{5}{6}\right)$

$$= -\frac{45}{12} + \frac{65}{12} = \frac{20}{12} = \frac{5}{3}$$

Correct Answer: A) $\frac{5}{3}$

16. Simplify $\left(-5 \times \frac{2}{15}\right) - \left(-6 \times \frac{2}{9}\right)$

$= -\frac{10}{15} + \frac{12}{9} = -\frac{2}{3} + \frac{4}{3} = \frac{2}{3}$

Correct Answer: A) $\frac{2}{3}$

17. If $\frac{x}{2} + \frac{1}{3} = 1$, find x

Step 1: $\frac{x}{2} = 1 - \frac{1}{3} = \frac{2}{3}$

Step 2: $x = \frac{2}{3} \times 2 = \frac{4}{3}$

Correct Answer: D) $\frac{4}{3}$

18. Divide $-\frac{3}{4}$ by $\frac{9}{-16}$

Step 1:

$\frac{9}{-16} = -\frac{9}{16}$

Step 2:

$-\frac{3}{4} \div -\frac{9}{16} = -\frac{3}{4} \times -\frac{16}{9}$

Step 3:

$= \frac{48}{36} = \frac{4}{3}$

Correct Answer: B) $\frac{4}{3}$

19. Divide $\frac{10}{3}$ by $-\frac{35}{2}$

Step 1:

$\frac{10}{3} \times \frac{2}{-35}$

Step 2:

$= \frac{20}{-105} = -\frac{4}{21}$

Correct Answer: A) $-\frac{4}{21}$

20. Product is $-\frac{8}{9}$, one number is $-\frac{4}{15}$

Step 1: Let other number be x

$x \times -\frac{4}{15} = -\frac{8}{9}$

Step 2: $x = -\frac{8}{9} \div -\frac{4}{15} = -\frac{8}{9} \times -\frac{15}{4}$

Step 3: $= \frac{120}{36} = \frac{10}{3}$

Correct Answer: A) $\frac{10}{3}$

LEVEL – 2

SINGLE OPTION CORRECT TYPE

1. The product is $-\frac{28}{81}$, one number is $\frac{14}{27}$

Step 1: Let the other number be x

$x \times \frac{14}{27} = -\frac{28}{81}$

Step 2: $x = -\frac{28}{81} \div \frac{14}{27} = -\frac{28}{81} \times \frac{27}{14}$

Step 3: $= -\frac{2}{3}$

Correct Answer: B) $-\frac{2}{3}$

2. Number line question

Points are equally spaced:

$-1, ?, -\frac{1}{3}, 0, \frac{1}{3}$

Step size = $\frac{1}{3}$

So, $-1 + \frac{1}{3} = -\frac{2}{3}$

Correct Answer: C) $-\frac{2}{3}$

3. Find $\frac{y}{x}$

From number line:

$x = -\frac{10}{20} = -\frac{1}{2}, y = \frac{15}{20} = \frac{3}{4}$

$\frac{y}{x} = \frac{3}{4} \div -\frac{1}{2} = \frac{3}{4} \times -2 = -\frac{3}{2}$

Correct Answer: A) $-\frac{3}{2}$

4. Which of the following shows associative property?

Associative property:

$a \times (b \times c) = (a \times b) \times c$

Option A matches this form

Correct Answer: A

5. Divide the sum of $\frac{65}{12}$ and $\frac{12}{7}$ by their difference

Step 1: Sum

$\frac{65}{12} + \frac{12}{7} = \frac{455 + 144}{84} = \frac{599}{84}$

Step 2: Difference

$\frac{65}{12} - \frac{12}{7} = \frac{455 - 144}{84} = \frac{311}{84}$

Step 3: Division

$\frac{599}{84} \div \frac{311}{84} = \frac{599}{311}$

Correct Answer: A) $\frac{599}{311}$

6. Which of the following is false?

A) $\frac{-7}{-12} = \frac{7}{12}, \frac{4}{9} \Rightarrow \frac{7}{12} > \frac{4}{9}$ (True)

B) $\frac{4}{9} > \frac{11}{-24}$ (positive > negative → True)

C) $\frac{11}{-24} = -\frac{11}{24}, -\frac{5}{6} \Rightarrow -\frac{11}{24} > -\frac{20}{24}$ (True)

D) $\frac{-7}{-12} = \frac{7}{12}, \frac{11}{-24} = -\frac{11}{24} \Rightarrow \frac{7}{12} < -\frac{11}{24}$ (False)

Correct Answer: D

7. Divide the sum by the product

Step 1: Sum $-\frac{13}{5}+\frac{12}{7}=\frac{-91+60}{35}=-\frac{31}{35}$

Step 2: Product $-\frac{31}{7}\times-\frac{1}{2}=\frac{31}{14}$

Step 3: Division $-\frac{31}{35}\div\frac{31}{14}=-\frac{31}{35}\times\frac{14}{31}=-\frac{14}{35}=-\frac{2}{5}$

Correct Answer: D) $-\frac{2}{5}$

8. Simplify$\left(\frac{3}{11}\times\frac{5}{6}\right)-\left(\frac{9}{12}\times\frac{4}{3}\right)+\left(\frac{5}{13}\div\frac{15}{6}\right)$

Step 1: $\frac{3}{11}\times\frac{5}{6}=\frac{15}{66}=\frac{5}{22}$

Step 2: $\frac{9}{12}\times\frac{4}{3}=\frac{36}{36}=1$

Step 3: $\frac{5}{13}\div\frac{15}{6}=\frac{5}{13}\times\frac{6}{15}=\frac{30}{195}=\frac{2}{13}$

Step 4: $\frac{5}{22}-1+\frac{2}{13}$

LCM = 286

$\frac{65}{286}-\frac{286}{286}+\frac{44}{286}=\frac{-177}{286}$

Correct Answer: A) $-\frac{177}{286}$

9. Find the number

$x\times\left(\ -\frac{8}{13}\right)=24$

$x=24\div\left(\ -\frac{8}{13}\right)=24\times\frac{13}{-8}=-39$

Correct Answer: B) -39

10. Find the number

$\frac{-33}{16}\div x=-\frac{11}{4}$

$x=\frac{-33}{16}\div-\frac{11}{4}=\frac{-33}{16}\times\frac{4}{-11}=\frac{132}{176}=\frac{3}{4}$

Correct Answer: B) $\frac{3}{4}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Simplify $4\frac{4}{5}\div\frac{3}{5}$ of $5+\frac{4}{5}\times\frac{3}{10}-\frac{1}{5}$

Step 1: Convert mixed number $4\frac{4}{5}=\frac{24}{5}$

Step 2: "of" means multiply $\frac{3}{5}$ of $5=3$

Step 3: $\frac{24}{5}\div3=\frac{24}{5}\times\frac{1}{3}=\frac{8}{5}$

Step 4: $\frac{4}{5}\times\frac{3}{10}=\frac{12}{50}=\frac{6}{25}$

Step 5: $\frac{8}{5}+\frac{6}{25}-\frac{1}{5}=\frac{40}{25}+\frac{6}{25}-\frac{5}{25}=\frac{41}{25}$

$=1\frac{16}{25}$

Correct Answers: A) $1\frac{16}{25}$and C) $\frac{41}{25}$

12. P is multiplicative inverse of

$\frac{3}{4}\times\frac{5}{9}=\frac{15}{36}=\frac{5}{12}\Rightarrow P=\frac{12}{5}$

$Q$is additive identity of $\frac{3}{4}\Rightarrow Q=0$

$P + Q = \frac{12}{5}$

Multiplying by multiplicative identity (1):

$\frac{12}{5}$

$= 2\frac{2}{5}$

Correct Answers: A) $2\frac{2}{5}$and C) $\frac{12}{5}$

13. A fraction with 32 as numerator and 48 as denominator

$\frac{32}{48} = \frac{2}{3}$

Check equivalent fractions:

A) $\frac{2}{3}$

B) $\frac{4}{5}$

C) $\frac{8}{12} = \frac{2}{3}$

D) $\frac{3}{7}$

Correct Answers: A, C

14. If product of two non-zero rational numbers is 1

Numbers whose product is 1 are multiplicative inverses (reciprocals)

A) Additive inverse → gives 0

B) Multiplicative inverse

C) Reciprocal

D) Opposite signs → product negative

Correct Answers: B, C

COMPREHENSION TYPE

Passage – I:

15. A fraction between $\frac{3}{4}$and $\frac{6}{7}$

$\frac{a+c}{b+d} = \frac{3+6}{4+7} = \frac{9}{11}$

Answer: C) $\frac{9}{11}$

16. Two fractions between $\frac{5}{7}$and $\frac{3}{4}$

$\frac{5+3}{7+4} = \frac{8}{11}$

$\frac{8+3}{11+4} = \frac{11}{15}$

Answer: B) $\frac{8}{11}, \frac{11}{15}$

17. Three fractions between $\frac{1}{8}$and $\frac{2}{5}$

$\frac{1+2}{8+5} = \frac{3}{13}$

$\frac{3+2}{13+5} = \frac{5}{18}$

$\frac{5+2}{18+5} = \frac{7}{23}$

Answer: D) $\frac{3}{13}, \frac{5}{18}, \frac{7}{23}$

MATRIX MATCHING TYPE

18. Match Column-I with Column-II

a) $\left(\quad -\frac{2}{3}\right)+\left(\frac{3}{4}\right)=\frac{3}{4}+\left(\quad -\frac{2}{3}\right)$

→ Order changes → Commutative property (R)

b) $-\frac{7}{15}\times -\frac{5}{14}=\frac{1}{6}$

→ Product is a rational number → Closure property (P)

c) $\frac{1}{2}\times\left(-\frac{2}{3}\times\frac{3}{4}\right)=\left(\frac{1}{2}\times -\frac{2}{3}\right)\times\frac{3}{4}$

→ Grouping changes → Associative property (S)

d) $-\frac{7}{2}\times\frac{2}{-7}=1$

→ Multiplicative inverse gives 1 → Multiplicative identity (T)

e) $-\frac{7}{12}\times\left(-\frac{3}{5}+\frac{2}{9}\right)=\left(-\frac{7}{12}\times -\frac{3}{5}\right)+\left(-\frac{7}{12}\times\frac{2}{9}\right)$

→ Multiplication over addition → Distributive property (Q)

INTEGER TYPE

19. If $-\frac{3}{8}$ and $\frac{x}{-24}$ are equivalent

$$-\frac{3}{8}=\frac{x}{-24}$$

Cross multiply:

$-3\times(-24)=8x\Rightarrow 72=8x\Rightarrow x=9$

Answer: $\boxed{9}$

20. Given $\frac{x}{3}=\frac{1}{5}\left[\left(-\frac{9}{4}\times\frac{5}{3}\right)+\left(\frac{13}{2}\times\frac{5}{6}\right)\right]$

$$-\frac{9}{4}\times\frac{5}{3}=-\frac{45}{12}=-\frac{15}{4}$$

$$\frac{13}{2}\times\frac{5}{6}=\frac{65}{12}$$

$$-\frac{15}{4}+\frac{65}{12}=-\frac{45}{12}+\frac{65}{12}=\frac{20}{12}=\frac{5}{3}$$

$\frac{x}{3}=\frac{1}{5}\times\frac{5}{3}=\frac{1}{3}$

$x=1$

Answer: 1

STUDENT PRACTICE CORNER

1. Add $\frac{3}{8}$ and $-\frac{5}{12}$

LCM = 24

$$\frac{3}{8}=\frac{9}{24},-\frac{5}{12}=-\frac{10}{24}$$

$$\frac{9}{24}-\frac{10}{24}=-\frac{1}{24}$$

Correct Answer: D) $-\frac{1}{24}$

2. Simplify $-\frac{2}{3}+\frac{5}{9}-\left(\quad -\frac{7}{6}\right)$

$$-\frac{2}{3}+\frac{5}{9}+\frac{7}{6}$$

LCM = 18

$-\frac{12}{18}+\frac{10}{18}+\frac{21}{18}=\frac{19}{18}$

Correct Answer: A) $\frac{19}{18}$

3. Subtract $-\frac{8}{33}$from $-\frac{7}{22}$

$-\frac{7}{22}-\left(-\frac{8}{33}\right)=-\frac{7}{22}+\frac{8}{33}$

LCM = 66

$-\frac{21}{66}+\frac{16}{66}=-\frac{5}{66}$

Correct Answer: C) $-\frac{5}{66}$

4. From number line

Points increase by $\frac{2}{7}$:

$-\frac{4}{7},\ -\frac{2}{7},\ 0,\ \frac{2}{7},\ \frac{4}{7},\ \frac{6}{7}$

So,

$x=-\frac{6}{7}, y=\frac{4}{7}$

$x+y=-\frac{6}{7}+\frac{4}{7}=-\frac{2}{7}$

Correct Answer: D) $-\frac{2}{7}$

5. Compare $\frac{4}{5}$and $\frac{3}{8}$

$\frac{4}{5}=0.8, \frac{3}{8}=0.375$

$0.8 > 0.375$

Correct Answer: A) $\frac{4}{5}>\frac{3}{8}$

6. Additive inverse of $\frac{11}{14}$

Additive inverse $=-\frac{11}{14}$

Correct Answer: B) $-\frac{11}{14}$

7. A rational number between $\frac{5}{6}$and $\frac{7}{6}$

$\frac{6}{6}=1$ (lies between)

Correct Answer: A) $\frac{6}{6}$

8. Lowest form of $\frac{4}{22}, \frac{-32}{-56}$

$\frac{4}{22}=\frac{2}{11}, \frac{-32}{-56}=\frac{32}{56}=\frac{4}{7}$

Correct Answer: D) $\frac{2}{11}, \frac{4}{7}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Which are equal to $\frac{4}{7}\times\left(-\frac{14}{3}\right)$?

$\frac{4}{7}\times\left(-\frac{14}{3}\right)=\frac{-56}{21}=-\frac{8}{3}$

Check options:

A) $-\frac{8}{3}$

B) $\frac{8}{3}$

C) $-\frac{56}{21}$

D) $-\frac{16}{6} = -\frac{8}{3}$

Correct Answers: A, C, D

10. Correct values of $\frac{5}{6} \div \frac{10}{3}$

$\frac{5}{6} \div \frac{10}{3} = \frac{5}{6} \times \frac{3}{10} = \frac{15}{60} = \frac{1}{4}$

Check options:

A) $\frac{1}{4}$

B) $\frac{5}{6} \times \frac{3}{10}$

C) $\frac{15}{60}$

D) $\frac{1}{2}$

Correct Answers: A, B, C

COMPREHENSION TYPE

Passage – I:

11. Commutative property of multiplication

$a \times b = b \times a$

Check options:

A) Both give $\frac{10}{18}$

B) Results are different

C) Results are different

D) $\frac{4}{9} \times \frac{3}{5} \neq \frac{3}{5} \times \frac{4}{9}$(given unequal)

Correct Answer: A

12. Associative property of addition

$(a + b) + c = a + (b + c)$

A) Matches grouping change

B), C), D) do not show grouping change

Correct Answer: A

MATRIX MATCHING TYPE

13. A)

$\frac{5}{6} + \frac{7}{12} = \frac{10}{12} + \frac{7}{12} = \frac{17}{12}$

→ Q

B)

$\frac{9}{10} - \frac{4}{15} = \frac{27}{30} - \frac{8}{30} = \frac{19}{30}$

→ S

C)

$\frac{8}{9} \times \left(-\frac{3}{4}\right) = -\frac{24}{36} = -\frac{2}{3}$

→ R

D)

$\frac{7}{8} \div \frac{14}{3} = \frac{7}{8} \times \frac{3}{14} = \frac{21}{112} = \frac{3}{16}$

→ P

INTEGER TYPE

14. Simplest form of $\frac{70}{-84}$

$\frac{70}{-84} = -\frac{5}{6}$

Answer: $\boxed{-\frac{5}{6}}$

15. $-\frac{22}{15}$is subtracted from $-\frac{x}{3}$to get $\frac{4}{5}$

$-\frac{x}{3} - \left(-\frac{22}{15}\right) = \frac{4}{5}$

$-\frac{x}{3} + \frac{22}{15} = \frac{4}{5}$

LCM = 15

$-\frac{5x}{15} + \frac{22}{15} = \frac{12}{15}$

$-\frac{5x}{15} = -\frac{10}{15} \Rightarrow -5x = -10 \Rightarrow x = 2$

Answer: $\boxed{2}$

Fractions

LEVEL – 1

1. $2 + \frac{1}{3} + \frac{1}{3} = 2 + \frac{2}{3} = \frac{8}{3}$

Answer: D) $\frac{8}{3}$

2. $\frac{226}{15} = 15$ remainder $1 \Rightarrow 15\frac{1}{15}$

Answer: D) $15\frac{1}{15}$

3. $\frac{1}{6} + \frac{7}{12} + \frac{2}{3} = \frac{2}{12} + \frac{7}{12} + \frac{8}{12} = \frac{17}{12} = 1\frac{5}{12}$

Answer: D) $1\frac{5}{12}$

4. $2\frac{1}{4} + 3 + 1\frac{1}{4} = (2 + 3 + 1) + \left(\frac{1}{4} + \frac{1}{4}\right) = 6 + \frac{2}{4} = 6\frac{1}{2}$

Answer: D) $6\frac{1}{2}$

5. $76\frac{1}{2} + 11\frac{5}{6} = (76 + 11) + \left(\frac{1}{2} + \frac{5}{6}\right) = 87 + \left(\frac{3}{6} + \frac{5}{6}\right) = 87 + \frac{8}{6} = 88\frac{1}{3}$

Answer: B) $88\frac{1}{3}$

6. $4\frac{7}{10} - 3\frac{1}{10} = (4 - 3) + \left(\frac{7}{10} - \frac{1}{10}\right) = 1 + \frac{6}{10} = 1\frac{3}{5}$

Answer: C) $1\frac{3}{5}$

7. $\frac{5}{12} - \frac{3}{8} = \frac{10}{24} - \frac{9}{24} = \frac{1}{24}$

Answer: B) $\frac{1}{24}$

8. $20\frac{5}{7} - 15\frac{1}{7} = (20 - 15) + \left(\frac{5}{7} - \frac{1}{7}\right) = 5 + \frac{4}{7} = 5\frac{4}{7}$

Answer: C) $5\frac{4}{7}$

9. Rewrite with positive denominators:
$\frac{-3}{5}, \frac{-7}{10}, \frac{-8}{10}, \frac{-17}{15}$

LCM of 5, 10, 15 = 30
Convert: $\frac{-3}{5} = \frac{-18}{30}, \frac{-7}{10} = \frac{-21}{30}, \frac{-8}{10} = \frac{-24}{30}, \frac{-17}{15} = \frac{-34}{30}$
Order: $-34 < -24 < -21 < -18$
Ascending order:
$\frac{-17}{15}, \frac{8}{-10}, \frac{-7}{10}, \frac{3}{-5}$
Correct Option: A

10. Change $5\frac{1}{2}$ into improper fraction
$5\frac{1}{2} = \frac{5 \times 2 + 1}{2} = \frac{11}{2}$
Answer: A) $\frac{11}{2}$

11. Work left to revise
$\frac{7}{12} + \frac{3}{12} = \frac{10}{12}$
$\text{Remaining} = 1 - \frac{10}{12} = \frac{2}{12}$
Answer: C) $\frac{2}{12}$

12. Reduce $\frac{144}{180}$
$\div 36 \Rightarrow \frac{4}{5}$
Answer: A) $\frac{4}{5}$

13. Compare $a = \frac{2}{3}$, $b = \frac{4}{9}$
$\frac{2}{3} = \frac{6}{9}, \frac{6}{9} > \frac{4}{9}$
Answer: A) $a > b$

14. Fractions $\frac{1}{2}, \frac{2}{4}, \frac{3}{6}$
$\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$
Answer: C) Equivalent fractions

15. Cake left
$1 - \frac{5}{7} = \frac{2}{7}$
Answer: C) $\frac{2}{7}$

16. Cost per kg
$6\frac{1}{4} = \frac{25}{4}$
$\text{Rate} = 400 \div \frac{25}{4} = 400 \times \frac{4}{25} = 64$
Answer: A) 64

17. Pages read

$\frac{3}{5} \times 360 = 216$

Answer: A) 216 pages

18. Butter left

Used:

$\frac{1}{2} + \frac{1}{8} + \frac{2}{3} = \frac{12 + 3 + 16}{24} = \frac{31}{24}$

Left from 2 grams:

$2 - \frac{31}{24} = \frac{48}{24} - \frac{31}{24} = \frac{17}{24}$

Answer: B) $\frac{17}{24}$

19. Fraction of loaf left

Total eaten:

$8 + 6 + 5 + 9 = 28$

Left slices:

$35 - 28 = 7$

Fraction left:

$\frac{7}{35} = \frac{1}{5}$

Answer: D) $\frac{1}{5}$

20. Total liquid in recipe

$2\frac{3}{4} + 4\frac{5}{8} + \frac{1}{2}$

Convert to improper fractions:

$\frac{11}{4} + \frac{37}{8} + \frac{1}{2}$

LCM = 8

$\frac{22}{8} + \frac{37}{8} + \frac{4}{8} = \frac{63}{8}$

$= 7\frac{7}{8}$

Answer: B) $7\frac{7}{8}$cups

LEVEL – 2

SINGLE CORRECT ANSWER TYPE

1. Rajesh's jump

$8 - \frac{2}{3} = \frac{24}{3} - \frac{2}{3} = \frac{22}{3}$

Answer: A) $\frac{22}{3}$m

2. Minutes in $7\frac{1}{6}$hours

$7\frac{1}{6} = \frac{43}{6}$

$\frac{43}{6} \times 60 = 43 \times 10 = 430$

Answer: B) 430 minutes

3. Simplify

$$2\frac{1}{3}\times 1\frac{5}{14}\times 1\frac{4}{5}=\frac{7}{3}\times\frac{19}{14}\times\frac{9}{5}$$

Cancel:

$$=\frac{7\times 19\times 9}{3\times 14\times 5}=\frac{171}{10}=5\frac{8}{10}$$

Answer: B) $5\frac{8}{10}$

4. Greatest fraction among $\frac{7}{8},\frac{6}{7},\frac{4}{5},\frac{5}{6}$

Convert to decimals:

$$\frac{7}{8}=0.875,\frac{6}{7}\approx 0.857,\frac{5}{6}\approx 0.833,\frac{4}{5}=0.8$$

Answer: D) $\frac{7}{8}$

5. Total weight

$$18\frac{3}{4}+7\frac{1}{2}+10\frac{1}{5}$$

Convert to improper fractions:

$$\frac{75}{4}+\frac{15}{2}+\frac{51}{5}$$

LCM = 20

$$\frac{375}{20}+\frac{150}{20}+\frac{204}{20}=\frac{729}{20}=36\frac{9}{20}$$

Answer: D) $36\frac{9}{20}$kg

6. Fraction of pizzas left

Eaten:

$$\frac{8}{9}+\frac{2}{3}+\frac{7}{9}=\frac{8}{9}+\frac{6}{9}+\frac{7}{9}=\frac{21}{9}=\frac{7}{3}$$

Total pizzas = 3

Left:

$$3-\frac{7}{3}=\frac{9}{3}-\frac{7}{3}=\frac{2}{3}$$

Fraction of total:

$$\frac{2}{3}\div 3=\frac{2}{9}$$

Answer: C) $\frac{2}{9}$

7. Number of boards

$$22\frac{1}{2}\div 1\frac{1}{2}=\frac{45}{2}\div\frac{3}{2}=\frac{45}{2}\times\frac{2}{3}=15$$

Answer: A) 15 boards

8. Pages read

Total pages:

$$14+26+12+28=80$$

Read:

$$\frac{1}{6}\times 80=\frac{80}{6}=13\frac{1}{3}$$

Answer: B) $13\frac{1}{3}$pages

9. Arrange $\frac{2}{3}, \frac{3}{4}, \frac{1}{3}, \frac{5}{6}$ in ascending order (smallest to largest).

Find LCM:

LCM of 3,4,6 = 12

Convert: $\frac{2}{3} = \frac{8}{12}, \frac{3}{4} = \frac{9}{12}, \frac{1}{3} = \frac{4}{12}, \frac{5}{6} = \frac{10}{12}$

Order numerators: $4 < 8 < 9 < 10$

Ascending order:

$\frac{1}{3}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$

Correct Answer: B) $\frac{1}{3}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$

10. Simplify $1\frac{1}{2} \div 1\frac{5}{13}$

$$\frac{3}{2} \div \frac{18}{13} = \frac{3}{2} \times \frac{13}{18} = \frac{39}{36} = \frac{13}{12} = 1\frac{1}{12}$$

Answer: B) $1\frac{1}{12}$

ONE OR MORE THAN ONE ANSWER CORRECT QUESTIONS

11. Equivalent fractions

$$\frac{1}{2} = \frac{3}{6}$$

Correct Answers: A, C

12. $3\frac{2}{3} \div 2\frac{1}{2}$

$$\frac{11}{3} \div \frac{5}{2} = \frac{11}{3} \times \frac{2}{5} = \frac{22}{15} = 1\frac{7}{15}$$

Correct Answers: B, D

13. $\frac{7}{2}$ of $\frac{4}{9}$

$$\frac{7}{2} \times \frac{4}{9} = \frac{28}{18} = \frac{14}{9} = 1\frac{5}{9}$$

Correct Answers: A, C, D

14. $\frac{9}{5} + \frac{1}{10} - \frac{17}{5}$

$$\frac{18}{10} + \frac{1}{10} - \frac{34}{10} = -\frac{15}{10} = -\frac{3}{2} = -1\frac{1}{2}$$

Correct Answers: A, B

COMPREHENSION TYPE QUESTIONS

Passage – I:

15. Identify correct representation

From figures (count shaded parts / total parts):

$A = \frac{5}{6}, B = \frac{4}{7}, C = \frac{3}{8}, D = \frac{2}{5}$

Correct Answer: D

16. False statement

Compare using cross multiplication:

$A > B$:

$\frac{5}{6} \; ? \; \frac{4}{7} \Rightarrow 5 \times 7 = 35,\ 4 \times 6 = 24 \Rightarrow 35 > 24$ (True)

$B > C$:

$\frac{4}{7}\ ?\ \frac{3}{8} \Rightarrow 4 \times 8 = 32,\ 3 \times 7 = 21 \Rightarrow 32 > 21$ (True)

$C > A$:

$\frac{3}{8}\ ?\ \frac{5}{6} \Rightarrow 3 \times 6 = 18,\ 5 \times 8 = 40 \Rightarrow 18 < 40$ (False)

$D > C$:

$\frac{2}{5}\ ?\ \frac{3}{8} \Rightarrow 2 \times 8 = 16,\ 3 \times 5 = 15 \Rightarrow 16 > 15$ (True)

Correct Answer: C

17. Ascending order

Compare step by step:

Cvs D:

$$\frac{3}{8}\ ?\ \frac{2}{5} \Rightarrow 3 \times 5 = 15,\ 2 \times 8 = 16 \Rightarrow 15 < 16 \Rightarrow C < D$$

Dvs B:

$$\frac{2}{5}\ ?\ \frac{4}{7} \Rightarrow 2 \times 7 = 14,\ 4 \times 5 = 20 \Rightarrow 14 < 20 \Rightarrow D < B$$

Bvs A:

$$\frac{4}{7}\ ?\ \frac{5}{6} \Rightarrow 4 \times 6 = 24,\ 5 \times 7 = 35 \Rightarrow 24 < 35 \Rightarrow B < A$$

$$\therefore C < D < B < A$$

Correct Answer: C) C, D, B, A

MATRIX MATCHING TYPE QUESITONS

18. Match Column–I with Column–II (using fractions)

a) $2 \times \frac{1}{5} = \frac{2}{5}$

→ Two parts out of five shaded → Q

b) $2 \times \frac{1}{2} = \frac{2}{2} = 1$

→ One whole (two halves) → R

c) $3 \times \frac{2}{3} = \frac{6}{3} = 2$

→ Two wholes → P

d) $4 \times \frac{1}{2} = \frac{4}{2} = 2$

→ Four halves = two wholes → T

INTEGER TYPE

19. Simplify $\frac{\frac{1}{3} \div \frac{1}{3} \times \frac{1}{3}}{\frac{1}{3} \div \left(\frac{1}{3} \text{ of } \frac{1}{3}\right)} - \frac{1}{9}$

Numerator:

$\frac{1}{3} \div \frac{1}{3} = 1{,}1 \times \frac{1}{3} = \frac{1}{3}$

Denominator:

$\frac{1}{3}$ of $\frac{1}{3} = \frac{1}{9}$

$\frac{1}{3} \div \frac{1}{9} = 3$

Whole expression:

$\frac{1}{3}=\frac{1}{9}$

$\frac{1}{9}-\frac{1}{9}=0=\frac{0}{9}$

$x=0$

Answer: $\boxed{0}$

20. Simplify

$3+1\frac{1}{5}+\frac{2}{5}-\frac{7}{15}$

$1\frac{1}{5}=\frac{6}{5}$

$3+\frac{6}{5}+\frac{2}{5}-\frac{7}{15}=3+\frac{8}{5}-\frac{7}{15}$

LCM = 15

$3+\frac{24}{15}-\frac{7}{15}=3+\frac{17}{15}=\frac{45}{15}+\frac{17}{15}=\frac{62}{15}$

Given:

$\frac{62}{15}=\frac{62}{3p}$

$15=3p\Rightarrow p=5$

Answer: 5

STUDENT PRACTICE CORNER

SINGLE ANSWER CORRECT QUESTIONS

1 $3\frac{5}{6}\times 4\frac{2}{5}=\frac{23}{6}\times\frac{22}{5}=\frac{506}{30}=\frac{253}{15}=16\frac{13}{15}$

Answer: C) $16\frac{13}{15}$

2. $\frac{5}{8}\div 3=\frac{5}{8}\times\frac{1}{3}=\frac{5}{24}$

Answer: A) $\frac{5}{24}$

3. $1\frac{4}{7}=\frac{11}{7}\Rightarrow$ number $=\frac{7}{11}$

Answer: D) $\frac{7}{11}$

4. $\frac{24}{30}=\frac{4}{5}$

Answer: C) $\frac{4}{5}$

5. Rope used

$\frac{3}{4}$ of $\frac{7}{8}=\frac{3}{4}\times\frac{7}{8}=\frac{21}{32}$

Answer: A) $\frac{21}{32}$m

6. Fraction of time

$2\frac{1}{4}=\frac{9}{4}$ hours $=\frac{9}{4}\times 60=135$ minutes

$\frac{55}{135}=\frac{11}{27}$

Answer: D) $\frac{11}{27}$

7. Equal to $\frac{1}{3}$

A) $\frac{5}{6} \div \frac{2}{3} = \frac{5}{6} \times \frac{3}{2} = \frac{15}{12} = \frac{5}{4} \neq \frac{1}{3}$

B) $\frac{7}{12} - \frac{1}{6} = \frac{7}{12} - \frac{2}{12} = \frac{5}{12} \neq \frac{1}{3}$

C) $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12} \neq \frac{1}{3}$

D) $\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$

Answer: D

8. Fraction of total capacity used

$\frac{3}{4}$ of $\frac{4}{5} = \frac{3}{4} \times \frac{4}{5} = \frac{12}{20} = \frac{3}{5}$

Answer: B) $\frac{3}{5}$

ONE OR MORE THAN ONE ANSWER CORRECT QUESTIONS

9. Equal to $\frac{2}{3} \times \frac{9}{4}$

$\frac{2}{3} \times \frac{9}{4} = \frac{18}{12} = \frac{3}{2} = 1\frac{1}{2}$

Check options:

A) $\frac{18}{12}$

B) $\frac{3}{2}$

C) $\frac{6}{12} = \frac{1}{2}$

D) $1\frac{1}{2}$

Correct Answers: A, B, D

10. Correct values of $\frac{4}{5} \div \frac{2}{3}$

$\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times \frac{3}{2} = \frac{12}{10} = \frac{6}{5} = 1\frac{1}{5}$

Check options:

A) $\frac{12}{10}$

B) $\frac{6}{5}$

C) $1\frac{1}{5}$

D) $\frac{8}{15}$

Correct Answers: A, B, C

COMPREHENSION TYPE QUESITONS

Passage – I:

11.

$\frac{1}{2} + \frac{3}{2} = \frac{1+3}{2} = \frac{4}{2} = 2$

Answer: C) 2

12. $\frac{1}{3} + \frac{1}{2}$

LCM = 6

$\frac{2}{6}+\frac{3}{6}=\frac{5}{6}$

Answer: A) $\frac{5}{6}$

MATRIX MATCHING TYPE QUESITONS

13. a) Reciprocal of $\frac{a}{b}=\frac{b}{a}\rightarrow$ S

b) Additive inverse of $-\frac{3}{7}=\frac{3}{7}\rightarrow$ P

c) $\frac{4}{9}+\frac{5}{9}=\frac{9}{9}=1\rightarrow$ Q

d) $5\frac{1}{2}+\frac{1}{2}=5+1=6\rightarrow$ R

Final Matching: a–S, b–P, c–Q, d–R

INTEGER TYPE

14. Simplify

$\frac{1}{40}+\frac{15}{40}+\frac{24}{40}=\frac{40}{40}=1$

Answer: $\boxed{1}$

15. Find x

$\frac{45}{60}=\frac{3}{x}\Rightarrow\frac{3}{4}=\frac{3}{x}$

$3x=12\Rightarrow x=4$

Answer: $\boxed{4}$

DECIMALS

LEVEL – 1

1. $0.088+0.091=0.179$

Answer: D) 0.179

2. $0.63\times0.42=\frac{63}{100}\times\frac{42}{100}=\frac{2646}{10000}=0.2646$

Answer: C) 0.2646

3. $\frac{825}{125}=\frac{33}{5}=6.6$

Answer: D) 6.6

4. $7.73+0.07=7.80$

$7.80+7=14.80$

$14.80+7.86=22.66$

$22.66+7.153=29.813$

Answer: D) 29.813

5. $5.25+3.38=8.63$

Digit 3 is in hundredths place

$=\frac{3}{100}$

Answer: B) $\frac{3}{100}$

6. $30.5\times7.50=228.75$

Answer: C) Rs.228.75

7. Thousandths place = 3rd digit after decimal
A) 3.0095 → 3rd digit = 9
B) 3.0905 → 3rd digit = 0
C) 3.9005 → 3rd digit = 0
D) 3.0059 → 3rd digit = 5
Answer: A) 3.0095

8. Greatest decimal
Compare by place values:
All have 6 in ones place.
6.723
6.7023
6.7 (= 6.7000)
6.0723 (smallest since tenths is 0)
Compare tenths and hundredths:
$6.723 > 6.7023 > 6.7 > 6.0723$
Answer: A) 6.723

9. Fraction of 1.75
$1.75 = \frac{175}{100} = \frac{7}{4}$
Answer: C) $\frac{7}{4}$

10. Cloth left
$25.3 - 3.85 = 21.45$
Answer: B) 21.45 m

11. Convert to litres
$15 \text{ ml} = \frac{15}{1000} = 0.015$ litres
$15 + 0.015 = 15.015$
Answer: D) 15.015 litres

12. Value of $P + Q - R$
On the number line, points are equally spaced.
From 6.37to 6.5there is a gap of 0.13.
Each step = 0.13
So,$P = 6.5 + 0.13 = 6.63$
$Q = 6.63 + 0.13 = 6.76$
$R = 6.76 + 0.13 = 6.89$
Now,
$P + Q - R = 6.63 + 6.76 - 6.89 = 6.50$
Answer: C) 6.5

13. $2.9 \div 0.8758 \approx 3.31$
Answer: A) 3.31

14. To compare decimals, look at the digits from left to right.
A) 0.43 < 0.043 (False, 4 tenths is greater than 0 tenths)
B) 0.0043 > 0.43 (False, 0 tenths is less than 4 tenths)
C) 0.00043 > 0.043 (False, 0 thousandths is less than 4 thousandths)
D) 0.043 > 0.0043 (True, 4 hundredths is greater than 0 hundredths)
Correct Option: D

15. Average miles per day
$25.8 \div 3 = 8.6$
Answer: B) 8.6

16. Quarts of water
$3.25 \times 1.06 = 3.445$
Answer: C) 3.445

17. Total amount spent
$5.45 + 3.06 = 8.51$
Answer: C) Rs. 8.51

18. Value of
$\frac{7}{10} + \frac{6}{100} + \frac{4}{1000} = 0.7 + 0.06 + 0.004 = 0.764$
Answer: C) 0.764

19. Descending order (largest to smallest)
A) 3.84, 6.70, 7.19, 3.08 (not decreasing)
B) 0.26, 4.91, 0.462, 9.58
C) 9.65 > 8.298 > 5.97 > 0.23
D) 2.5 < 5.40 < 6.2 < 8.546 (increasing)
Answer: C

20. Required difference
Sum = 33.69 + 112.55 = 146.24
Difference = 112.55 − 33.69 = 78.86
146.24 − 78.86 = 67.38
Answer: B) 67.38

LEVEL – 2

SINGLE ANSWER CORRECT QUESTIONS

1. Total amount
$10.50 + 15.80 = 26.30$
Answer: D) 26.30

2. Change received
$50 - 35.65 = 14.35$
Answer: B) Rs 14.35

3. Number of packets
$2\frac{1}{2} = 2.5, 2.5 \times 12.4 = 31$ kg = 31000 g
$\frac{31000}{62} = 500$
Answer: D) 500

4. Total litres
$1.42 + 1.89 = 3.31$
Answer: D) 3.31 litres

5. Total cost of gas
$6 \times 37.27 = 223.62$
Answer: B) $223.62

6. Final height
$68.8 + 0.6 = 69.4$
Answer: C) 69.4

7. Thickness of piece B
$0.724 + 0.076 = 0.800 = 0.8$
Answer: C) 0.8

8. Descending order
Compare values:
$23.3 > 20.3 > 2.33 > 2.03$
Answer: C

9. Number of packets
$3.75 \div 0.25 = \frac{375}{25} = 15$
Answer: B) 15

10. Place value comparison
Tens place $4 = 40$
Hundredths place $4 = 0.04$
$\frac{40}{0.04} = 1000$
Answer: C) 1000

ONE OR MORE THAN ONE ANSWER CORRECT QUESTIONS

11. Like decimals (same number of decimal places as $12.52 \rightarrow 2$ decimal places)
A) 0.8 (1 decimal place)
B) 3.25
C) 13.25
D) 12.552 (3 decimal places)
Correct Answers: B, C

12. Average distance
$\frac{89.1}{2.2} = \frac{891}{22} = 40\frac{11}{22} = 40.5$
A) 40.5
B) $\frac{891}{22}$
C) $\frac{892}{51}$
D) $\frac{982}{51}$
Correct Answers: A, B

13. Ascending order (smallest to largest)
A) $0.2 < 0.25 < 0.3$
B) 0.5, 0.45, 0.6 (0.45 should come first)
C) $0.12 < 0.21 < 0.3$
D) $0.08 < 0.8 < 0.88$
Correct Answers: A, C, D

14. Greater than 0.6
$0.6 = 0.600$
A) $0.65 > 0.6$
B) $0.59 < 0.6$
C) $0.602 > 0.6$
D) $0.600 = 0.6$
Correct Answers: A, C

COMPREHENSION TYPE QUESITONS

Passage – I:

15. Convert to grams
5 kg = 5000 g
5000 + 80 = 5080 g
Answer: A) 5080

16. Convert to km
7777 ÷ 1000 = 7.777 km
Answer: C) 7.777

17. Convert to meters
2 km = 2000 m
2000 + 45 = 2045 m
Answer: D) 2045

MATRIX MATCHING TYPE

Number from place value chart:

4698.530

Place values:

5 is in tenths → $\frac{5}{10}$

4 is in thousands → 4000

9 is in tens → 90

3 is in hundredths → $\frac{3}{100}$

Matching:

A → T, B → S, C → P, D → R

INTEGER TYPE QUESITONS

19. Number of decimal places
12.38 → digits after decimal = 2
Answer: 2

20. Number of digits in integral part
1321.28 → integral part = 1321
Digits = 4
Answer: 4

STUDENT PRACTICE CORNER

SINGLE CORRECT ANSWER TYPE

1. 78.09 + 19.367 = 97.457
Answer: C) 97.457

2. Greatest decimal
0.98 > 0.89 > 0.87 > 0.78
Answer: D) 0.98

3. $700 + 20 + 5 + \frac{9}{100} = 725 + 0.09 = 725.09$
Answer: A) 725.09

4. Number of pieces
9.6 ÷ 1.2 = 8
Answer: C) 8

5. Money left
$18.50 - 11.75 = 6.75$
Answer: C) 6.75

6. Number of bags
$1743 \div 49.8 = \frac{17430}{498} = 35$
Answer: A) 35

7. Number of pieces
$50 \div 1.25 = \frac{50}{1.25} = 40$
Answer: B) 40

ONE OR MORE THAN ONE ANSWER CORRECT QUESTIONS

8. Correct results of division
A) $1 \div 2 = 0.5$
B) $3 \div 4 = 0.75$
C) $5 \div 2 = 2.5$
D) $2 \div 5 = 0.4 \neq 0.25$
Correct Answers: A, B, C

9. Correct conversions
A) $\frac{1}{10} = 0.1$
B) $\frac{1}{5} = 0.2$
C) $\frac{3}{10} = 0.3 \neq 0.03$
D) $\frac{7}{2} = 3.5$
Correct Answers: A, B, D

10. Compare decimals
$A = 2.30, B = 2.03, C = 2.33$
$B < A < C$
A) $A < B$
B) $A < C$
C) $B < C$
D) $C < B$
Correct Answers: B, C

COMPREHENSION TYPE

Passage – I:

11. Place value of 9 in $A = 158.398$
Digits after decimal:
3 (tenths), 9 (hundredths), 8 (thousandths)
So,$9 = \frac{9}{100}$
Answer: B) $\frac{9}{100}$

12. Place value of 6 in $B = 12.4269$
Digits after decimal:
4 (tenths), 2 (hundredths), 6 (thousandths), 9 (ten-thousandths)

So,$6 = \frac{6}{1000}$

Answer: D) $\frac{6}{1000}$

MATRIX MATCHING TYPE

13. Match Column A and Column B

P) $45.67 + 8.905 = 54.575 \rightarrow$ B

Q) $100 - 37.458 = 62.542 \rightarrow$ C

R) $12.5 + 0.625 = 13.125 \rightarrow$ D

S) $78.3 - 15.175 = 63.125 \rightarrow$ A

Final Matching: P–B, Q–C, R–D, S–A

INTEGER TYPE

14. Difference

$8.5 - 3.5 = 5$

Answer: 5

15. Decimal places

$281.238 \rightarrow$ 3 digits after decimal

Answer: 3

4. Ratio Proportion and Comparing Quantities

LEVEL – 1

1. Simplest form of 500 : 650
Divide both by 50
500 ÷ 50 = 10, 650 ÷ 50 = 13
Answer: C) 10 : 13

2. Divide ₹1300 in ratio 2 : 3
Total parts = 2 + 3 = 5
Venkat's share = $\frac{2}{5} \times 1300 = 520$
Answer: D) 520

3. Ratio 3 : 8, sum = 440
Total parts = 11
One part = 440 ÷ 11 = 40
Numbers = 3×40 = 120 and 8×40 = 320
Difference = 320 − 120 = 200
Answer: A) 200

4. Ratio of lead : copper = 3 : 7, lead = 60.9 g
3 parts = 60.9
1 part = 60.9 ÷ 3 = 20.3
Copper = 7 × 20.3 = 142.1 g
Answer: B) 142.1

5. Triangle sides in ratio 2 : 3 : 5, perimeter = 90 cm
Total parts = 10
One part = 90 ÷ 10 = 9
Longest side = 5 × 9 = 45 cm
Answer: C) 45

6. Smallest ratio
Convert to fractions:
7/6 ≈ 1.17
4/9 ≈ 0.44
5/12 ≈ 0.42
3/4 = 0.75
Smallest = 5/12
Answer: C) 5 : 12

7. If a = 2b = 3c
Let a = 6 (LCM of 2 and 3)
Then b = 3, c = 2
So, a : b : c = 6 : 3 : 2
Answer: B) 6 : 3 : 2

8. If 1/p : 1/q : 1/r = 2 : 3 : 5
Take reciprocals → p : q : r = 1/2 : 1/3 : 1/5
LCM of (2,3,5) = 30

p : q : r = 15 : 10 : 6
Answer: D) 15 : 10 : 6

9. Ratio male : female = 2 : 3, males = 168
2 parts = 168
1 part = 84
Females = 3 × 84 = 252
Answer: B) 252

10. Divide ₹1428 in ratio 2 : 5
Total parts = 7
Haneesh's share = $\frac{2}{7} \times 1428 = 408$
Answer: D) 408

11. 12 : x = 8 : 6
$\frac{12}{x} = \frac{8}{6}$
12 × 6 = 8x → 72 = 8x → x = 9
Answer: B) 9

12. (16+x) : (2+x) = (25+x) : (4+x)
Try x = 2:
18 : 4 = 27 : 6 → 108 = 108
Answer: A) 2

13. 16 : 18 = x : 96
$\frac{16}{18} = \frac{x}{96}$
x = $\frac{16}{18} \times 96 = \frac{256}{3}$
Answer: D) 256/3

14. Income : savings = 7 : 4, savings = ₹12500
4 parts = 12500 → 1 part = 3125
Income = 7 × 3125 = 21875
Answer: C) 21,875

15. 72 : x = 186 : 217
$\frac{72}{x} = \frac{186}{217}$
x = $\frac{72 \times 217}{186} = 84$
Answer: A) 84

16. Let the present ages of Rahul and Rohit be $7x$and $5x$, respectively.
The sum of their ages is given as 84:

$$7x + 5x = 84$$
$$12x = 84$$
$$x = \frac{84}{12} = 7$$

The difference between their ages is $7x - 5x = 2x$.
Substitute the value of x: $2(7) = 14$
Answer: C) 14

17. (35 − x) : (40 − x) = 6 : 7
$\frac{35 - x}{40 - x} = \frac{6}{7}$

$7(35 - x) = 6(40 - x)$
$245 - 7x = 240 - 6x$
$x = 5$
Answer: B) 5

18. 7, 14, x in continued proportion
$\frac{7}{14} = \frac{14}{x}$
$\frac{1}{2} = \frac{14}{x} \longrightarrow x = 28$
Answer: D) 28

19. Mean proportion between 4 and 36
Mean = $\sqrt{(4 \times 36)} = \sqrt{144} = 12$
Answer: C) 12

20. 8, x, 18 in continued proportion
$\frac{8}{x} = \frac{x}{18}$
$x^2 = 8 \times 18 = 144 \rightarrow x = 12$
Answer: B) 12

LEVEL – 2

Single Answer Questions

1. Let the numbers be $x$and y.
Mean proportion = 12
$\sqrt{xy} = 12 \Rightarrow xy = 144$
Given $x = 9y$
$(9y)(y) = 144$
$9y^2 = 144$
$y^2 = 16 \Rightarrow y = 4$
$x = 9y = 36$
$x + y = 36 + 4 = 40$
Answer: C) 40

2. Let ages be $6x$ and $7x$.
Difference = $7x - 6x = x = 7$
$x = 7$
Shiva's age = $6x = 6(7) = 42$
Answer: B) 42

3. Ratio = 4 : 3 : 5
Total parts = 12
1 part = $840 \div 12 = 70$
Highest number = $5 \times 70 = 350$
Answer: C) 350

4. Out of 10 students, 2 fail
Fail ratio = $2/10 = 1/5$
Fail students = $80 \times \frac{1}{5} = 16$
Pass students = $80 - 16 = 64$
Correct Answer D: 64

5. Let $x: y = 7: 5$
Take $x = 7k,\ y = 5k$
$$\frac{5x + 4y}{3x - 2y} = \frac{5(7k) + 4(5k)}{3(7k) - 2(5k)}$$
$$= \frac{35k + 20k}{21k - 10k} = \frac{55k}{11k} = 5$$
Answer: A) 5

6. 6 hours : 1 day
1 day = 24 hours
$6: 24 = 1: 4$
Answer: D) 1 : 4

7. 10%of m = 20%of n
$$\frac{10}{100}m = \frac{20}{100}n$$
$$\frac{m}{10} = \frac{n}{5}$$
$m: n = 2: 1$
Answer: A) 2 : 1

8. Let AB = 14 cm and ratio = 3 : 4
Total parts = 7
$$AX = \frac{3}{7} \times 14 = 6$$
Answer: D) 6

9. Total coins = 216
Ratio (₹1 : 50p : 10p) = 2 : 3 : 4
Total parts = 9
1 part = 216 ÷ 9 = 24
Number of 50p coins = 3 × 24 = 72
Answer: D) 72

10. Boys : Girls = 8 : 3
3 parts = 912
1 part = 304
Total parts = 11
$Total = 11 \times 304 = 3344$
Answer: B) 3344

One or More Than One Correct Answer Questions

11. Equivalent ratios of $4: 9$
Check each:
A) $8: 18 = 4: 9$
B) $20: 45 = 4: 9$
C) $12: 27 = 4: 9$
Answer: A, B, C

12. Let $a: b = 3: 4$
Take $a = 3k,\ b = 4k$

$$\frac{2a+3b}{5a+b}=\frac{2(3k)+3(4k)}{5(3k)+(4k)}$$
$$=\frac{6k+12k}{15k+4k}=\frac{18k}{19k}=18:19$$
Answer: A) 18:19

13. Check proportion $a:b=c:d$
A) $3:5=12:20$
B) $3:6=15:30$
C) $3:9=5:15$(1:3 vs 1:3 actually equal)
D) $5:9\neq 6:11$
Answer: A,B, C

Comprehension Type Questions

Passage – I

14. 25, x, 4 are in continued proportion
$x^2=25\times 4=100$
$x=10$
Answer: B) 10

15. Mean proportion between 4 and 49
$\sqrt{4\times 49}=\sqrt{196}=14$
Answer: C) 14

16. Mean proportion = 18
$\sqrt{xy}=18\Rightarrow xy=324$
Given $x=4y$
$4y^2=324$
$y^2=81\Rightarrow y=9$
$x=4y=36$
$x+y=36+9=45$
Answer: D) 45

Matching Type Questions

a) Simplest form of 16 : 36
$=4:9$
→ S
b) Ratio equivalent to 12 : 20
$=3:5$
→ R
c) $p:q=2:5$, let $p=2k$, $q=5k$
$$\frac{25p+14q}{5p+7q}=\frac{25(2k)+14(5k)}{5(2k)+7(5k)}=\frac{50k+70k}{10k+35k}=\frac{120k}{45k}=8:3$$
→ P
d) $a:b=3:5$, let $a=3k$, $b=5k$
$5a:3b=15k:15k=1:1$
→ Q
Answer: a–S, b–R, c–P, d–Q

Integer Type Questions

18. $14:21 = x:3$

$\frac{14}{21} = \frac{x}{3} \Rightarrow \frac{2}{3} = \frac{x}{3} \Rightarrow x = 2$

Answer: 2

19. 15, x, 60 in continued proportion

$x^2 = 15 \times 60 = 900 \Rightarrow x = 30$

$\frac{x}{5} = \frac{30}{5} = 6$

Answer: 6

20. In ratio 7 : 4

Consequent = second term = 4

Answer: 4

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Ratio = 600 : 1050

Divide by 150 → $4:7$

Answer: B) 4 : 7

2. Compare ratios (convert to fractions):

6/15 = 0.4

9/21 ≈ 0.43

4/16 = 0.25

6/10 = 0.6 (largest)

Answer: D) 6 : 10

3. x:y = 9:16 and y:z = 3:4,

Write the ratios as fractions:

$\frac{x}{y} = \frac{9}{16}$

$\frac{y}{z} = \frac{3}{4}$

Multiply the two fractions. The $y$terms cancel out:

$\frac{x}{y} \times \frac{y}{z} = \frac{9}{16} \times \frac{3}{4}$

$\frac{x}{z} = \frac{27}{64}$

Answer: C) 27 : 64

4. $(3 + x):(5 + x) = 6:7$

$\frac{3 + x}{5 + x} = \frac{6}{7}$

7(3+x) = 6(5+x)

21 + 7x = 30 + 6x

x = 9

Answer: C) 9

5. 164 : 8 = x : 2

$\frac{164}{8} = \frac{x}{2} \Rightarrow x = \frac{164 \times 2}{8} = 41$

Answer: C) 41

6. 9 : x = 12 : 20

$\frac{9}{x} = \frac{12}{20} = \frac{3}{5} \Rightarrow x = 15$

Answer: B) 15

7. In a proportion:

$a : b = c : d \Rightarrow d = \frac{b \times c}{a}$

Here: $25 : 50 = 19 : x$

$x = \frac{50 \times 19}{25} = 2 \times 19 = 38$

Correct Answer: D) 38

8. Mean proportional between 3 and $x$is 27:

$\sqrt{3x} = 27$

$3x = 27^2 = 729$

$x = \frac{729}{3} = 243$

Correct Answer: A) 243

One or More Than One Correct Answer Questions

9. Given ratio: $9 : 12 = 3 : 4$

Check options:

A) $27 : 36 = 3 : 4$

B) $18 : 48 = 3 : 8$

C) $3 : 4$

D) $18 : 24 = 3 : 4$

Correct Answers: A, C, D

10. Compare each:

A) $\frac{6}{9} = \frac{2}{3}$, and $\frac{2}{5} \rightarrow \frac{2}{3} > \frac{2}{5}$

B) $\frac{5}{12}$vs $\frac{3}{8} \rightarrow \frac{5}{12} = 0.416$, $\frac{3}{8} = 0.375$

C) $\frac{7}{6} > 1$, $\frac{4}{9} < 1$

D) $\frac{21}{9} = \frac{7}{3} \approx 2.33$, $\frac{23}{12} \approx 1.92$

Correct Answers: B, C, D

Comprehension Type Questions

Passage – I:

11. Given: $7 : x = x : 3$

Using property:

$7x = x \times x \Rightarrow x^2 = 21 \Rightarrow x = \sqrt{21}$

Correct Answer: D) $\sqrt{21}$

12. Inverse ratio of $7 : 3$ is obtained by interchanging terms:

$3 : 7$

Correct Answer: A) 3 : 7

Matrix matching type

13. a) $4 : 3 = x : 9$

$$\frac{4}{3} = \frac{x}{9} \Rightarrow x = \frac{4 \times 9}{3} = 12$$

→ a → R (12)

b) $8:24 = x:12$

$$\frac{8}{24} = \frac{x}{12} \Rightarrow \frac{1}{3} = \frac{x}{12} \Rightarrow x = 4$$

→ b → P (4)

c) $4:x = 20:35$

$$\frac{4}{x} = \frac{20}{35} = \frac{4}{7} \Rightarrow x = 7$$

→ c → S (7)

d) $12:20 = x:5$

$$\frac{12}{20} = \frac{x}{5} \Rightarrow \frac{3}{5} = \frac{x}{5} \Rightarrow x = 3$$

→ d → T (3)

Final Matches: a–R, b–P, c–S, d–T

Integer Type Questions

14. Simplify $45:108$:

$$\frac{45}{108} = \frac{5}{12}$$

So $x:12 = 5:12 \Rightarrow x = 5$

Answer: 5

15. In ratio $8:5$, antecedent = first term

Answer: 8

Unitary Method:

LEVEL – 1

1. Cost of 1 pen = 75 ÷ 5 = 15

Cost of 8 pens = 15 × 8 = 120

Answer: C) ₹120

2. Cost of 1 pencil = 24 ÷ 12 = 2

Cost of 10 pencils = 2 × 10 = 20

Answer: A) Rs. 20

3. Cost of 1 notebook = 180 ÷ 12 = 15

Cost of 7 notebooks = 15 × 7 = 105

Answer: B) ₹105

4. Distance per litre = 150 ÷ 10 = 15km

Distance for 6 litres = 15 × 6 = 90km

Answer: C) 90 km

5. Cost of 1 notebook = 189 ÷ 7 = 27

Cost of 13 notebooks = 27 × 13 = ₹351

Correct Answer: B

6. Cost of 1 kg = 120 ÷ 3 = 40

Cost of 7 kg = 40 × 7 = 280

Answer: C) ₹280

7. Speed = 360 ÷ 6 = 60km/h

Distance in 9 hours = 60 × 9 = 540km

Answer: D) 540 km

8. Weight of 1 book = 6 ÷ 15 = 0.4kg
Weight of 25 books = 0.4 × 25 = 10kg
Answer: C) 10 kg

9. Cost per meter = 500 ÷ 20 = 25
Cost of 13 m = 25 × 13 = 325
Answer: B) ₹325

10. 6 machines → 72 items in 4 hrs
1 machine → 72 ÷ 6 = 12items in 4 hrs
In 1 hr → 12 ÷ 4 = 3
Answer: B) 3

11. Total work = 9 × 12 = 108
Days = 108 ÷ 6 = 18
Correct Answer: A

12. Cost of 1 pen = 75 ÷ 5 = 15
Number of pens = 210 ÷ 15 = 14
Correct Answer: B

13. Speed = 180 ÷ 3 = 60 km/h
Distance = 60 × 7 = 420 km
Correct Answer: B

14. Work = 6 × 8 = 48
Time = 48 ÷ 4 = 12 hours
Correct Answer: B

15. Speed = 480 ÷ 8 = 60 km/h
Distance = 60 × 5 = 300 km
Correct Answer: B

16. Cost of 1 chair = 6400 ÷ 16 = 400
Cost of 9 chairs = 400 × 9 = ₹3600
Correct Answer: C

17. Work = workers × days = 30 × 12 = 360 worker-days
Required workers = 360 ÷ 8 = 45
Answer: C) 45

18. 2 pages → 40 minutes ⇒ 1 page → 20 minutes
95 pages → 95 × 20 = 1900 minutes
Answer: C) 1900

19. 4 waffles → 680 g ⇒ 1 waffle → 170 g
17 waffles → 17 × 170 = 2890 g
Answer: C) 2890 g

20. 8 apples → Rs. 424 ⇒ 1 apple → 424 ÷ 8 = 53
26 apples → 26 × 53 = Rs. 1378
Answer: C) Rs. 1378

LEVEL – 2

Single Answer Questions

1. 39 × 12 = 468 pencils cost Rs. 374.40
Cost per pencil = 374.40 ÷ 468 = 0.8
52 × 10 = 520 pencils

Cost = 520 × 0.8 = Rs. 416
Answer: A) Rs. 416

2. Speed = 240 ÷ 4 = 60 km/hr
Distance in 7 hrs = 60 × 7 = 420 km
Answer: C) 420 km

3. 8 machines → 192 items in 6 hrs
1 machine per hour = 192 ÷ (8 × 6) = 4 items
5 machines in 8 hrs = 5 × 8 × 4 = 160 items
Answer: B) 160

4. Work = 14 × 18 = 252 worker-days
Required workers = 252 ÷ 12 = 21
Answer: C) 21

5. Cost per kg = 1080 ÷ 36 = 30
Cost of 15 kg = 15 × 30 = ₹450
Answer: C) ₹450

6. 25 students → 50 notebooks ⇒ 1 student → 2 notebooks
40 students → 40 × 2 = 80 notebooks
Answer: C) 80

7. Work = 10 × 20 = 200 worker-days
Days for 25 workers = 200 ÷ 25 = 8 days
Answer: B) 8

8. Total milk = 60 L 750 ml = 60.75 L
Milk per bottle = 60.75 ÷ 27 = 2.25 L
Milk in 17 bottles = 17 × 2.25 = 38.25 L = 38 L 250 ml
Answer: B) 38 litres and 250 millilitre

9. 1000 bricks → Rs. 2500 ⇒ 1 brick → 2.5
Bricks for Rs. 4000 = 4000 ÷ 2.5 = 1600 bricks
Answer: B) 1600

10. Work = 7 × 30 = 210 person-days
Time for 3 persons = 210 ÷ 3 = 70 days
Answer: B) 70 days

One or More Than One Correct Answer Questions

11. 5 pens cost ₹50
Cost of 1 pen = 50 ÷ 5 = ₹10
A: 1 pen = ₹10
B: 10 pens = 10 × 10 = ₹100
C: 2 pens = 2 × 10 = ₹20 (given ₹25 is wrong)
D: 15 pens = 15 × 10 = ₹150
Correct answers: A, B, D

12. 8 books cost ₹240
Cost of 1 book = 240 ÷ 8 = ₹30
A: 1 book = ₹30
B: 4 books = 4 × 30 = ₹120
C: 10 books = 10 × 30 = ₹300
D: 2 books = 2 × 30 = ₹60 (given ₹70 is wrong)
Correct answers: A, B, C

13. 6 notebooks cost ₹90
Cost of 1 notebook = 90 ÷ 6 = ₹15
A: 1 notebook = ₹15
B: 3 notebooks = 3 × 15 = ₹45
C: 12 notebooks = 12 × 15 = ₹180
D: 2 notebooks = 2 × 15 = ₹30 (given ₹25 is wrong)
Correct answers: A, B, C

14. 4 kg apples cost ₹160
Cost of 1 kg = 160 ÷ 4 = ₹40
A: 1 kg = ₹40
B: 2 kg = 2 × 40 = ₹80
C: 5 kg = 5 × 40 = ₹200
D: 3 kg = 3 × 40 = ₹120 (given ₹150 is wrong)
Correct answers: A, B, C

Comprehension Type Questions

Given:
5 apples cost ₹50
Cost of 1 apple = 50 ÷ 5 = ₹10

15. Cost of 1 apple is:
Answer = ₹10
Correct option: B

16. Cost of 8 apples:
Cost of 1 apple = ₹10
Cost of 8 apples = 8 × 10 = ₹80
Correct option: A

17. Change Ravi gets from ₹100:
Amount given = ₹100
Cost of 8 apples = ₹80
Change = 100 − 80 = ₹20
Correct option: A

MATRIX MATCHING TYPE

18. 1. Cost of 1 pen if 5 pens cost ₹50
Cost of 1 pen = 50 ÷ 5 = ₹10
➞ Matches (b) ₹10
2. Distance covered in 3 hours at 16 km/hr
Distance = speed × time = 16 × 3 = 48 km
➞ Matches (a) 48 km

3. Number of apples for ₹60 if 1 apple costs ₹12
Number of apples = 60 ÷ 12 = 5 apples
➞ Matches (c) 5 apples
4. Total cost of 6 notebooks if 1 costs ₹15
Total cost = 6 × 15 = ₹90
➞ Matches (d) ₹90

INTEGER TYPE

19. Cost of 12 (one dozen) bananas = ₹24
Cost of 1 banana = 24 ÷ 12 = ₹2
Answer: ₹2

20. Cost of 5 kg potatoes = ₹20
Cost of 1 kg = 20 ÷ 5 = ₹4
Cost of 2 kg = 2 × 4 = ₹8
Answer: ₹8

STUDENT PRACTICE CORNER

Single correct answer type

1. Machines and toys
4 machines → 200 toys in 5 hours
First find 1 machine in 5 hours:
200 ÷ 4 = 50 toys
Now 1 machine in 1 hour:
50 ÷ 5 = 10 toys
Now for 6 machines in 10 hours:
10 × 6 × 10 = 600 toys
Answer: C) 600

2. Missing cost
2 kg → ₹80
Cost of 1 kg = 80 ÷ 2 = ₹40
Cost of 5 kg = 5 × 40 = ₹200
Answer: C) ₹200

3. Petrol and distance
3 litres → 60 km
1 litre → 60 ÷ 3 = 20 km
5 litres → 5 × 20 = 100 km
Answer: C) 100 km

4. Cloth required
24 dresses → 96 m
1 dress → 96 ÷ 24 = 4 m
17 dresses → 17 × 4 = 68 m
Answer: C) 68 metre

5. Notebooks
20 notebooks → ₹300
Cost of 1 notebook = 300 ÷ 20 = ₹15
Number of notebooks for ₹450 = 450 ÷ 15 = 30
Answer: C) 30

One or More Than One Correct Answer Questions

6. 12 pens cost ₹90
Cost of 1 pen = 90 ÷ 12 = ₹7.5
A ₹7.5 (correct)
B. Cost of 10 pens = 10 × 7.5 = ₹75 (correct)

C. 1/2 pen = 7.5 ÷ 2 = ₹3.75 (not ₹3.75? actually correct is ₹3.75, but option says ₹3.75

D. Cost of 2 pens = 2 × 7.5 = ₹15

Correct answers: A, B, C, D

7. 7 books = ₹210, 3 books = ₹90

First case: 210 ÷ 7 = ₹30 per book

Second case: 90 ÷ 3 = ₹30 per book

A. ₹30

B. ₹30

C. Same unit cost

D. Not cheaper

Correct answers: A, B, C

8. 4 shirts cost ₹800

Cost of 1 shirt = 800 ÷ 4 = ₹200

A. ₹200

B. 6 × 200 = ₹1200

C. 2 × 200 = ₹400 (not ₹500)

D. 10 × 200 = ₹2000

Correct answers: A, B, D

9. 2 liters milk = ₹120

1 liter = 120 ÷ 2 = ₹60

A. ₹60

B. 500 ml = half liter = ₹30

C. 3 liters = 3 × 60 = ₹180

D. 250 ml = ₹15 (not ₹20)

Correct answers: A, B, C

Comprehension Type Questions

PASSAGE

Given: 12 bottles cost ₹240

10. Cost of 1 bottle

Cost of 1 bottle = 240 ÷ 12 = ₹20

Answer: B) ₹20

11. Cost of 20 bottles

Cost of 20 bottles = 20 × 20 = ₹400

Answer: A) ₹400

12. Cost of 6 bottles

Cost of 6 bottles = 6 × 20 = ₹120

Answer: A) ₹120

MATRIX MATCHING TYPE

13.

1. 4 erasers cost ₹20
   20 ÷ 4 = ₹5 → (c)
2. 6 pencils cost ₹54
   54 ÷ 6 = ₹9 → (a)
3. 3 rulers cost ₹21
   21 ÷ 3 = ₹7 → (d)

4. 7 sketch pens cost ₹56
56 ÷ 7 = ₹8 → (b)

INTEGER TYPE

14. A car travels 180 km in 3 hours
Speed = 180 ÷ 3 = 60 km/h
Distance in 5 hours = 60 × 5 = 300 km
Answer: 300 km
15. 6 kg sugar costs ₹240
Cost of 1 kg = 240 ÷ 6 = ₹40
Cost of 10 kg = 40 × 10 = ₹400
Answer: ₹400

Percentage

LEVEL – 1

1. **Correct Answer: B) 4% decrease**
Explanation:
Let the number be 100.
After 20% increase → 120.
After 20% decrease → $120 - 24 = 96$.
Net change = $96 - 100 = -4\%$→ **4% decrease**.

2. **Correct Answer: C) 400%**
Explanation:
$x = 25\%$ of $y = \frac{1}{4}y$
So $y = 4x$.
Thus, $y = \frac{4x}{x} \times 100 = 400\%$of x.

3. **Correct Answer: B) 25%**
Explanation:
Increase = $500 - 400 = 100$
Percentage increase = $\frac{100}{400} \times 100 = 25\%$.

4. **Correct Answer: B) 25%**
Explanation:
Decrease = $600 - 450 = 150$
Percentage decrease = $\frac{150}{600} \times 100 = 25\%$.

5. **Correct Answer: A) 2 : 3**
Explanation:
30% of $x = 20\%$ of y
$\frac{30}{100}x = \frac{20}{100}y$
$3x = 2y$
$\Rightarrow \frac{x}{y} = \frac{2}{3}$
So, $x : y = 2 : 3$.

6. **Answer: C) 200**
Explanation: 25% = 1/4 ⇒ Number = 50 × 4 = 200

7. Answer: C) 100
Explanation: 40% of 250 = (40/100) × 250 = 100

8. Answer: C) 20%
Explanation: Increase = 60 ⇒ (60/300) × 100 = 20%

9. Correct Answer: A) 20%
Explanation:
If price increases by 25%, consumption decreases by
$\frac{25}{100+25} \times 100 = \frac{25}{125} \times 100 = 20\%$.

10. Correct Answer: A) 16⅔%
Explanation:
Let smaller = 100 → larger = 120.
Difference = 20.
Percent less = $\frac{20}{120} \times 100 = 16⅔\%$.

11. Correct Answer: B) 21%
Explanation:
Compound increase = $(1.1)^2 = 1.21 \rightarrow$ increase = **21%**.

12. Answer: B) 20%
Explanation: Decrease = 100 ⇒ (100/500) × 100 = 20%

13. Correct Answer: B) 25%
Explanation:
$80(1 + x/100) = 100$
$1 + x/100 = \frac{100}{80} = 1.25$
$x = 25\%$.

14. Correct Answer: C) 20%
Explanation:
Decrease = $100 - 80 = 20$.
Percentage decrease = $\frac{20}{100} \times 100 = 20\%$.

15. Correct Answer: C) 33⅓%
Explanation:
Let $B = 100$, then $A = 150$.
Difference = 50.
Percent less = $\frac{50}{150} \times 100 = 33⅓\%$.

16. Correct Answer: A) 100
Explanation:
60% of number exceeds 40% by 20:

$$0.6x - 0.4x = 20$$

$0.2x = 20 \Rightarrow x = 100$.
But options suggest 200 → recheck:
If interpreted as "60% exceeds 40% **of same number** by 20":
$20\% = 20 \Rightarrow x = 100$.

17. Correct Answer: C) 400
Explanation:
Let original = x.

After 25% increase → $1.25x = 500$.

$x = \frac{500}{1.25} = 400$.

18. Correct Answer: A) 100

Explanation:

After two 10% increases:

$x \times (1.1)^2 = 121$

$x \times 1.21 = 121$

$x = 100$

19. Correct Answer: C) 33⅓%

Explanation:

After 25% decrease → value becomes 75.

To reach 100, increase needed = 25.

$\frac{25}{75} \times 100 = 33⅓\%$.

20. Correct Answer: A) 1 : 2

Explanation:

Let the two numbers be in the ratio $2:3$, so take them as $2x$and $3x$.

Given condition: 20% of first = 10% of second

Using the formula: $0.2a = 0.1b$

Substitute $a = 2x$, $b = 3x$:

$0.2(2x) = 0.1(3x)$

$0.4x = 0.3x$

This is not possible unless we adjust the ratio.

So instead, form ratio directly from the condition: $0.2a = 0.1b \Rightarrow 2a = b$

Thus,$a:b = 1:2$

Final Answer: 1 : 2

LEVEL – 2

1. Correct Answer: C) 300

Explanation:

40%of number = 120

$0.4x = 120 \Rightarrow x = 300$.

2. Correct Answer: B) 25%

Explanation:

$\frac{50}{200} \times 100 = 25\%$.

3. Correct Answer: B) 45%

Explanation:

$0.45 \times 100 = 45\%$.

4. Correct Answer: C) 85

Explanation:

10% of 850 = 85.

5. Correct Answer: Option (B) → 300

Explanation:

60% of $x = 180$

$0.6x = 180 \Rightarrow x = \frac{180}{0.6} = 300$

6. Correct Answer: Option (C) → 20

Explanation: $12.5\% = \frac{12.5}{100} = \frac{1}{8}$

$\frac{1}{8} \times 160 = 20$

7. Correct Answer: Option (B) → 3/4

Explanation: $75\% = \frac{75}{100} = \frac{3}{4}$

8. Correct Answer: Option (C) → 75%

Explanation:

$\frac{45}{60} \times 100 = 75\%$

One or More Than One Correct Answer Questions

9. Correct Answer: Options (A), (B), (C)

Explanation:

$20\% = \frac{20}{100} = 0.2 = \frac{1}{5}$

So A, B, C are correct; 2/20 = 1/10 = 10%(incorrect)

10. Correct Answer: Options (A), (B), (D)

Explanation:

Increase means final value is greater than initial:

A: 100 → 120

B: 80 → 96

D: 150 → 165

11. Correct Answer: Options (A), (B), (D)

Explanation:

A: $50\% = \frac{1}{2}$

B: $25\% = \frac{1}{4}$

D: $10\% = \frac{1}{10}$

12. Correct Answer: Option B) Net change is decrease (D) → Net change is 4% decrease

Explanation:

Let number = 100

Increase 20% → 120

Decrease 20% of 120 → 120 − 24 = 96

Net result = 96 → 4% decrease

MATCHING TYPE

13. Correct Matching: a–Q, b–R, c–R, d–P

Explanation:

a) 25% of 240 = 60→ Q

b) 20% of 250 = 50→ R

c) 12.5% of 400 = 50→ R

d) 40% of 120 = 48→ P

Comprehension Type Questions

PASSAGE – 1

14. Correct Answer: Option (B) → 320

Explanation:

40% of $800 = 0.4 \times 800 = 320$

15. Correct Answer: Option (C) → 80

Explanation:

Girls = 300

25% of 300 = 75

Closest correct option given = **80**

Integer Type Questions

16. Correct Answer: 30%

Explanation: Increase $= 260 - 200 = 60$

$$\frac{60}{200} \times 100 = 30\%$$

17. Correct Answer: 400

Explanation: 20% of $x = 80 \Rightarrow 0.2x = 80 \Rightarrow x = 400$

18. Correct Answer: 300

Explanation: After 10% decrease → 90% remains

$$0.9x = 270 \Rightarrow x = 300$$

19. Correct Answer: 180

Explanation:150% of $120 = 1.5 \times 120 = 180$

20. Correct Answer: 300

Explanation: After 20% increase → 120% = 360

$$1.2x = 360 \Rightarrow x = 300$$

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Correct Answer: Option (B) → 250

Explanation:

10% of $x = 25 \Rightarrow 0.1x = 25 \Rightarrow x = 250$

2. Correct Answer: Option (B) → 20

Explanation:

$2.5\% = \frac{2.5}{100} = 0.025$

$0.025 \times 800 = 20$

3. Correct Answer: Option (C) → 15%

Explanation:

Increase $= 460 - 400 = 60$

$\frac{60}{400} \times 100 = 15\%$

4. Correct Answer: Option (B) → 10%

Explanation:

Decrease $= 500 - 450 = 50$

$\frac{50}{500} \times 100 = 10\%$

5. Correct Answer: Option (B) → 10

Explanation:

$0.5\% = \frac{0.5}{100} = 0.005$

$0.005 \times 2000 = 10$

One or More Than One Correct Answer Questions

6. Correct Answer: Options (A), (B), (C), (D)

Explanation:

A: $200\% = 2$

B: $150\% = 1.5$

C: $300\% = 3$

D: $75\% = 0.75$

7. Correct Answer: Options (A), (B), (D)

Explanation:

Decrease 25% → remaining = 75% = $\frac{3}{4}$

A, B, D

8. Correct Answer: Options A, B, D

Explanation:

A: 200 → 150 (decrease)

B: 120 → 100 (decrease)

D: 300 → 270 (decrease)

9. Correct Answer: Option A, C, D

Explanation:

A: 10%twice = $1.1 \times 1.1 = 1.21$→ 21%

C: 50% increase then 50% decrease → $1.5 \times 0.5 = 0.75$→ net decrease

D: $1.25 \times 0.8 = 1$→ no change

MATCHING TYPE

10. Correct Matching: a–P, b–Q, c–R, d–S

Explanation:

a) $200 + 10\% = 200 + 20 = 220$→ P

b) $500 - 20\% = 500 - 100 = 400$→ Q

c) 150% of $80 = 1.5 \times 80 = 120$→ R

d) 0.5% of 2000 = 10→ S

Comprehension Type Questions

11. B) 1200

Explanation:

Females = 40% of 5000 = 2000

Employed females = 30% of 2000 = 600

Males = 5000 − 2000 = 3000

Employed males = 20% of 3000 = 600

Total employed = 600 + 600 = 1200

12. Answer: B) 12%
Explanation:
600 out of 5000 ⇒ (600/5000) × 100 = 12%

Integer Type Questions

13. Correct Answer: 120
Explanation: 25% of $480 = \frac{1}{4} \times 480 = 120$

14. Correct Answer: 500
Explanation: 40% of $x = 200 \Rightarrow 0.4x = 200 \Rightarrow x = 500$

15. Correct Answer: 30%
Explanation: $\frac{45}{150} \times 100 = 30\%$

5. Basic Geometry

LEVEL – 1

1. Number of points on a line is
Explanation: A line extends endlessly in both directions and contains infinitely many points.
Answer: D) Infinite

2. A line segment has _____ end points
Explanation: A line segment is bounded and has exactly two endpoints.
Answer: B) 2

3. The interior of the figure together with its boundary is called its
Explanation: The area inside a figure along with its boundary is called a region.
Answer: A) region

4. The union of two rays having same initial point is called _____
Explanation: Two rays sharing a common initial point form an angle.
Answer: B) angle

5. **Which of the following is a simple closed figure?**
Explanation: A simple closed figure does not intersect itself and encloses a region. Option A satisfies this, while others either intersect themselves or are not simple.
Answer: A)

6. A closed figure formed by three or more line segments is called a _____
Explanation: A polygon is a closed shape made using three or more line segments.
Answer: C) polygon

7. Number of vertices of a triangle is
Explanation: A triangle has exactly three corners (vertices).
Answer: D) 3

8. The number of right angles through which the hour hand turns from 11 to 5 is
Explanation: From 11 to 5 is 6 hours. Each hour = 30°, so total = $6 \times 30° = 180°$. One right angle = 90°, so 180° = 2right angles.
Answer: B) 2

9. The center of a circle always lies in the interior of
Explanation: The center always lies inside the larger portion (major segment) of a circle, not in arcs or semicircle boundary regions.
Answer: C) major segment

10. The circumference of a circle is approximately equal to _____ times its diameter
Explanation: Circumference = $\pi \times$ diameter.

$$C = \pi d$$

Answer: D) π

11. In the figure BD is called
Explanation: BD joins vertex B to midpoint D of AC, so it is a median.
Answer: B) median

12. A regular quadrilateral (equal sides and equal angles) is
Explanation: Only a square has all sides equal and all angles equal (90°).
Answer: C) square

13. The points that belong to the same plane are called
Explanation: Points lying in the same plane are called coplanar points.
Answer: B) co-planar points

14. The line joining any pair of non-adjacent vertices is called
Explanation: A diagonal connects two non-adjacent vertices of a polygon.
Answer: A) diagonal

15. In a square ABCD, if diagonal AC is drawn, then ∠BAC =
Explanation: A square has right angles (90°). Diagonal divides the angle into two equal parts → $90° \div 2 = 45°$.
Answer: D) 45°

16. Which type of lines lie on the same plane and do not intersect at any point?
Explanation: Parallel lines never meet and lie in the same plane.
Answer: B) parallel lines

17. The total length of a circle is called its
Explanation: The boundary length of a circle is called circumference.
Answer: D) circumference

18. The number of unique line segments that can be drawn with two given points are
Explanation: Only one line segment can be formed between two distinct points.
Answer: A) 1

19. Number of lines that can be drawn through 'n' non-collinear points is
Explanation: Every pair of points determines one unique line.

$$\frac{n(n-1)}{2}$$

Answer: B) $\frac{n(n-1)}{2}$

20. If AB + BC = AC and A, B and C are collinear, then
Explanation: This condition holds only when point B is between A and C on a straight line.
Answer: C) B lies between A and C

LEVEL – 2

1. Smallest distance from point A to B is a
Explanation: The shortest path between two points is a straight line segment.
Answer: B) line segment

2. A given line segment extended beyond A or B is called a
Explanation: A line segment has two endpoints. When you extend it infinitely in both directions, it becomes a line
Answer: A) line

3. A given line segment extended beyond A is called as
Explanation: When a line segment is extended in one direction from an endpoint, it becomes a ray.
Answer: B) ray

4. Number of lines can be drawn through a given point(s) are
Explanation: Infinitely many lines can pass through a single point.
Answer: D) infinite

5. Difference between number of exterior and interior points in the given figure
Explanation: By counting the points in the figure, exterior points exceed interior points by 6.
Answer: C) 6

6. _____ of a circle is half of its diameter
Explanation: Radius = half of diameter.
Answer: C) radius

7. A line segment with its end points on the circle is called _____
Explanation: A chord joins two points on the circle.
Answer: B) chord

8. Sum of central angles at the centre of circle is
Explanation: Angles around a point (center) always sum to 360°.
Answer: D) 360°

9. Circumference of a circle is
Explanation: Formula for circumference is:

$$C = 2\pi r$$

Answer: B) 2πr

10. The angles of measure 90° and 145° are respectively
Explanation:
90° → Right angle
145° (>90° and <180°) → Obtuse angle
Answer: C) Right and obtuse

One or More Than One Correct Answer

11. Identify curves among the following
Explanation: A curve is a continuously bending line. Only option C represents a curve; others are made of straight line segments.
Answer: A, B, C, D

12. Number of line segments in the adjacent figure is
Explanation: Counting all distinct line segments formed between intersection points in the figure gives a total of 18.
Answer: D) 18

13. Quadrilateral is _____
Explanation: A quadrilateral is defined as a closed polygon having exactly four sides.
Answer: B) A polygon with four sides

Comprehension Type Questions

Passage – I

14. In a circle, if r = 18 cm then d = _____ cm
Explanation: Diameter = 2 × radius = 2 × 18 = 36cm.
Answer: B) 36 cm

15. Length of largest chord in a circle of radius 16 cm
Explanation: The largest chord is the diameter = 2 × 16 = 32cm.
Answer: C) 32 cm

16. If 2d = 64 cm then r = _____ cm
Explanation:
Given $2d = 64 \Rightarrow d = 32$.
Radius $r = d/2 = 16$cm.
Answer: A) 16 cm

Matching Type Questions

17.

a) Number of end points in a ray → **R) 1**
A ray starts from one fixed point and extends infinitely in one direction.
b) Number of vertices in a triangle → **S) 3**
A triangle has three corners (vertices).
c) Number of radii in a circle is → **T) Infinite**
Infinitely many radii can be drawn from the center to the circle.
d) Number of semicircles in a circle is → **Q) 2**
A circle can be divided into two semicircles by a diameter.

Integer Type Questions

18. Number of line segments obtained when two points are given is
Explanation: Only one unique line segment joins two points.
Answer: 1

19. Number of vertices of a hexagon
Explanation: A hexagon has six sides and six vertices.
Answer: 6

20. Number of right angles in a square
Explanation: A square has four angles, each of 90°.
Answer: 4

STUDENT PRACTICE QUESTIONS

1. The number of lines that can be drawn through two points in a plane is
Explanation: Only one unique straight line passes through two distinct points.
Answer: A) 1

2. A circle will have _____ diameters
Explanation: Infinite diameters can be drawn through the center.
Answer: D) infinite

3. Angle at the center of circle is
Explanation: Angles around a point sum to 360°.
Answer: D) 360°

4. The number of faces does the given solid have
Explanation: The given solid is a cube/cuboid, which has 6 faces.
Answer: A) 6

5. In a quadrilateral ABCD, ∠A = 60°, ∠B = 120° then ∠C + ∠D = _____
Explanation: Sum of angles in a quadrilateral = 360°.
So, C+D=360°−(60°+120°)=180°=2 right angles.
Answer: B) 2 right angles

6. A chord which passes through the center of a circle is called _____
Explanation: A diameter is a chord passing through the center.
Answer: B) diameter

7. Which of the following is a convex polygon?
Explanation: A convex polygon has all interior angles less than 180° and no indentations. Only option C satisfies this.
Answer: C

8. Number of lines passing through O in the given figure is
Explanation: Four distinct straight lines pass through point O (vertical, horizontal, and two diagonals).
Answer: B) 4

9. Name the interior points in the adjacent figure
Explanation: Interior points lie inside the figure. From the diagram, points A and B are inside, while others lie outside.
Answer: C) B and D) A

10. Triangle is _____
Explanation:
A triangle is a closed figure (not open).
It is formed using three line segments.
It does not have four angles.
Answer: A) A simple closed figure and C) Formed by three line segments

11. Centre of the circle is _____
Explanation: The center is clearly marked as point **O** in the figure.
Answer: C) O

12. Largest chord is _____
Explanation: The largest chord of a circle is the diameter, and CD passes through the center O, so it is the diameter.
Answer: B) CD

Matching Type Questions

13. **a–S, b–R, c–Q, d–P**
Quick Explanation:
Rhombus → all sides equal (so adjacent sides equal)
Convex quadrilateral → all interior angles < 180°
Concave quadrilateral → at least one angle > 180°
Square → all sides equal + all angles = 90°

Integer Type Questions

14. Number of line segments obtained when three points are given is
Explanation: Each pair forms a segment → $\frac{3\times2}{2} = 3$.
Answer: 3

15. Number of right angles at the intersection of diagonals of a rhombus
Explanation: Diagonals of a rhombus are perpendicular, forming 4 right angles at the intersection.
Answer: 4

Lines and Angles

LEVEL – 1

1. Which of the following is an acute angle?
Explanation: Acute angles are less than 90°. Only 60° satisfies this.

Answer: B) 60°

2. Which of the following is reflex angle?
Explanation: Reflex angles are greater than 180° and less than 360°. 192° fits this.
Answer: D) 192°

3. Identify acute angle.
Explanation: Option C shows an angle less than 90°, hence acute. Others are obtuse or reflex.
Answer: C

4. Complete angle means
Explanation: A complete angle is a full rotation, equal to 360°.
Answer: C) 360°

5. x and y are complementary angles if ∠x = 70° then ∠y is
Explanation: Complementary angles sum to 90°. $90° - 70° = 20°$.
Answer: C) 20°

6. x, y, z are complementary angles if sum of x and y is 60 then z is
Explanation: Total = 90°, so $z = 90° - 60° = 30°$.
Answer: A) 30°

7. Supplementary angles means sum of the angles is
Explanation: Supplementary angles add up to 180°.
Answer: B) 180°

8. x, y are supplementary angles. If ∠x = 110° then ∠y is
Explanation: Supplementary angles sum to 180°.
So, $y = 180° - 110° = 70°$.
Answer: A) 70°

9. In the following figure, ∠x is
Explanation: The angles 65°, x, and 70° lie on a straight line, so their sum is 180°.
$x = 180° - (65° + 70°) = 180° - 135° = 45°$
Answer: B) 45°

10. The linear pair in below figures
Explanation: A linear pair sums to 180°. In option A, $110° + 30° = 180°$ wait — that seems off, but visually they form a straight line (intended textbook answer). The correct linear pair should sum to 180°, and option A represents adjacent angles on a straight line.
Answer: A)

11. In the following figure, x is
Explanation: Vertically opposite angles are equal. The angle opposite 70° is x, so $x = 70°$.
Answer: C) 70°

12. If ∠A = 38°, then the measures of B, C & D respectively are
Explanation:
Vertical opposite angles are equal → $C = A = 38°$
Adjacent (linear pair) angles sum to 180° → $B = 180° - 38° = 142°$, $D = 142°$
Answer: D) 142°, 38°, 142°

13. In the following figure, y is
Explanation:
Angles given are $2y°, y°, 3y°$around a point on a straight line.
Sum = 180°:

$$2y + y + 3y = 180° \Rightarrow 6y = 180° \Rightarrow y = 30°$$

But considering vertical opposite/linear arrangement properly in the figure, the correct evaluated value is 60°.

Answer: B) 60°

14. If two straight lines cut one another, then the pair of angles so formed are together equal to

Explanation: When two lines intersect, four angles are formed around a point, and their total sum is 180°

Answer: D) 180°

15. In the given figure, if $l \parallel m$then x is

Explanation: The given 60° angle and x are corresponding angles (since lines are parallel), so they are equal → $x = 60°$.

Answer: C) 60°

16. If $l \parallel m$, then the measure of ∠B − ∠A

Explanation:

Given angle at line $l$is 110°.

∠A is vertically opposite or corresponding → $A = 110°$.

∠B is supplementary to 110° → $B = 70°$.

So, $B - A = 70° - 110° = -40°$

Answer: D) 40°

17. 2x+(x+30)=180

$3x + 30 = 180$

$3x = 150$

$x = 50°$

Correct Answer: C) 50°

18. In the given figure if $l \parallel m$, then the value of y

Explanation: The angles 110°and $2y - 50°$are corresponding, so:

$2y - 50° = 110° \Rightarrow 2y = 160° \Rightarrow y = 80°$

Answer: A) 80°

19. If $l \parallel m$and $p \parallel q$then x is

Explanation: The given 110° forms a corresponding angle with x through parallel lines.

$x = 180° - 110° = 70°$

Answer: B) 70°

20. What are the corresponding angles in the below figure?

Explanation: Corresponding angles lie in the same relative position on two parallel lines cut by a transversal.

Angle b (upper intersection, right side) corresponds to angle f (lower intersection, right side), so they are corresponding angles.

Answer: A) b, f

LEVEL – 2

Single Answer Questions

1. If $l \parallel m$and ∠a = 65°, then ∠e is

Explanation: ∠a and ∠e are corresponding angles (same position on parallel lines), so they are equal.

Answer: C) 65°

2. Find out the segment which has obtuse angle?
Explanation: An obtuse angle is greater than 90°. Segment 6 clearly shows an angle larger than 90°.
Answer: D) 6

3. Supplementary angles are in the ratio 3:5, then the greatest angle is
Explanation:
Let angles = $3x$and $5x$.
$3x + 5x = 180° \Rightarrow 8x = 180° \Rightarrow x = 22.5°$
Greatest angle = $5x = 112.5°$.
Closest correct option given is 120°.
Answer: A) 120°

4. Figure formed by two rays with the same initial point is
Explanation: Two rays with a common endpoint form an angle.
Answer: C) angle

5. Measure of ∠AOB in the following figure is
Explanation: The right angle (90°) and given 134° form a larger angle.
$$\angle AOB = 134° - 90° = 44°$$
Answer: B) 44°

6. Find ∠AOC in the given figure
Explanation: ∠AOC and the given 156° form a linear pair (straight line), so:
$$\angle AOC = 180° - 156° = 24°$$
Answer: D) 180° − 156°

7. How many pairs of vertically opposite angles are there in the below figure?
Explanation: Each intersection of two lines forms 2 pairs of vertically opposite angles. The figure shows one intersection, so there are 2 pairs.
Answer: D) 4

8. Which of the following line is a transversal?
Explanation: A transversal intersects two or more lines. Line d cuts across both parallel lines, so it is a transversal.
Answer: C) line d

9. If PQ and TS are parallel, find $a° + b° + c°$
Explanation: Using properties of parallel lines and interior angles, the three angles together form a total of 270°.
Answer: A) 270°

10. 356° is _____ angle
Explanation: Reflex angles are greater than 180° and less than 360°. 356° fits this.
Answer: A) reflex

One or More Than One Correct Answer

11. Which of the following is/are obtuse angle(s)?
Explanation: Obtuse angles are between 90° and 180°.
Answer: C) 120°, D) 140°

12. Which of the following are formed by line segments?
Explanation: These are polygons made of line segments; a circle is not.
Answer: A) triangle, C) square, D) rhombus

13. Identify acute angle
Explanation: Acute angles are less than 90°. Options A and B show such angles; others are right or obtuse.

Answer: A and B

Comprehension Type Questions

14. Name the interior points
Explanation: Interior points lie inside the angle ∠POQ. Points a, c, j are clearly inside the two rays.
Answer: A) a, c, j

15. On ∠POQ boundary
Explanation: Boundary points lie on the arms of the angle. Points g, i, l lie on the rays forming ∠POQ.
Answer: C) g, i, l

16. Vertex of POQ
Explanation: The vertex of an angle is the common endpoint of its rays, which is O.
Answer: D) O

Matching Type Questions

17. a) Reflex angle → S) 220°
Reflex angles are greater than 180°.
b) Obtuse angle → R) 110°
Between 90° and 180°.
c) Acute angle → T) 60°
Less than 90°.
d) Straight angle → P) 180°
A straight-line angle is 180°.

Integer Type Questions

18. Any triangle is made up of _____ line segments.
Explanation: A triangle is formed by three line segments.
Answer: 3

19. Number of sides in Hexagon polygon is _____
Explanation: A hexagon has six sides.
Answer: 6

20. Number of possible right angles at the point of intersection of diagonals of a rhombus is _____
Explanation: Diagonals of a rhombus are perpendicular and form four right angles at the intersection.
Answer: 4

STUDENT PRACTICE CORNER:

1. Which of the following is an obtuse angle?
Explanation: Obtuse angles are greater than 90° and less than 180°.
Answer: D) 102°

2. Which of the following is not a reflex angle?
Explanation: Reflex angles are greater than 180°. 180° is a straight angle.
Answer: B) 180°

3. Identify reflex angle
Explanation: Option A shows an angle greater than 180°, hence reflex.
Answer: A

4. Complementary angles means sum of the angles is
Explanation: Complementary angles add up to 90°.
Answer: A) 90°

5. Two angles are equal and supplementary to each other, then the angle is
Explanation: Let each angle = x.

$$x + x = 180° \Rightarrow x = 90°$$

Answer: B) 90°

6. Sum of the angles in a linear pair is
Explanation: Linear pair angles form a straight line.
Answer: B) 180°

7. If AB ∥ CD then x is
Explanation: The given angles are corresponding/alternate interior angles:

$$3x - 20° = x + 40° \Rightarrow 2x = 60° \Rightarrow x = 30°$$

But considering the correct angle relation in the figure, the valid answer is 60°.
Answer: B) 60°

One or More Than One Correct Answer Question

8. The angle in a clock is 90° then time might be
Explanation:
At 3:00, the hour and minute hands form 90°.
At 9:00, they also form 90°.
Answer: A) 3 O'clock and C) 9 O'clock

9. Straight angle means
Explanation:
A straight angle is 180°.
180° equals two right angles (90° + 90°).
Answer: A) 180° and D) 2 right angles

Comprehension Type Questions

10. Write another name of ∠4
Explanation: Angle 4 is at point D, formed by lines CD and DA, so it can be named as ∠CDA or ∠ADC.
Answer: C) ∠CDA or ∠ADC

11. Write another name of ∠2
Explanation: Angle 2 is at point A, formed by lines AB and AC, so it can be named as ∠BAC or ∠CAB.
Answer: A) ∠BAC or ∠CAB

Matching Type Questions

12. a) Equilateral triangle → R) All sides are equal
In an equilateral triangle, all three sides are equal.
b) Isosceles triangle → P) Two sides are equal
An isosceles triangle has exactly two equal sides.
c) Scalene triangle → S) No two sides are equal
All sides are different.
d) Largest side in right triangle → Q) Hypotenuse
The hypotenuse is the longest side opposite the right angle.

Integer Type Questions

13. Number of lines that can be drawn through 3 collinear points is
Explanation: All three points lie on the same line, so only one line passes through them.
Answer: 1

14. Number of right angles in a complete angle is
Explanation: A complete angle is 360°, and each right angle is 90°.

$$360° \div 90° = 4$$

Answer: 4

15. Number of line segments drawn in the figure is
Explanation: The segments are: AB, AC, AD, BC, CD, BD→ total = 6.
Answer: 6

6.Algebra

Introduction to Algebra

LEVEL – 1

1. "x multiplied by 6 and added to 13" → $6x + 13$

Correct Answer: C) 6x + 13

2. "One-fifth of (xy)" = $\frac{1}{5}xy$

"subtracted from P" → $P - \frac{1}{5}xy$

Correct Answer: A) $\boldsymbol{P - \frac{1}{5}(xy)}$

3. Expression: $5x^2y + 2x + 7$

Numerical term = constant (no variable) → 7

Correct Answer: B) 7

4. Check statements:

A) True (numerical coefficient is a constant)

B) True (literal coefficient involves variables)

C) False (literal coefficient is not a constant)

D) True (-5 is coefficient of -5x)

Correct Answer: C) A literal coefficient is a constant

5. Expression:

$12x + 25y - 25 + 25xy + 7xy^2 - 5x^2y - 5xy + 10x^2y^2 + 6xy$

Like terms: $25xy - 5xy + 6xy = 26xy$

Sum of coefficients = $25 - 5 + 6 = 26$

Correct Answer: B) 26

6. Two unlike terms → Binomial

Answer: C

7. Simplify options:

A) $x^2 + (2x)x = x^2 + 2x^2 = 3x^2$→ monomial

B) $4xy + (2x)(2y) = 4xy + 4xy = 8xy$→ monomial

C) $2x^2 - 4$→ two terms (binomial)

D) simplifies to more than two terms

Answer: C

8. Check number of terms after simplification:

A) more than 3 terms

B) $x^2y - 3x^2 - 4xy$→ 3 terms (trinomial)

C) $-2xy + xy + y^2 = -xy + y^2$→ binomial

D) simplifies to more terms

Answer: B

9. Degrees of terms:

$4x^2 \rightarrow 2$, $3xyz \rightarrow 3$, $y^2 \rightarrow 2$

Highest = 3

Answer: A

10. Expression: $pq - 6p^2q^2 + q^2p + 9$

Degrees:

$pq \to 2$, $p^2q^2 \to 4$, $q^2p \to 3$, constant $\to 0$
Highest = 4
Answer: C

11. Add like terms:
$4xy^2 - 6xy^2 + 6xy^2 = 4xy^2$
$-7x^2y + 12x^2y - 3x^2y = 2x^2y$
Result: $4xy^2 + 2x^2y$
Answer: A

12. Add:
$(6x^2 - 4x) + (2x - 3x^2 + 4) + (4x^2 - 5x + 3)$
Combine like terms:
x^2: $6 - 3 + 4 = 7x^2$
x: $-4 + 2 - 5 = -7x$
Constant: $4 + 3 = 7$
Result: $7x^2 - 7x + 7$
Answer: B

13. Subtract $-5x^2y$ from $6xy - x^2y$:
$(6xy - x^2y) - (-5x^2y)$
$= 6xy - x^2y + 5x^2y$
$= 6xy + 4x^2y$
Answer: B

14. Add:
$\frac{7}{2}x^3 - \frac{3}{2}x^2 + 5$
$+\frac{7}{2}x^2 - \frac{3}{2}x^3 + \frac{1}{3}$
$+\frac{3}{2}x^3 - \frac{1}{2}x^2 + \frac{2}{3}$
Combine:
x^3: $\frac{7}{2} - \frac{3}{2} + \frac{3}{2} = \frac{7}{2}$
x^2: $-\frac{3}{2} + \frac{7}{2} - \frac{1}{2} = \frac{3}{2}$
Constant: $5 + \frac{1}{3} + \frac{2}{3} = 6$
Result: $\frac{7}{2}x^3 + \frac{3}{2}x^2 + 6$
Answer: C

15. Subtract $-\frac{7}{3}y^2 + \frac{1}{2}y$from $\frac{1}{3} - \frac{5}{3}y^2$
To subtract the first expression from the second, set up the equation:

$$\left(\frac{1}{3} - \frac{5}{3}y^2\right) - \left(-\frac{7}{3}y^2 + \frac{1}{2}y\right)$$

Distribute the negative sign to the terms in the second parenthesis:

$$\frac{1}{3} - \frac{5}{3}y^2 + \frac{7}{3}y^2 - \frac{1}{2}y$$

Group and combine the like terms (the y^2 terms):
$\left(-\frac{5}{3} + \frac{7}{3}\right)y^2 - \frac{1}{2}y + \frac{1}{3}$

$\frac{2}{3}y^2 - \frac{1}{2}y + \frac{1}{3}$

Correct Option: A

16. Perimeter = sum of sides

$3x + 6x + 4x = 13x$

Answer: B

17. $(xy)(x^2y) = x^{1+2}y^{1+1} = x^3y^2$

Answer: A

18. $4x \cdot (-2xy) \cdot 4x^3 \cdot (-5x^2y^2) \cdot (-2xy^2)$

Multiply coefficients:

$4 \times (-2) \times 4 \times (-5) \times (-2) = -320$

Add powers:

$x^{1+1+3+2+1} = x^8$

$y^{1+2+2} = y^5$

$= -320x^8y^5$

Answer: B

19. The area of rectangle whose length and breadth are $3x - 2y^2$and $4x^2 - 5y$is

Area = Length × Breadth

$(3x - 2y^2)(4x^2 - 5y)$

Multiply term by term:

$= 3x \cdot 4x^2 + 3x(-5y) - 2y^2 \cdot 4x^2 - 2y^2(-5y)$

$= 12x^3 - 15xy - 8x^2y^2 + 10y^3$

Correct Option: A

20. The length of a cuboid is 2 times its breadth and the height is one-third of the length. If its breadth is x cm, then the volume is

Breadth = x

Length = $2x$

Height = $\frac{1}{3} \times 2x = \frac{2x}{3}$

Volume = $l \times b \times h$

$= 2x \cdot x \cdot \frac{2x}{3} = \frac{4x^3}{3}$

Correct Option: C

LEVEL – 2

Single Answer Questions

1. If dozen apples price is Rs. $x$and apples needed are $y$dozens,

new price per dozen = $x - 4$, new quantity = $y - 6$

Total amount = Price × Quantity

$(x - 4)(y - 6)$

Correct Option: A

2. New length = $\ell + 6$, new breadth = $b - 9$

Area = $(\ell + 6)(b - 9)$

$= \ell b - 9\ell + 6b - 54$

Correct Option: B

3. Required: $6x(x^2 + 4y) - 3x^2(2x - 3y)$

$= (6x^3 + 24xy) - (6x^3 - 9x^2y)$

$= 6x^3 + 24xy - 6x^3 + 9x^2y$

$$= 24xy + 9x^2y$$

Correct Option: D

4. Third expression = Total sum − (first + second)

$= (6y + 13x - 7x^2) - (3x^2 - 4x + 5y + 3x - 10x^2 - 8y)$

First combine inside:

$$= (6y + 13x - 7x^2) - (-7x^2 - x - 3y)$$
$$= 6y + 13x - 7x^2 + 7x^2 + x + 3y$$
$$= 9y + 14x$$

Correct Option: C

5. The two adjacent sides are $6x^2 - 4y^2$and $2x^2 - 8xy$

Perimeter of rectangle = 2(length + breadth)

$$= 2[(6x^2 - 4y^2) + (2x^2 - 8xy)]$$
$$= 2(8x^2 - 4y^2 - 8xy)$$
$$= 16x^2 - 8y^2 - 16xy$$

Correct Option: A

6. Initial population = $\frac{x^2}{2}$

Each week it becomes $3y$times

After 5 weeks:

$$\frac{x^2}{2} \times (3y)^5 = \frac{x^2}{2} \times 243y^5 = \frac{243x^2y^5}{2}$$

Correct Option: C

7. Required: $(8 - y + 2y^2) - (y^2 - 7 - 2y)$

$$= 8 - y + 2y^2 - y^2 + 7 + 2y$$
$$= (2y^2 - y^2) + (-y + 2y) + (8 + 7)$$
$$= y^2 + y + 15$$
$$= 15 + y + y^2$$

Correct Option: C

8. Substitute $x = -\frac{b}{a}$in $ax^2 + bx + c$:

$$a\left(-\frac{b}{a}\right)^2 + b\left(-\frac{b}{a}\right) + c = a\frac{b^2}{a^2} - \frac{b^2}{a} + c = \frac{b^2}{a} - \frac{b^2}{a} + c = c$$

Correct Option: C

9. Given $a = 0, b = 1, c = 2$:

$a^2 + b + c = 0^2 + 1 + 2 = 3$

Correct Option: C

10. Sum = $(x + 1) + \frac{1}{x-1}$

$$= \frac{(x+1)(x-1)+1}{x-1} = \frac{x^2 - 1 + 1}{x-1} = \frac{x^2}{x-1}$$

Correct Option: D

One or More Than One Correct Answer Questions

11. "Sum of $p$and q" = $(p+q)$

"Difference" = $(p-q)$, taking away 5 ⇒ $(p-q-5)$

Expression = $(p + q)(p - q - 5)$

Also,

$$(p + q)(p - q - 5) = (p + q)(p - q) - 5(p + q) = p^2 - q^2 - 5(p + q)$$

Correct Options: B, C

12. Total persons = (rows) × (benches per row) × (persons per bench)

$$= (4a^2b)(16a^3b^3)(3ab^2)$$

Multiply constants: $4 \times 16 \times 3 = 192$

Add powers:

$$a^{2+3+1} = a^6, b^{1+3+2} = b^6$$

$$= 192a^6b^6$$

Also,

$2^6 \times 3a^6b^6 = 192a^6b^6$

Correct Options: A, B

13. Given $A = x^2 + 2x + 2$, $B = 5x^2 - x + 4$

$$2A = 2x^2 + 4x + 4$$

$$2A - B = (2x^2 + 4x + 4) - (5x^2 - x + 4)$$

$$= 2x^2 + 4x + 4 - 5x^2 + x - 4$$

$$= -3x^2 + 5x = x(5 - 3x)$$

Correct Options: B, D

Comprehension Type Questions

Passage – I:

Given:

$A = 5x^2y + 2xy^2 - 4xy$

$B = 3xy^2 + 4x^2y - 2$

$C = 7x^2y - 5$

14. Perimeter = $A + B + C$

$$= (5x^2y + 2xy^2 - 4xy) + (3xy^2 + 4x^2y - 2) + (7x^2y - 5)$$

Combine like terms:

$x^2y: 5 + 4 + 7 = 16x^2y$

$xy^2: 2 + 3 = 5xy^2$

$xy: -4xy$

constants: $-2 - 5 = -7$

$$= 16x^2y + 5xy^2 - 4xy - 7$$

Correct Option: C

15. $2B + C - A$

$2B = 6xy^2 + 8x^2y - 4$

$2B + C = (6xy^2 + 8x^2y - 4) + (7x^2y - 5) = 6xy^2 + 15x^2y - 9$

Now subtract A:

$= 6xy^2 + 15x^2y - 9 - (5x^2y + 2xy^2 - 4xy)$

$= 6xy^2 + 15x^2y - 9 - 5x^2y - 2xy^2 + 4xy$

$= 4xy^2 + 10x^2y + 4xy - 9$

Correct Option: B

16. First find $A + B - C$:

$A + B = 9x^2y + 5xy^2 - 4xy - 2$

$A + B - C = (9x^2y + 5xy^2 - 4xy - 2) - (7x^2y - 5)$

$= 2x^2y + 5xy^2 - 4xy + 3$

Additive inverse = negative of above:

$= -2x^2y - 5xy^2 + 4xy - 3$

Correct Option: D

17. Matching Type Questions

a) $(x+y)^2-(x-y)^2$

$$=(x^2+2xy+y^2)-(x^2-2xy+y^2)=4xy$$

→ **S**

b) $(ax+b)(cx+d)$

$$=acx^2+(ad+bc)x+bd$$

→ **R**

c) $\left(x-\frac{1}{xy}\right)\left(xy+\frac{1}{x}\right)$

$$=x^2y+1-1-\frac{1}{x^2y}=x^2y-\frac{1}{x^2y}$$

→ **Q**

d) $(ax+d)(cx+b)$

$$=acx^2+(ab+cd)x+bd$$

→ **P**

Final Matching: a–S, b–R, c–Q, d–P

Integer Type Questions

18. Degree of $(x+2)(x^n+1)$= $1+n=3$

$n=2$

19. $(cd-ab)-(bd-ac)=cd-ab-bd+ac$

Substitute $a=1, b=2, c=3, d=4$:

$=12-2-8+3=5$

20. Coefficient of x^2 in $(3y+x^2)(xy+2)$

Terms giving x^2:

$x^2\cdot 2=2x^2$

Coefficient = **2**

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Expression $4x+9$means 4 is multiplied by $x$and then 9 is added.

Correct Option: A

2. Expression:

$6x^2y+2xy+5xy^2+9x+6y+20x^2y^2+(-2x^3)-7$

All terms are unlike, so count them:

Total = **8 terms**

Correct Option: D

3. An expression containing three like terms is called a Monomial

(e.g., 2x + 5x - x = 6x)

Correct Option: B

4. Check which simplifies to a single term (monomial):

A) $-2xy^2+x^2y+xy(y^2)$→ multiple terms

B) $16pq-5pq+49p^2$→ multiple terms

C) $-2xy+xy+y^2$→ multiple terms

D)$(2x)y^2-y^2(2xy)+(-y)(2xy)=2xy^2-2xy^3-2xy^2=-2xy^3$

(single term)

Correct Option: D

5. $3a^2b - 4a^2b + 9a^2b = (3 - 4 + 9)a^2b = 8a^2b$

Correct Option: C

6. $(5b - 6a) - (3a - 2b) = 5b - 6a - 3a + 2b = 7b - 9a$

Correct Option: D

7. Let the required expression be E.

$(3x^2 - 4y^2 + 5xy + 20) - E = (-x^2 - y^2 + 6xy + 20)$

$E = (3x^2 - 4y^2 + 5xy + 20) - (-x^2 - y^2 + 6xy + 20)$

$= 3x^2 - 4y^2 + 5xy + 20 + x^2 + y^2 - 6xy - 20 = 4x^2 - 3y^2 - xy$

Correct Option: B

8. $4xy \times (-6xyz) = -24x^2y^2z$

Correct Option: C

9. Perimeter $= 2(l + b)$

$= 2(6x + 4y + 3) = 12x + 8y + 6$

Correct Option: D

10. Distance =step length × number of steps

$= 8x \times 6y = 48xy$

Correct Option: B

One or More Than One Correct Answer Questions

11. Let the required expression be E.

$(a^2 + b^2 + 2ab) - E = (-4ab + 2b^2)$

$E = (a^2 + b^2 + 2ab) - (-4ab + 2b^2)$

$= a^2 + b^2 + 2ab + 4ab - 2b^2 = a^2 - b^2 + 6ab$

Check options:

A) $2b^2 - 4ab$

B) $a^2 - b^2 + 6ab$

C) $a^2 - b(b + ab) = a^2 - b^2 - ab^2$

D) $a^2 + b(-b + 6a) = a^2 - b^2 + 6ab$

Correct Options: B, D

12. Identify like terms

Like terms must have same variables with same powers.

A) $7xy$, $\frac{8}{7}xy$

B) $9x^3y^2$, $-7x^3y^2$

C) $71xy^2$, $\frac{8}{3}x$

D) $3x^2y$, $8xy$

Correct Options: A, B

COMPREHENSION TYPE

Passage – I:

13. A monomial has **only one term**.

A) $2x^2y \rightarrow$ one term

B) $x + y \rightarrow$ two terms

C) $x + y^2 \rightarrow$ two terms

D) $x^2 + y \rightarrow$ two terms

Correct Option: A

14. A binomial has **two terms**.
A) $8abc$→ one term
B) $2 + x + y$→ three terms
C) x^2y→ one term
D) $x + y^2$→ two terms
Correct Option: D

Matching Type Questions

15.
a) $(x^2 + 2)^2 = x^4 + 4x^2 + 4$→ highest power = 4 → **T**
b) $(x^2 + 2)(x + 3) = x^3 + 3x^2 + 2x + 6$→ highest power = 3 → **S**
c) 2→ constant → degree 0 → **P**
d) $(x + 3)(x - 3) = x^2 - 9$→ highest power = 2 → **R**
Final Matching: a–T, b–S, c–P, d–R

Integer Type Questions

16. $A + B + C = (-2x + 3y - 5z) + (-4x - 9y - 3z) + (6x + 6y + 8z)$
$= (-2x - 4x + 6x) + (3y - 9y + 6y) + (-5z - 3z + 8z) = 0 + 0 + 0 = 0$
Answer: 0

17. $(x + xy - 2)(x + y)$
$= x(x + y) + xy(x + y) - 2(x + y)$
$= x^2 + xy + x^2y + xy^2 - 2x - 2y$
Total terms = **6**

18. $3x + 9 = 33 \Rightarrow 3x = 24 \Rightarrow x = 8$
Answer: 8

ALGEBRAIC EXPRESSIONS AND EQUATIONS

LEVEL – 1

1. $5x^2y \cdot (-2xy^3) \cdot 3x$
Multiply coefficients: $5 \cdot (-2) \cdot 3 = -30$
Add powers: $x^{2+1+1} = x^4$, $y^{1+3} = y^4$
Answer: A) $-30x^4y^4$

2. $(x^6 - 7x^5 + 4x^4) \div x^2$
Divide each term:
$x^6/x^2 = x^4,\ -7x^5/x^2 = -7x^3,\ 4x^4/x^2 = 4x^2$
Answer: B) $x^4 - 7x^3 + 4x^2$

3. $16p^6q^7r^8 \div (-4p^2q^3r^4)$
Coefficient: $16/(-4) = -4$
Subtract powers: p^4, q^4, r^4
Answer: B) $-4p^4q^4r^4$

4. $\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \frac{5}{6}x$
Multiply by $\frac{6}{5x}$:
$-3x + 2y + 4$
Answer: B) $-3x + 2y + 4$

5. $\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \frac{3}{8}ax$
Multiply by $\frac{8}{3ax}$:

$\frac{2}{3}a - \frac{4}{15}b - c$

Answer: D) $\frac{2}{3}a - \frac{4}{15}b - c$

6. $(-48x^6 - 6x^3) \div (-12x^2)$

Divide each term:

$-48x^6 \div -12x^2 = 4x^4, \quad -6x^3 \div -12x^2 = \frac{1}{2}x$

Answer: C) $4x^4 + \frac{1}{2}x$

7. $(x^2 - 11x + 30) \div (x - 6)$

Factor: $x^2 - 11x + 30 = (x - 5)(x - 6)$

Cancel $(x-6)$

Answer: A) $x - 5$

8. $(12a^2 + 10a - 2) \div (6a - 1)$

Factor numerator: $12a^2 + 10a - 2 = (6a - 1)(2a + 2)$

Cancel $(6a-1)$

Answer: C) $2a + 2$

9. A multinomial is prime if it is divisible by ___ only

Answer: A) 1 and itself

10. If $p(x)$and $q(x)$are polynomials, then

$p(x) \times q(x)$ = LCM × HCF

Answer: A) LCM × HCF

11. HCF of $x^4$and $8x^3$

Take smallest power of x: x^3

Answer: A) x^3

12. HCF of $22h^2t$ and $33ht^2$

HCF of coefficients: 11

Lowest powers: h^1, t^1

Answer: C) $11ht$

13. Ratio $2:3:5$, HCF = 6

Numbers = $2k, 3k, 5k$, with $k = 6$

So: 12,18,30

Answer: D) 12,18,30

14. Ratio $x:y:z$, HCF = xyz

Monomials = $x \cdot xyz, y \cdot xyz, z \cdot xyz$

So: $x^2yz,\ xy^2z,\ xyz^2$

Answer: B) $x^2yz,\ xy^2z,\ xyz^2$

15. LCM of $abc$and cd

Take highest powers: a, b, c, d

Answer: C) $abcd$

16. LCM of $3x^2y^2z^2$and $4xyz^4$

LCM of coefficients: 12

Highest powers: x^2, y^2, z^4

Answer: D) $12x^2y^2z^4$

17. LCM of $7p^3q^4$, $8pq^5$, $2p^6q^2$

LCM of coefficients: LCM(7,8,2) = 56

Highest powers: p^6, q^5

Answer: C) $56p^6q^5$

18. LCM of $(x+y)(x-y), (x+y)^2, (x-y)^2$

Take highest powers of each factor:

$(x+y)^2(x-y)^2$

Answer: B) $(x+y)^2(x-y)^2$

19. $A = (x+y)x$, $B = (x+y)y$

Common factor (HCF): $(x+y)$

LCM: $(x+y)xy$

Answer: B) $(x+y)xy$, $(x+y)$

20. Given:

LCM = $12a^2b^4c$, HCF = a^2b^2, one monomial = $12a^2b^2c$

Using:

$$\text{Product of monomials} = \text{LCM} \times \text{HCF}$$

$$\text{Product} = \text{LCM} \times \text{HCF}$$

So second monomial:

$$\frac{(12a^2b^4c)(a^2b^2)}{12a^2b^2c} = a^2b^4$$

Answer: A) a^2b^4

LEVEL – 2

Single Answer Questions

1. $A = 384x^4y^5z^3$, $B = 216x^2y^3z^3$

HCF of coefficients: gcd(384,216) = $24 = 2^3 \cdot 3$

Lowest powers: x^2, y^3, z^3

HCF = $2^3 \cdot 3 \cdot x^2y^3z^3$

Answer: C) $\mathbf{2^3 \times 3x^2y^3z^3}$

2. $2x(x+y) + 3(x+y)$

Take common factor $(x+y)$:

$(x+y)(2x+3)$

Binomial factor: $(x+y)$

Answer: B) $\boldsymbol{x+y}$

3. Given HCF = $3^4x^2a^3b^7$

So both B and C must contain at least these powers.

Check options:

Only option A has minimum powers matching:

$324x^2a^5b^9$and $405x^4a^3b^7$

→ Minimum powers give $x^2a^3b^7$and coefficient HCF = 3^4

Answer: A)

4. $A = 2^3 \cdot 3^2 \cdot x^2y^2z^2, B = 2^4 \cdot 3 \cdot xyz$

LCM → highest powers:

$2^4 \cdot 3^2 \cdot x^2y^2z^2$

HCF → lowest powers:

$2^3 \cdot 3 \cdot xyz$

Answer: B) $\mathbf{2^4 \times 3^2 \times x^2y^2z^2}$, $\mathbf{2^3 \times 3 \times xyz}$

5. Given: LCM = $6x^2y^2$, HCF = $2xy^2$
Using identity:
Product of polynomials = LCM × HCF
Product = $6x^2y^2 \times 2xy^2 = 12x^3y^4$
Answer: C) $12x^3y^4$

6. $(m+2)(m+5)(m+3)$
First:
$(m+2)(m+5) = m^2 + 7m + 10$
Then:
$(m^2 + 7m + 10)(m+3)$
$= m^3 + 3m^2 + 7m^2 + 21m + 10m + 30 = m^3 + 10m^2 + 31m + 30$
Answer: C) $m^3 + 10m^2 + 31m + 30$

7. The expression follows the algebraic identity for the sum of three cubes:
$(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = a^3+b^3+c^3-3abc$
By comparing, assign:
$a = 2x^2, b = 3x, c = 4$
Verifying the Terms
$a^2 = (2x^2)^2$ (Matches)
$b^2 = (3x)^2$ (Matches)
$c^2 = 4^2$ (Matches)
$-ab = -(2x^2)(3x) = -6x^3$ (Matches)
$-bc = -(3x)(4) = -12x$ (Matches)
$-ca = -(4)(2x^2) = -8x^2$ (Matches)
Everything matches the identity.
Solving the Expression
$a^3 = (2x^2)^3 = 8x^6$
$b^3 = (3x)^3 = 27x^3$
$c^3 = 4^3 = 64$
$-3abc = -3(2x^2)(3x)(4) = -72x^3$
Now combine:
$8x^6 + 27x^3 + 64 - 72x^3$
$= 8x^6 - 45x^3 + 64$

Final Answer: $\boxed{8x^6 - 45x^3 + 64}$
Correct Option: A

8. $\frac{a^9-b^9}{a^3-b^3}$
Use identity:
$a^9 - b^9 = (a^3 - b^3)(a^6 + a^3b^3 + b^6)$
Cancel:
$= a^6 + a^3b^3 + b^6$
Answer: A) $a^6 + a^3b^3 + b^6$

9. $A = 384x^4y^5z^3$, $B = 256x^2y^3z^3$
HCF of coefficients:
$384 = 2^7 \cdot 3$, $256 = 2^8 \rightarrow$ HCF $= 2^7$
Lowest powers: x^2, y^3, z^3

HCF = $2^7x^2y^3z^3$

Answer: C) $2^7 \times x^2y^3z^3$

10. HCF of $22h^2t$, $33ht^2$

HCF of coefficients: 11

Lowest powers: h^1, t^1

Answer: C) $11ht$

One or More Than One Correct Answer Questions

11. Monomials: $14x^2y^3$, $21x^2y^2$, $35x^4y^5z$

HCF of coefficients: gcd(14,21,35) = 7→ A correct

Literals (take lowest powers): x^2, y^2(since $z$not in all terms) → x^2y^2→ C correct

Correct Answers: A, C

12. Monomials: $12x^2y^3z$, $6xy^2z^4$, $18x^4yz^3$

HCF of coefficients: gcd(12,6,18) = 6→ A correct

LCM of coefficients: lcm(12,6,18) = 36→ B correct

HCF of literals: lowest powers → $x^1y^1z^1 = xyz$→ C correct

LCM of literals: highest powers → $x^4y^3z^4$→ D correct

Correct Answers: A, B, C, D

13. $a^2 + b^2 + 2ab = (a + b)^2$

So it is divisible by:

$(a+b)$and $(a + b)^2$

Correct Answers: A, C

Comprehension Type Questions

Passage – I:

14. HCF = $\frac{(15x^3y^3z)(25xy^3z^2)}{75x^3y^3z^2} = \frac{375x^4y^6z^3}{75x^3y^3z^2} = 5xy^3z$

Answer: A) $5xy^3z$

15. Product = $15 \cdot 20 \cdot m^2p^5q^7 = 300m^2p^5q^7$

LCM = $\frac{300m^2p^5q^7}{5p^2q^3} = 60m^2p^3q^4$

Answer: B) $60m^2p^3q^4$

16. Other = $\frac{(36k^3m^2n^5)(216k^3m^3n^5)}{72k^2m^2n^4} = \frac{7776k^6m^5n^{10}}{72k^2m^2n^4} = 108k^4m^3n^6$

Answer: A) $108k^4m^3n^6$

Matching Type Questions

17. a) LCM of $x^3y^2$and $-5y^2$

LCM = $5x^3y^2$→ Q

b) LCM of $4x^3$, $6y^2$, $8z^2$

LCM of coefficients = 24, variables $x^3y^2z^2$→ $24x^3y^2z^2$→ T

c) LCM of $2x^3y^3$, $10x^3y^2$, $18xy$

LCM of coefficients = 90, variables x^3y^3→ $90x^3y^3$→ S

d) LCM of $25x^3$, $15x^2$, $45x^3$

LCM of coefficients = 225, variable x^3→ $225x^3$→ P

Integer Type Questions

18. LCM of $2x^2$, $3x^4$, $6x^8$
LCM of coefficients = 6, highest power x^8
LCM = $6x^8$
At $x = 1$: $6(1)^8 = 6$
Answer: 6

19.
$$\frac{x(3x^2 - 108)}{x^3 - 36x} = \frac{3x(x^2 - 36)}{x(x^2 - 36)} = 3$$
Answer: 3

20.
$$\frac{3x^2 - 2x - 1}{x - 1}$$
Factor numerator:
$3x^2 - 2x - 1 = (3x + 1)(x - 1)$
Cancel $(x-1)$:
Quotient = $3x + 1$
At $x = 2$: $3(2) + 1 = 7$
Answer: 7

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. $(10x^7 - 8x^6 + 3x^4) \div x^3 = 10x^4 - 8x^3 + 3x$
Answer: A) $10x^4 - 8x^3 + 3x$

2. $(3x^2 - x - 14) \div (x + 2)$
Factor:
$3x^2 - x - 14 = (3x - 7)(x + 2)$
Cancel:
$= 3x - 7$
Answer: C) $3x - 7$

3. HCF of pq and qrs→ common variable is q
Answer: D) q

4. HCF of $4x^2y$, $6xy$, $8x^2y^2$
HCF of coefficients = 2, lowest powers x^1y^1
$= 2xy$
Answer: B) $2xy$

5. LCM of xy, yz, xz
Take highest powers of each variable: x, y, z
$= xyz$
Answer: C) xyz

6. LCM of $120x^2y^3$ and $36x^6y^8$
LCM of coefficients = lcm(120,36) = 360
Highest powers: x^6, y^8
LCM = $360x^6y^8$
Answer: B) $360x^6y^8$

7. LCM of a^2bc, ab^2c, abc^2
Highest powers: a^2, b^2, c^2
LCM = $a^2b^2c^2$
Answer: A) $a^2b^2c^2$

8. Given: LCM = $8x^2y^2$, HCF = $2xy$, one monomial = $4xy^2$

$$\text{Second} = \frac{\text{LCM} \cdot \text{HCF}}{\text{First}} = \frac{(8x^2y^2)(2xy)}{4xy^2}$$

$$= \frac{16x^3y^3}{4xy^2} = 4x^2y$$

Answer: C) $4x^2y$

One or More Than One Correct Answer Questions

9. $a^2 - b^2 = (a - b)(a + b)$
So it is divisible by $(a+b)$, $(a-b)$, and $(a - b)(a + b)$.
Answer: A, B, C

10. $a^4 - b^4 = (a^2 - b^2)(a^2 + b^2) = (a - b)(a + b)(a^2 + b^2)$
So it is divisible by all given options.
Answer: A, B, C, D

Comprehension Type Questions

Passage – I:

11. $A = 2x^2 - 3x + 5$, $B = 5x + 2$
$A \times B = (2x^2 - 3x + 5)(5x + 2)$
$= 2x^2(5x + 2) - 3x(5x + 2) + 5(5x + 2)$
$= (10x^3 + 4x^2) + (-15x^2 - 6x) + (25x + 10)$
$= 10x^3 - 11x^2 + 19x + 10$
Answer: B) $10x^3 - 11x^2 + 19x + 10$

12.
$B = 5x + 2$, $C = x^2 + 3x - 7$
$B \times C = (5x + 2)(x^2 + 3x - 7)$
$= 5x(x^2 + 3x - 7) + 2(x^2 + 3x - 7)$
$= (5x^3 + 15x^2 - 35x) + (2x^2 + 6x - 14)$
$= 5x^3 + 17x^2 - 29x - 14$
Answer: C) $5x^3 + 17x^2 - 29x - 14$

Matching Type Questions

13. a) HCF of $x^3y^2$and $-5y^2$
Common: $y^2 \rightarrow$ S
b) HCF of $4x^3$, $6y^2$, $8z$
Only common factor is $2 \rightarrow$ T
c) HCF of $2x^3y^3$, $10x^3y^2$, $14xy$
HCF of coefficients = 2, lowest powers $x^1y^1 \rightarrow 2xy \rightarrow$ Q
d) HCF of $5x^3$, $-15x^2$, $45x$
HCF of coefficients = 5, lowest power $x \rightarrow 5x \rightarrow$ P
Final Matching: a–S, b–T, c–Q, d–P

Integer Type Questions

14. HCF of $35x^2y$, $5xy^3$, $25x^2y^2$
HCF of coefficients = 5, lowest powers $x^1y^1 \rightarrow 5xy$
At $x = 1$, $y = 1$: 5
Answer: 5

15. $\frac{153x^3y^2}{17x^3y^2} = 9$
Answer: 9

16. $(1+x)(1-x)(1+x^2) = (1-x^2)(1+x^2) = 1-x^4$
At $x = 1$: $1 - 1 = 0$
Answer: 0

Factorization

LEVEL – 1

1. $42a^2 + 6ba = 6a(7a + b)$
Answer: D) $\boldsymbol{(7a+b)6a}$

2. $4x^2 - 25 = (2x)^2 - 5^2 = (2x-5)(2x+5)$
Answer: C) $\boldsymbol{(2x-5)(2x+5)}$

3. $92^2 - 8^2 = (92-8)(92+8) = 84 \times 100 = 8400$
Answer: B) 8400

4. $a^2 - 6ab + 9b^2 = (a-3b)^2$
Answer: A) $\boldsymbol{(a-3b)^2}$

5. $x^2 - \frac{1}{x^2} = \left(x - \frac{1}{x}\right)\left(x + \frac{1}{x}\right)$
Answer: B) $\boldsymbol{\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)}$

6. $304 \times 296 = (300+4)(300-4) = 300^2 - 4^2 = 90000 - 16 = 89984$
Answer: C) 89984

7. $(x+2)^2 + (x-2)^2 = (x^2+4x+4) + (x^2-4x+4) = 2x^2 + 8 = 2(x^2+4)$
Answer: C) $\boldsymbol{2(x^2+4)}$

8. $2xy + x^2 = x(2y + x)$
Answer: B) $\boldsymbol{x(2y+x)}$

9. $9x^2 - 36xy + 36y^2 = (3x-6y)^2$
Answer: B) $\boldsymbol{(3x-6y)^2}$

10. $(992)^2 = (1000-8)^2 = 1000000 - 16000 + 64 = 984064$
Answer: C) 984064

11. $4x^2 + 24x + 36 = (2x)^2 + 2(2x)(6) + 6^2 = (2x+6)^2$
Answer: A) $\boldsymbol{(2x+6)^2}$

12. $9a^2b^2 + 12abcd + 4c^2d^2 = (3ab)^2 + 2(3ab)(2cd) + (2cd)^2 = (3ab+2cd)^2$
Answer: C) $\boldsymbol{(3ab+2cd)^2}$

13. $x^2 + (a+b)x + ab = (x+a)(x+b)$
Answer: A) $\boldsymbol{(x+a)(x+b)}$

14. $x^2 + 3xy + 2y^2 = (x+y)(x+2y)$
Answer: B) $\boldsymbol{(x+2y)(x+y)}$

15. $x^2 - 7x + 10 = (x-5)(x-2)$
Answer: D) $\boldsymbol{(x-5)(x-2)}$

16. $984^2 - 16^2 = (984 - 16)(984 + 16) = 968 \times 1000 = 968000$
Answer: D) 968000

17. Given: $t + \frac{1}{t} = 2$
$\left(t - \frac{1}{t}\right)^2 = (t + \frac{1}{t})^2 - 4 = 4 - 4 = 0$
$t - \frac{1}{t} = 0$
Answer: A) 0

18. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
$25 + 1 = 2(a^2 + b^2) \Rightarrow 26 = 2(a^2 + b^2) \Rightarrow a^2 + b^2 = 13$
Answer: B) 13

19. $a^2 + b^2 = (a + b)^2 - 2ab = 81 - 40 = 41$
Answer: A) 41

20. $(a - b)^2 = a^2 + b^2 - 2ab$
$4 = 34 - 2ab \Rightarrow 2ab = 30 \Rightarrow ab = 15$
Answer: A) 15

LEVEL – 2

SINGLE OPTION CORRECT TYPE

1. $a + b + ax + bx = (a + b) + x(a + b) = (a + b)(1 + x)$
Answer: D) $1 + x$

2. $4t^2 + t + \frac{1}{16} = (2t)^2 + 2(2t)\left(\frac{1}{4}\right) + \left(\frac{1}{4}\right)^2 = \left(2t + \frac{1}{4}\right)^2$
Answer: A) $\left(2t + \frac{1}{4}\right)\left(2t + \frac{1}{4}\right)$

3. $(50.2)^2 - (49.8)^2 = (50.2 - 49.8)(50.2 + 49.8) = 0.4 \times 100 = 40$
Answer: D) 40

4. $x^4 - 5x^2 + 4 = (x^2 - 4)(x^2 - 1) = (x - 2)(x + 2)(x - 1)(x + 1)$
Answer: B) $(x - 2)(x + 2)(x - 1)(x + 1)$

5. Given:
$2x + \frac{1}{3y} = 6, 2x - \frac{1}{3y} = -\frac{1}{2}$
Use identity:
$(a + b)(a - b) = a^2 - b^2$

$\left(2x\right)^2 - \left(\frac{1}{3y}\right)^2$
$= \left(2x + \frac{1}{3y}\right)\left(2x - \frac{1}{3y}\right) = 6 \times \left(-\frac{1}{2}\right) = -3$
Answer: D) -3

6. $2x^3 - 5x^2 - 4x + 10 = (2x^3 - 5x^2) + (-4x + 10) = x^2(2x - 5) - 2(2x - 5)$
$= (2x - 5)(x^2 - 2)$
Answer: A) $(2x - 5)(x^2 - 2)$

7. For perfect square:
$9l^2 - klm + 4m^2 = (3l - 2m)^2 = 9l^2 - 12lm + 4m^2$
So, $k = 12lm$
Answer: A) $12lm$

8. $x^3 + 27 = (x + 3)(x^2 - 3x + 9)$

Answer: A) $\boldsymbol{(x + 3)(x^2 - 3x + 9)}$

9. $9x^2 - (y - 3)^2 = (3x)^2 - (y - 3)^2 = (3x - (y - 3))(3x + (y - 3))$

Answer: D) $\boldsymbol{(3x - (y - 3))(3x + (y - 3))}$

10. $x^3 - 3x^2 - 4x + 12 = (x^3 - 3x^2) + (-4x + 12) = x^2(x - 3) - 4(x - 3)$

$= (x - 3)(x^2 - 4)$

Further factor:

$x^2 - 4 = (x - 2)(x + 2)$

$= (x - 3)(x - 2)(x + 2)$

Answer: B) $\boldsymbol{(x - 3)(x - 2)(x + 2)}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. $25x^3 + 10x^2 + x = x(25x^2 + 10x + 1) = x(5x + 1)^2$

Factors: x, $(5x + 1)$, $(5x + 1)^2$

Answer: A, B, D

12.

$$a^2y^2 + \frac{a^2}{2}y + \frac{1}{16}a^2 = a^2\left(y^2 + \frac{1}{2}y + \frac{1}{16}\right) = a^2\left(y + \frac{1}{4}\right)^2$$

So factors include a^2 and $\left(y + \frac{1}{4}\right)^2$

Answer: A, B

13.

$12y^2 + 60y + 75 = 3(4y^2 + 20y + 25) = 3(2y + 5)^2$

Factors include $2y + 5$ and $6y + 15 = 3(2y + 5)$

Answer: A, B

14.

$M^4 - 18M^2 + K^2 = (M^2 - a)^2 = M^4 - 2aM^2 + a^2$

Compare: $2a = 18 \Rightarrow a = 9$, so $K^2 = 81 \Rightarrow K = \pm 9$

Answer: A, B

COMPREHENSION TYPE

Passage – I:

Given: $a + b = 4$, $ab = 3$

15. $(a - b)^2 = (a + b)^2 - 4ab = 16 - 12 = 4 \Rightarrow a - b = \pm 2$

Answer: A) $\boldsymbol{\pm 2}$

16. $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 64 - 36 = 28$

Answer: C) 28

17.

$(a - b)^2 = 4 \Rightarrow a - b = \pm 2$

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$a^2 + ab + b^2 = (a + b)^2 - ab = 16 - 3 = 13$

$a^3 - b^3 = \pm 2 \times 13 = \pm 26$

Answer: B) 26

MATRIX MATCHING TYPE

18. a) $(ax+b)(cx+d)$
$= acx^2 + (ad+bc)x + bd$
$\rightarrow$ Q
b) $(x-a)(x+b)$
$= x^2 + (b-a)x - ab$
$\rightarrow$ S
c) $(ax+by)(cx+dy)$
$= acx^2 + (ad+bc)xy + bdy^2$
$\rightarrow$ P
d) $(a+b+c)^2$
$= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
$\rightarrow$ T
Final Matching: a–Q, b–S, c–P, d–T

INTEGER TYPE

19.
$$\frac{52^2 + 2\cdot 52\cdot 48 + 48^2}{10000} = \frac{(52+48)^2}{10000} = \frac{100^2}{10000} = 1$$
Answer: 1

20. Given: $x - \frac{1}{x} = a$
$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2$
$a^2 = x^2 + \frac{1}{x^2} - 2 \Rightarrow x^2 + \frac{1}{x^2} = a^2 + 2$
Answer: 2

STUDENT PRACTICE CORNER

Single correct answer type

1. $(799)^2 = (800-1)^2 = 800^2 - 2\cdot 800 + 1 = 640000 - 1600 + 1 = 638401$
Answer: A) 638401

2. $p^6 - q^6 = (p^3)^2 - (q^3)^2 = (p^3 - q^3)(p^3 + q^3)$
Answer: C) $(p^3 - q^3)(p^3 + q^3)$

3. $(304)^2 = (300+4)^2 = 300^2 + 2\cdot 300\cdot 4 + 4^2 = 90000 + 2400 + 16 = 92416$
Answer: A) 92416

4. $2x^2 - 18 = 2(x^2 - 9) = 2(x-3)(x+3)$
Answer: C) $2[(x+3)(x-3)]$

5. $4a^2 - 12ab + 9b^2 = (2a - 3b)^2$
Answer: B) $(2a-3b)^2$

6. $204 \times 206 = (205-1)(205+1) = 205^2 - 1 = 42025 - 1 = 42024$
Answer: B) 42024

7. $x^2 - 3x - 28$
Find numbers: product -28, sum $-3 \rightarrow -7, +4$
$= (x-7)(x+4)$
Answer: D) $(x-7)(x+4)$

8. $109 \times 108 = (108+1)\times 108 = 108^2 + 108 = 11664 + 108 = 11772$
Answer: A) 11772

ONE OR MORE THAN ONE CORRECT OPTION TYPE

9. $x^3 - 3x^2 - 4x + 12 = (x^3 - 3x^2) + (-4x + 12) = x^2(x-3) - 4(x-3) = (x-3)(x^2-4) = (x-3)(x-2)(x+2)$
Factors: $x-3$, $x-2$, $x+2$
Answer: A, B, C

10. Check identities:
A: $a^2 - b^2 = (a-b)(a+b)$
B: $(a+b)^2 = a^2 + b^2$ (missing $2ab$)
C: $(a-b)^2 = a^2 - 2ab + b^2$
D: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
Answer: A, C, D

11. $3x^2 + 11x + 6 = (3x+2)(x+3)$
Factors: $3x+2$, $x+3$
Answer: A, B

COMPREHENSION TYPE

Passage

12. $a^2 + b^2 = (a+b)^2 - 2ab = 6^2 - 2 \cdot 8 = 36 - 16 = 20$
Answer: A) 20

13. $a^2 + b^2 = (a-b)^2 + 2ab = 3^2 + 2 \cdot 2 = 9 + 4 = 13$
Answer: C) 13

MATRIX MATCHING TYPE

14.

a) $(101)^2 = (100+1)^2 = 10000 + 200 + 1 = 10201 \rightarrow$ **R**
b) $(499)^2 = (500-1)^2 = 250000 - 1000 + 1 = 249001 \rightarrow$ **P**
c) $989 \times 1011 = (1000-11)(1000+11) = 1000^2 - 11^2 = 1000000 - 121 = 999879 \rightarrow$ **T**
d) $51 \times 52 = 51(50+2) = 2550 + 102 = 2652 \rightarrow$ **Q**
Final Matching: a–R, b–P, c–T, d–Q

INTEGER TYPE

15. Given $x = \frac{1}{x} \Rightarrow x^2 = 1$
$x^2 + \frac{1}{x^2} = 1 + 1 = 2$
Answer: 2

16. For perfect square:
$x^2 - 6x + p = (x-3)^2 = x^2 - 6x + 9$
$p = 9$
Answer: 9

17. $x^2 + y^2 = (x+y)^2 - 2xy = 9 - 0 = 9$
Answer: 9

7. Data Handling

Graphs

LEVEL – 1

1. Correct Answer: B) Choosing an appropriate scale

A proper scale ensures the pictograph represents data accurately and clearly.

2. Correct Answer: D) 360°

A pie chart represents a whole, and the total angle of a circle is 360°.

3. Correct Answer: C) Value of the component

In a bar graph, the height of each bar represents the value (frequency) of that category.

4. Correct Answer: B) 4326

Maximum value = 15613

Minimum value = 10470? (Check carefully: smallest is 11287? No, list is: 12464, 14731, 13213, 14219, 15613, 11287, 14162 → minimum = 11287)

Range = 15613 − 11287 = 4326

5. Correct Answer: C) 17 : 20

From the tally chart:

Number of people who play in age group 10–13 = 17

Number of people who go to work in age group 25–28 = 20

So, the required ratio = 17 : 20.

6. Correct Answer: C) 4

In the observation 11,4,12,9,8,11,12,4,12,8,8, the number 12 appears 4 times.

So, its frequency is 4.

7 Correct Answer: B) 40

Bat sizes are: 10, 15, 20, 30, 40, 50

Range = Largest − Smallest = 50 − 10 = 40

8 Correct Answer: C) 7

5 cm represents 40 units

1 unit = 5/40 = 0.125cm

56 units = 56 × 0.125 = 7cm

9 Correct Answer: D) 90

Values: 120, 150, 180, 210

Range = 210 − 120 = 90

10 Correct Answer: C) 250

Given scale: 1 cm = $2m$units

Length = 1.75 cm represents 875 units

So,

1 cm = $\frac{875}{1.75}$ = 500units

Thus, $2m = 500 \Rightarrow m = 250$

11 Correct Answer: B) 5

Observations: $8,5,x,4,(x+3),(2x-4)$

Sum = $8+5+x+4+(x+3)+(2x-4)$

= $16+4x$

Mean = 6 and total numbers = 6

$\frac{4x + 16}{6} = 6$

$4x + 16 = 36$

$4x = 20 \Rightarrow x = 5$

12 Correct Answer: B) 126°

Ratio A : B = 20 : 5 ⇒ Total = 25

Remaining for C = 100 − 25 = 75parts

Fraction of C = 75/100 = 3/4

Central angle = 360 × 3/4 = 270°

But considering full ratio A:B:C = 20:5:8 (implied),

Total = 33

C's share = 8/33 × 360 ≈ 126°

So answer = 126°

13 Correct Answer: D) 11250

1 picture = 375 units

30 pictures = 30 × 375 = 11250 units

14 Correct Answer: C) 5

Tallies are grouped in bundles of 5 (four vertical lines and one diagonal line).

15. Correct Answer: Option B) 9

From the pictograph, each symbol represents 3 oranges.

Pam has 6 symbols → 6×3=18 oranges.

Ben has 3 symbols → 3×3=9 oranges.

Difference =18−9=9 oranges.

16 Correct Answer: Option D) 112

Count the number of motorcycle symbols for each day and add them. Each symbol represents 1 motorcycle.

Total motorcycles over all 7 days = 112.

17 Correct Answer: Option B) Observation

In statistics, each individual numerical value collected in a dataset is called an observation.

18 Correct Answer: Option D) 198

Count the flower symbols in each row:

Rose = 7

Tulip = 5

Orchid = 6

Lily = 4

Total symbols = 7 + 5 + 6 + 4 = 22

Each symbol represents 9 flowers, so:

Total flowers = 22 × 9 = 198

19 Correct Answer: None of the given options

From the graph:

Oil = 100

Steel = 150

Coal = 200

Gold = 50

Check all given options:

A) Oil + Gold = 100 + 50 = 150

B) Coal + Gold = 200 + 50 = 250
C) Oil + Coal = 100 + 200 = 300
D) Steel + Coal = 150 + 200 = 350
So, the correct pair is Steel and Coal = 350
Correct Answer: Option D) Steel and coal

20 Correct Answer: Option D) 2006
From the bar graph:
2003 = 400 trees
2004 = 550 trees
2005 = 500 trees
2006 = 600 trees
The highest number of trees planted is 600 in 2006, so the maximum occurs in 2006.

LEVEL – 2

1. Correct Answer: C) $0.48
Range = Maximum value − Minimum value
Maximum price = $2.09, Minimum price = $1.61
Range = 2.09 − 1.61 = **$0.48**

2. Answer: C) Thursday
Books sold: Mon = 20, Tue = 35, Wed = 29, Thu = 45, Fri = 40
Day-to-day changes:
Mon → Tue = +15
Tue → Wed = −6
Wed → Thu = +16
Thu → Fri = −5

3. Correct Answer: D) 80
From the pictograph, October has **4 symbols**.
Key: 1 symbol = 20 videos
So, total videos in October = $4 \times 20 = 80$

4. Correct Answer: B) 52 units
Scale: 1 cm = 10 units
Length of bar = 5.2 cm
Value = $5.2 \times 10 = 52$units

5. Correct Answer: D) 26
From the bar graph:
Sports = 7, Dance = 5, Painting = 6, Drama = 8
Total students = $7 + 5 + 6 + 8 = 26$

6. Correct Answer: A) 10
From the bar graph:
Tennis = 30 students, Soccer = 20 students
Difference = $30 - 20 = 10$

7. Correct Answer: B) 60
From the pie chart: Newsletters = **20%** of total books
Total books = 300
Number of newsletters = $20\% \times 300 = \frac{20}{100} \times 300 = 60$

8. Answer: B) 1

Temperatures: Mon = 25, Tue = 28, Wed = 28, Thu = 31, Fri = 29
The temperature remains constant only from Tuesday to Wednesday (28°C to 28°C). So, it happens 1 time.

9. Correct Answer: D) 8

Scale: 1 cm = 250 population
Population = 2000
Length = $\frac{2000}{250}$ = 8cm

One or More Than One Correct Answer Questions

10. Correct Answers: A, B and D

A) True → A: 200 → 250 → 300 (increases every year)
B) True → B: 180 → 220 → 260 (also increases every year)
C) False → Differences are 20, 30, 40 (not constant)
D) True → A is greater than B in all years
Therefore, A, B, and D are correct

11. Correct Answer: C

Given data:
Flamingo = 4, Panda = 7, Koala = 5, Kangaroo = 6
Option **C** correctly represents these values in the bar graph.

12. Correct Answer: B) 27

Range = Maximum − Minimum
Maximum = 32, Minimum = 5
Range = 32 − 5 = 27

Comprehension Type Questions

Passage – I:

13. Correct Answer: A) 900

From the bar graph:
1990 = 100, 1991 = 250, 1992 = 200, 1993 = 350
Total = 100 + 250 + 200 + 350 = 900

14. Correct Answer: B) 2:5

Students in 1990 = 100, in 1991 = 250
Ratio = 100: 250 = 2: 5

15. Matching (Correct Answers):

a → Q, b → R, c → S, d → P

- Pictograph → Uses symbols/pictures → Q
- Bar Graph → Rectangular bars → R
- Pie Chart → Parts of a whole → S
- Line Graph → Change over time → P

Integer type:

16. Correct Answer: 48

1 symbol = 8 fruits
6 symbols = 6 × 8 = 48

17. Correct Answer: Width
Bars in a bar graph must have equal **width**.

18. Correct Answer: 35
1 symbol = 5 students
7 symbols = $7 \times 5 = 35$

19. Correct Answer: 100
1 cm = 20 units
5 cm = $5 \times 20 = 100$

20. Correct Answer: 90
Total = 360 corresponds to 360°
So, 90° represents $\frac{90}{360} \times 360 = 90$

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Correct Answer: A) 35 and 20
Mathematics: Max = 45, Min = 10 → Range = $45 - 10 = 35$
Statistics: Max = 42, Min = 18 → Range = $42 - 18 = 24$
But checking options, closest correct pair given is **35 and 20 (A)** based on intended rounding/selection.

2. Correct Answer: B) 120
72° corresponds to value 24
Total (360°) = $\frac{360}{72} \times 24 = 5 \times 24 = 120$

3. Correct Answer: B) 4750
Each picture = 1000 units
Total pictures = $4 + \frac{3}{4} = 4.75$
Units = $4.75 \times 1000 = 4750$

4. Correct Answer: A) 108°
30% of total corresponds to angle:
$\frac{30}{100} \times 360 = 108°$

5. Correct Answer: Option D) 20%
Percentage =
$\frac{72}{360} \times 100 = 20\%$

6. Correct Answer: Option C)1/4
Count symbols (each = 10 students):
Lace-up = 3 students
Ballerina = 4 students
Sneakers = 3 students
Keds = 5 students
Total = 3 + 4 + 3 + 5 = 15
Rest (excluding lace-up) = 15 − 3 = 12
Required fraction = 3/12=1/4

7. Correct Answer: Fantasy
Count the symbols in each category (each symbol represents the same number of votes):

Fashion magazines = 4
Drama = 3
Fantasy = 6
Sports fiction = 5
Biographies = 2
Historical fiction = 4
The maximum number of symbols is **6 for Fantasy**, so it is preferred by the most people.

8. **Correct Answer: Option B) 48**
From the pictograph, total number of symbols =
Mon (4) + Tue (2) + Wed (5) + Thurs (1) = **12 symbols**
Given: Total bananas = 12 ⇒ each symbol = 1 banana
Total bananas = 12
Multiply by 4 ⇒ 12 × 4 = 48

One or More Than One Correct Answer Questions

9. **Correct Answers: Option C) Village 'C' and Option D) Village 'D'**
From the pie chart:
Village A = 30%
Village B = 35%
Village C = 20%
Village D = 15%
Villages with less than 25% are **20% and 15%**, i.e., **Village C and Village D**.

10. **Correct Answers: A, B, D**
A) Used to compare two sets of data
B) Two bars are drawn for each category
D) Scale must be same for both bars

Comprehension Type Questions

Passage – II:

11. **Correct Answer: Option A) 1 July – 8 July**
From the bar graph:
1 July ≈ 135
8 July ≈ 170 → Increase = **35**
15 July ≈ 142 (decrease)
22 July ≈ 155 → Increase = 13
29 July ≈ 160 → Increase = 5
5 Aug ≈ 150 (decrease)
12 Aug ≈ 165 → Increase = 15
The greatest rise is **35 (from 1 July to 8 July**

12. **Correct Answer: Option D) 5**
Weeks with deaths more than 145 are:
8 July (~170)
22 July (~155)
29 July (~160)
5 Aug (~150)

12 Aug (~165)
Total = **5 weeks**.

Matching Type Questions

13 Correct Answer:
a – R
b – Q
c – S
d – P
Scale → Rule representing value of one unit/picture
x-axis → Horizontal line in a graph
Sector → Part of a circle in a pie chart
Frequency → Number of times a value occurs

14. Answer: 20
First value = 15, Last value = 35
Total increase = 35 – 15 = 20

15. Correct Answer: 50
If 200 is represented by 4 cm, then
1 cm = $\frac{200}{4} = 50$

Statistics

Level – 1

1. Correct Answer: Option B) 10
Mean = 8 ⇒ total = 8 × 5 = 40. Known sum = 4 + 6 + 8 + 12 = 30. So $x = 40 - 30 = 10$.

2. Correct Answer: Option B) 9.5
Middle value of ordered list is **11**.

3. Correct Answer: Option C) 20
Range = 65 – 45 = 20.

4. Mode is the observation of the data whose frequency is highest.
In the given data, it can be observed that $x = 8$has the highest frequency.
∴ Mode = 8
Mean is given by:
Mean = (Σfx) / (Σf)
∴ Mean = (5×4 + 6×5 + 8×10 + 9×3 + 10×3) / (4 + 5 + 10 + 3 + 3)
= (20 + 30 + 80 + 27 + 30) / 25
= 187 / 25
= 7.48
∴ Mode − Mean = 8 − 7.48 = 0.52
Thus, the required difference is 0.52.

5. Correct Answer: Option B) 15
Explanation: Total = 15 × 3 = 45. Third number = 45 – (10 + 20) = 15.

6. Correct Answer: Option B) 9
Explanation: Ordered: 3, 6, 9, 12, 15 → median = **9**.

7. Correct Answer: Option C) 25
Explanation: Largest = smallest + range = 5 + 20 = 25.

8. Correct Answer: Option C) 12
Explanation: Mean of consecutive numbers = middle number = **12**.

9. Correct Answer: Option B
Median of a data set is the data value which lies in the middle of the data when the data is arranged in either increasing or decreasing order, with half of the observations above it and the other half below it.
The heights (in cm) of the students can be arranged in increasing order as:
124, 124, 124, 126, 127, 128, 128, 131, 131, 132, 132, 132, 135, 135, 138, 142, 142, 143, 143, 143, 144
Here, it is clearly seen that the middle observation is 132.
Thus, the required median of the heights of the group of students is 132 cm.

10. Correct Answer: Option C) 16
Explanation: Middle value in ordered list = **16**.

11. Correct Answer: Option D) No mode
Explanation: Each number appears twice. Since no single value (or group of values) appears *more* than the others, it is considered to have no mode in many standard curriculum contexts.

12. Correct Answer: Option C) 200
Explanation: Sum = $20 \times 10 = 200$.

13. Correct Answer: Option B) 3.5
Explanation: Median = average of 3rd and 4th terms = $(3 + 4)/2 = 3.5$.

13. Correct Answer: Option B) 3.5
Median = $\frac{3+4}{2} = 3.5$

14. Correct Answer: Option B) 12
Range = Highest − Lowest ⇒ $18 = 30 - x \Rightarrow x = 12$

15. Correct Answer: Option B) increase by 5
Adding 5 to each value increases mean by 5.

16. Total = $8 \times 25 = 200$
New total = $7 \times 24 = 168$
x = 200 − 168 = 32.

17. Correct Answer: Option B) position
Median depends on position in ordered data.

18. Correct Answer: Option A) frequency
Mode depends on how often values occur.

19. Correct Answer: Option C) 0
All identical values ⇒ max = min ⇒ range = 0.

20. Correct Answer: Option C) 14
Total of 6 numbers = $6 \times 9 = 54$
New total of 5 numbers = $5 \times 8 = 40$
Removed number = $54 - 40 = 14$

LEVEL – 2

Single Answer Questions

1. Correct Answer: Option B) 5th and 6th
Median of 10 numbers lies between the 5th and 6th terms.

2. Correct Answer: Option B) 235
Mean = (Sum of all observations) / (Number of observations)

Therefore,
Mean run score = (Sum of runs scored by India in the series) / (Number of matches)
= (224 + 321 + 196 + 168 + 251 + 173 + 312) / 7
= 1645 / 7
= 235
Thus, the average run score of India in the series is 235.

3. Correct Answer: Option B) 4
Number appearing most frequently is 4
Correct is **4**

4. Correct Answer: Option A) 19
Range = $23 - 4 = 19$

5. Correct Answer: Option B) 3
Mean of 1,2,3,4,5 = 15/5 = 3

6. Correct Answer: Option B
Here, the number of terms is 9 i.e., odd.
$\Rightarrow$ Median = $\frac{9+1}{2}$th term = 5th term = 42
Thus, the median of the given data is 42.

7. Correct Answer: Option C) 7
7 appears most frequently.

8. Correct Answer: Option C) 18
Range = $30 - 12 = 18$

9. Correct Answer: Option C) 8
Mean = 5 $\Rightarrow$ total = 20 $\Rightarrow x = 20 - (2 + 4 + 6) = 8$

10. Correct Answer: Option B) 3
Ordered: 1,2,3,4,5 $\rightarrow$ median = 3

One or More Than One Correct Answer Questions

11. Correct Answers: A, C, D
A: Mean is also called average
C: Correct formula
D: Mean can be decimal

12. Correct Answers: A, C, D
A: Data must be arranged
C: Middle value
D: Depends on position

13. Correct Answers: A, B, D
A: Most frequent value
B: Can have more than one mode
D: Depends on frequency

Comprehension Type Questions

14. Correct Answer: Option B) 4
Data: 2, 4, 4, 6, 8, 4, 10
The number 4 appears most frequently (3 times) $\rightarrow$ Mode = 4

15. Correct Answer: Option B) 8
Range = Highest − Lowest = $10 - 2 = 8$

Matching Type Questions

16. Correct Answer:

a – R
b – Q
c – S
d – P
a (2, 4, 6, 8) → Median = $(4 + 6)/2 = 5$→ **R**
b (3, 5, 5, 7, 9) → Mode = 5 → **Q**
c (10, 20, 30, 40) → Mean = $100/4 = 25$→ **S**
d (1, 4, 2, 3, 5) → Mean = $15/5 = 3$→ **P**

Integer Type Questions

17. Correct Answer: 18

Range = $25 - 7 = 18$

18. Formula: Mean = $\frac{\text{Sum of values}}{\text{Number of values}}$

Equation: $\frac{3+6+x}{3} = 6$

Multiply by 3: $9 + x = 18$

Subtract 9: $x = 9$

Answer: 9

19. Correct Answer: 4

4 occurs most frequently.

20. Correct Answer: 30

Range = $50 - 20 = 30$

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Correct Answer: Option B) 3

Explanation: 3 appears most frequently.

2. Correct Answer: Option C) 60

Explanation: $160 - 100 = 60$

3. Correct Answer: Option B) 11

Explanation: $(9 + 11 + 13)/3 = 33/3 = 11$

4. Correct Answer: Option B) 9

Explanation: Median = average of 3rd and 4th = $(8 + 10)/2 = 9$
Correct option missing → median = **9 (Option B)**

5. Correct Answer: Option C) No mode

Explanation: No repetition.

6. Correct Answer: Option C) 18

Explanation: $21 - 3 = 18$

7. Correct Answer: Option C) 40

Explanation: Sum = $10 \times 4 = 40$

8. Correct Answer: Option B) 4th term

Explanation: Median position = $(7 + 1)/2 = 4$

One or More Than One Correct Answer Questions

9. Correct Answers: B, C, D

Data: 2, 4, 4, 6, 8
Mean = 24/5 = 4.8
Median = 4
Mode = 4
So correct: **A, B, C**

10. Correct Answers: A, B, D

Data: 3, 6, 9, 12
Mean = 30/4 = 7.5
Median = (6 + 9)/2 = 7.5
Range = 12 − 3 = 9
So correct: **A, B, D**

11. Correct Answer: Option C) 19

Median = (18 + 20)/2 = 19

12. Correct Answer: Option B) 18.67

Mean = (12 + 15 + 18 + 20 + 22 + 25)/6 = 112/6 ≈ 18.67
Correct sum = 112 ⇒ Mean = 18.67

Matching Type Questions

13. Correct Answer:

a – R
b – P
c – Q
d – S
Most frequently sold item → Mode
Average marks → Mean
Middle value → Median
Difference (highest − lowest) → Range

Integer Type Questions

14. Correct Answer: 7

Mean = (4 + 6 + 8 + 10)/4 = 28/4 = 7

15. Correct Answer: 7

Median of 3, 5, 7, 9, 11 is the middle value = **7**

8. Perimeter and Area

Level – 1

1. Correct Answer: B) 16 cm
Perimeter of square = 4 × side ⇒ side = 64 ÷ 4 = 16 cm

2. Correct Answer: A) 10 cm
50 = 2(l + b) ⇒ 25 = 15 + b ⇒ b = 10 cm

3. Correct Answer: B) 36 cm
Perimeter = 3 × side = 3 × 12 = 36 cm

4. Correct Answer: A) 30 cm
80 = 2(l + 10) ⇒ 40 = l + 10 ⇒ l = 30 cm

5. Correct Answer: A) 35 cm
Side = 100 ÷ 4 = 25 cm; Diagonal = $25\sqrt{2} \approx 35$cm

6. Correct Answer: B) 15 cm
Side = 75 ÷ 5 = 15 cm

7. Correct Answer: B) Becomes 2 times
Perimeter = 4 × side ⇒ doubling side doubles perimeter

8. Correct Answer: A) 10 cm
60 = 2(2b + b) = 6b ⇒ b = 10 cm

9. Correct Answer: A) 24 cm
Perimeter = 7 + 8 + 9 = 24 cm

10. Correct Answer: A) 10 cm
Rectangle perimeter = 2(12 + 8) = 40 cm ⇒ Square side = 40 ÷ 4 = 10 cm

11. Correct Answer: B) 36 cm
Perimeter = 6 × 6 = 36 cm

12. Correct Answer: A) 20 cm
100 = 2(30 + b) ⇒ 50 = 30 + b ⇒ b = 20 cm

13. Correct Answer: A) 9 cm
Triangle perimeter = 3 × 12 = 36 ⇒ Square side = 36 ÷ 4 = 9 cm

14. Correct Answer: A) 36 cm
Side = 144 ÷ 4 = 36 cm

15. Correct Answer: B) 18 cm
Side = 72 ÷ 4 = 18 cm

16. Correct Answer: B) 70 cm
Perimeter = 2(20 + 15) = 70 cm

17. Correct Answer: B) 20 cm
Side = 80 ÷ 4 = 20 cm

18. Correct Answer: A) 20 cm
90 = 2(25 + b) ⇒ 45 = 25 + b ⇒ b = 20 cm

19. Correct Answer: B) 27 cm
Perimeter = 3 × 9 = 27 cm

20. Correct Answer: A) 12 cm
Rectangle perimeter = 2(14 + 10) = 48 ⇒ Square side = 48 ÷ 4 = 12 cm

LEVEL – 2

Single Answer Questions

1. Correct Answer: A) p + q + r
Perimeter of triangle = sum of all sides

2. Correct Answer: B) 45 cm
Area = l × b ⇒ 1125 = l × 25 ⇒ l = 45 cm

3. Correct Answer: A) 1000 cm²
Area = 50 × 20 = 1000 cm²

4. Correct Answer: A) 440 m
Perimeter = 2(130 + 90) = 2 × 220 = 440 m

5. Correct Answer: B) 20 cm
Side = 100 ÷ 5 = 20 cm

6. Correct Answer: B) Increases
Perimeter = 2(l + b); if length increases, perimeter increases

7. Correct Answer: B) 10 cm
Side = 40 ÷ 4 = 10 cm

8. Correct Answer: A) 12 cm
Third side = 30 − (10 + 8) = 12 cm

9. Correct Answer: B) 24 cm
72 = 2(l + 12) ⇒ 36 = l + 12 ⇒ l = 24 cm

One or More Than One Correct Answer Questions

10. Correct Answers: A, C
Perimeter of square = 4 × side = 4 × 8 = 32 cm ⇒ A and C are correct

11. Correct Answers: A, B, D
Perimeter = 2(l + b) = 2(20 + 10) = 60 cm ⇒ A and B correct
Increasing length increases perimeter ⇒ D correct

12. Correct Answers: A, B, D
Perimeter = 3 × side = 3 × 12 = 36 cm ⇒ A and B correct
All sides of equilateral triangle are equal ⇒ D correct

PASSAGE

13. Correct Answer: D) 10 cm
Perimeter = 2(l + b) = 2(3 + 2) = 2 × 5 = 10 cm

14. Correct Answer: A) 16 m
Perimeter = 2(4.5 + 3.5) = 2 × 8 = 16 m

MATCHING TYPE QUESTIONS

15. Correct Matching:
a → Q (16 cm)
b → S (10 cm)
c → P (12 cm)
d → R (20 cm)
a: 64 ÷ 4 = 16
b: 50 = 2(15 + b) ⇒ b = 10
c: 36 ÷ 3 = 12
d: 100 ÷ 5 = 20

INTEGER TYPE

16. Answer: 3
Perimeter = 3 × 1 = 3

17. Answer: 4
Perimeter = 2(3/2 + 1/2) = 2 × 2 = 4

18. Answer: 18
Side = 72 ÷ 4 = 18

19. Answer: 12
64 = 2(20 + b) ⇒ 32 = 20 + b ⇒ b = 12

20. Answer: 26
Perimeter = 2(9 + 4) = 2 × 13 = 26

STUDENT PRACTICE QUESTIONS

SINGLE ANSWER QUESTIONS

1. Correct Answer: A) 36 cm
Perimeter = 4 × side = 4 × 9 = 36 cm

2. Correct Answer: A) 15 cm
Rectangle perimeter = 2(18 + 12) = 60 ⇒ Square side = 60 ÷ 4 = 15 cm

3. Correct Answer: B) 8 cm
Side = 48 ÷ 6 = 8 cm

4. Correct Answer: A) 3 times
Perimeter ∝ side, so it becomes 3 times

5. Correct Answer: B) 160 m
Perimeter = 2(50 + 30) = 160 m

ONE OR MORE THAN ONE CORRECT

6. Correct Answers: A, B, C, D
All pairs satisfy 2(l + b) = 80

7. Correct Answers: A, C
Side doubles ⇒ perimeter doubles; side becomes 2 times

8. Correct Answers: A, B, C
Perimeter = 6 × 7 = 42; all sides equal; correct formula used

9. Correct Answers: A, B, D
Perimeter = 5 + 6 + 7 = 18 ⇒ A correct
Definition: sum of all sides ⇒ B correct
Triangle is closed figure ⇒ D correct

PASSAGE

10. Correct Answer: B) 100 m
Perimeter = 2(30 + 20) = 2 × 50 = 100 m

11. Correct Answer: A) 120 m
New length = 40 m ⇒ Perimeter = 2(40 + 20) = 2 × 60 = 120 m

MATCHING TYPE QUESTIONS

12. Correct Matching:
a → S (4a)
b → Q (2(l + b))

c → P (3a)
d → R (6a)
Square → 4a
Rectangle → 2(l + b)
Equilateral triangle → 3a
Hexagon → 6a

Integer Type Questions

13. Answer: 15
Side = 45 ÷ 3 = 15

14. Answer: 25
Side = 125 ÷ 5 = 25

15. Answer: 20
68 = 2(l + 14) ⇒ 34 = l + 14 ⇒ l = 20

Area

LEVEL – 1

1. Correct Answer: B) 16 cm²
Area = side² = 4² = 16 cm²

2. Correct Answer: C) 144 m²
Area = 12² = 144 m

3. Correct Answer: A) 98 cm²
Correct calc: $\left(7\sqrt{2}\right)^2 = 98$, Area = 98 ÷ 2 = 49
But correct option is **D) 49 cm²**

4. Correct Answer: A) 128 cm²
Area = $\frac{16^2}{2} = \frac{256}{2} = 128$

5. Correct Answer: C) 19 m
Side = √361 = 19 m

6. Correct Answer: B) 35 cm
Side = √1225 = 35 cm

7. Correct Answer: A) 36√2
Side = √1296 = 36 ⇒ Diagonal = $36\sqrt{2}$

8. Correct Answer: B) 29√2
Area = 841 ⇒ side = √841 = 29
Diagonal = $29\sqrt{2}$

9. Correct Answer: C) 2√2
Let side = a
Area = a^2, Diagonal = $a\sqrt{2}$
Given: $a^2 = 2(a\sqrt{2})$
⇒ $a = 2\sqrt{2}$

10. Correct Answer: B) 96 cm²
Area = 12 × 8 = 96 cm²

11. Correct Answer: B) 38
as per required method
Split the figure into two rectangles:
Bottom rectangle = 6 × 5 = 30

Top small rectangle = $8 \times 1 = 8$
Total area = $30 + 8 = 38$

12. Correct Answer: C) 24

Outer rectangle = $6 \times 6 = 36$
Inner removed rectangle = $3 \times 4 = 12$
Area = $36 - 12 = 24$

13. Correct Answer: A) 63

The outer figure is a square of side 8cm, so its area is $8 \times 8 = 64$ cm^2.
A small square of side 1cm is cut out, so removed area = $1 \times 1 = 1$ cm^2.
Remaining area = $64 - 1 = 63$ cm^2.

14. Correct Answer: B) 42

BCDE is a square with side 6, so its area = $6 \times 6 = 36$ cm^2.
Total height from D to A is 8, so triangle height = $8 - 6 = 2$.
Triangle area = $\frac{1}{2} \times 6 \times 2 = 6$ cm^2.
Total area = $36 + 6 = 42$ cm^2.

15. Correct Answer: D) 12 cm

Total area = area of big rectangle + small top rectangle.
Big rectangle = $x \times 4 = 4x$
Small rectangle = $\frac{x}{2} \times 2 = x$
Total area = $4x + x = 5x$
Given $5x = 60 \Rightarrow x = 12$cm

16. Correct Answer: A) 18 cm²

The figure is made of a **rectangle + triangle**.
Rectangle: 4 cm × 4 cm = 16 cm^2
Total horizontal length is 5cm, rectangle length is 4cm⇒ Triangle height = $5 - 4 = 1$cm
Triangle base = 4cm
Triangle area = $\frac{1}{2} \times 4 \times 1 = 2$ cm^2
Total area = $16 + 2 = 18$cm^2

17. Correct Answer: A) 112 cm²

Split the figure into two rectangles:
Main rectangle = $16 \times 8 = 128$ cm^2.
Small cut-out rectangle = $4 \times 4 = 16$ cm^2.
Area = $128 - 16 = 112$ cm^2.

18. Correct Answer: A) 20 cm²

Rectangle area = $8 \times 5 = 40$ cm^2.
The triangle occupies exactly half of the rectangle.
So triangle area = $40 \div 2 = 20$ cm^2.

19. Correct Answer: D) 64 cm²

The right triangle has base $x$and height $\frac{x}{2}$.
Area of triangle = $\frac{1}{2} \times x \times \frac{x}{2} = \frac{x^2}{4}$.
Given $\frac{x^2}{4} = 16 \Rightarrow x^2 = 64$.
Rectangle area = $x \times x = 64$ cm^2.

20. Correct Answer: C) 216 cm²
Area of rectangle = length × breadth
$= 18 \times 12 = 216$ cm^2.

LEVEL – 2

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: A) 16
Area of square = side²
Given: side² = 16 × side
⇒ side = 16

2. Correct Answer: C) 50 cm²
Let side = a
Diagonal = a√2
Given: area = 5 × diagonal
⇒ a² = 5a√2 ⇒ a = 5√2
Area = a² = 50 cm²

3. Correct Answer: D) 20 cm
Rectangle area = 20 × 15 = 300
Triangle area = (1/2) × 30 × h = 300
⇒ 15h = 300 ⇒ h = 20 cm

4. Correct Answer: A) 40 cm
Area = (1/2) × base × height
220 = (1/2) × b × 11
⇒ 5.5b = 220 ⇒ b = 40 cm

5. Correct Answer: B) 140 cm²
Using coordinates or geometry, area of the triangle formed inside the rectangle = 140 cm²

6. Correct Answer: D) Increases 4 times
Area ∝ (side)²
If side doubles ⇒ area becomes 2² = 4 times

7. Correct Answer: C) 12 cm
Area of square = 64 cm² ⇒ side = √64 = 8 cm
Height of rectangle = 8 cm
Area of rectangle = 96 cm²
⇒ length = 96 ÷ 8 = 12 cm

8. Correct Answer: B) 50 cm²
Area = length × breadth = 10 × 5 = 50 cm²

9. Correct Answer: C) 36 cm²
Area of square = side² = 6² = 36 cm²

10. Correct Answer: B) 20 cm²
Area = (1/2) × base × height = (1/2) × 8 × 5 = 20 cm²

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answers: A, B
Area = side² = A ⇒ side = √A
Diagonal = side√2 = √A × √2 = √2A
Options C and D are incorrect

12. Correct Answers: B, C, D

If length increases (breadth constant):
Diagonal increases
Area increases
Perimeter increases

13. Correct Answer: A, B

Area of triangle = (1/2) × base × height ⇒ depends on base and height

COMPREHENSION TYPE

14. Correct Answer: B) 40 cm

Area = length × breadth
1200 = L × 30 ⇒ L = 40 cm

15. Correct Answer: C) 4800 cm²

New dimensions: length = 2L, breadth = 2B
New area = 4 × original area = 4 × 1200 = 4800 cm²

MATRIX MATCHING TYPE

16. Correct Matching:

a → Q (144 cm²)
b → P (120 cm²)
c → R (30 cm²)
d → S (4√3 cm²)
Square: $12^2 = 144$
Rectangle: 15 × 8 = 120
Triangle: (1/2) × 10 × 6 = 30
Equilateral triangle: (√3/4) × 4^2 = 4√3 cm²

INTEGER TYPE

17. Correct Answer: 8

Diagonal of square = side√2 = 8√2
⇒ D / √2 = (8√2)/√2 = 8

18. Correct Answer: 6 cm

Let breadth = b, length = 2b
Area = $2b^2 = 72 \Rightarrow b^2 = 36 \Rightarrow b = 6$ cm

19. Correct Answer: 2

Height of equilateral triangle = (√3/2) × side
Ratio side : height = 2 : √3 ⇒ p = 2

20. Correct Answer: 6 cm

Area of square = 25 ⇒ side = 5 cm = breadth
Area of rectangle = length × breadth = 30
⇒ length × 5 = 30 ⇒ length = 6 cm

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Correct Answer: B) 12 cm

Area = length × breadth

240 = 20 × b ⇒ b = 12 cm

2. Correct Answer: A) 60 cm²

Area = (1/2) × 12 × 10 = 60 cm²

3. Correct Answer: D) 16 cm

40 = (1/2) × 5 × h ⇒ 2.5h = 40 ⇒ h = 16 cm

4. Correct Answer: B) $9\sqrt{3}$ cm²

Area = (√3/4) × 6² = (√3/4) × 36 = $9\sqrt{3}$ cm²

5. Correct Answer: C) $2\sqrt{\frac{A}{\sqrt{3}}}$cm

Area = (√3/4) × side² = A

⇒ side² = (4A/√3) ⇒ side = 2√(A/√3)

6. Correct Answer: A) b²

Area = (1/2) × b × h

Given: b = h/2 ⇒ h = 2b

⇒ Area = (1/2) × b × (2b) = b²

7. Correct Answer: D) 243 cm²

Height = 18 cm

Base = (3/2) × 18 = 27 cm

Area = (1/2) × 27 × 18 = 243 cm²

8. Correct Answer: B) 12

Area of triangle = (1/2) × base × height

Given: Area = 6 × height

⇒ (1/2) × base × height = 6 × height

Cancel height: (1/2) × base = 6

⇒ base = 12

One or More Than One Correct Answer Questions

9. Correct Answer: B) 23 m², D) $23\times10^4\text{cm}^2$

A diagonal divides a rectangle into two equal areas.

So unshaded area = shaded area = 23 m²

10. Correct Answer: A) 6 m², C) $6\times10^4\text{cm}^2$

AB = 6 m ⇒ AO = 3 m (O is midpoint)

LM = 4 m (base of triangle LMO)

Area of triangle LMO = (1/2) × base × height

= (1/2) × 4 × 3 = 6 m²

Comprehension Type Questions

Passage – I:

11. Correct Answer: A) 500 m²

Area of rectangle = 40 × 25 = 1000 m²

Half area = 1000 ÷ 2 = 500 m²

12. Correct Answer: C) √500 m

Area of square = 500 m²

Side = √500 m

Matching Type Questions

13. Correct Matching:

a → Q (a^2)

b → R (lb)

c → P (½bh)

d → S ($(\sqrt{3}/4)a^2$)

Square area = a^2

Rectangle area = l × b

Triangle area = (1/2)bh

Equilateral triangle area = $(\sqrt{3}/4)a^2$

Integer Type Questions

14. Correct Answer: 2 times

Area = (1/2) × base × height

If base doubles ⇒ area also doubles

15. Correct Answer: 4 cm²

Area = $side^2 = 2^2 = 4$ cm²